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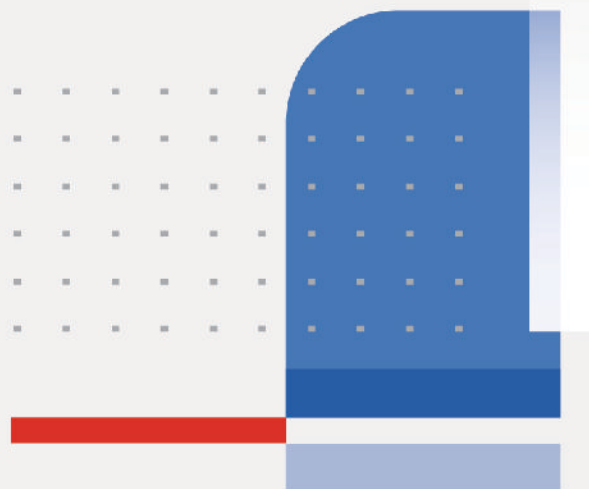
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FortiGate Administrator Lab Guide

FortiOS 7.6

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6/18/2025

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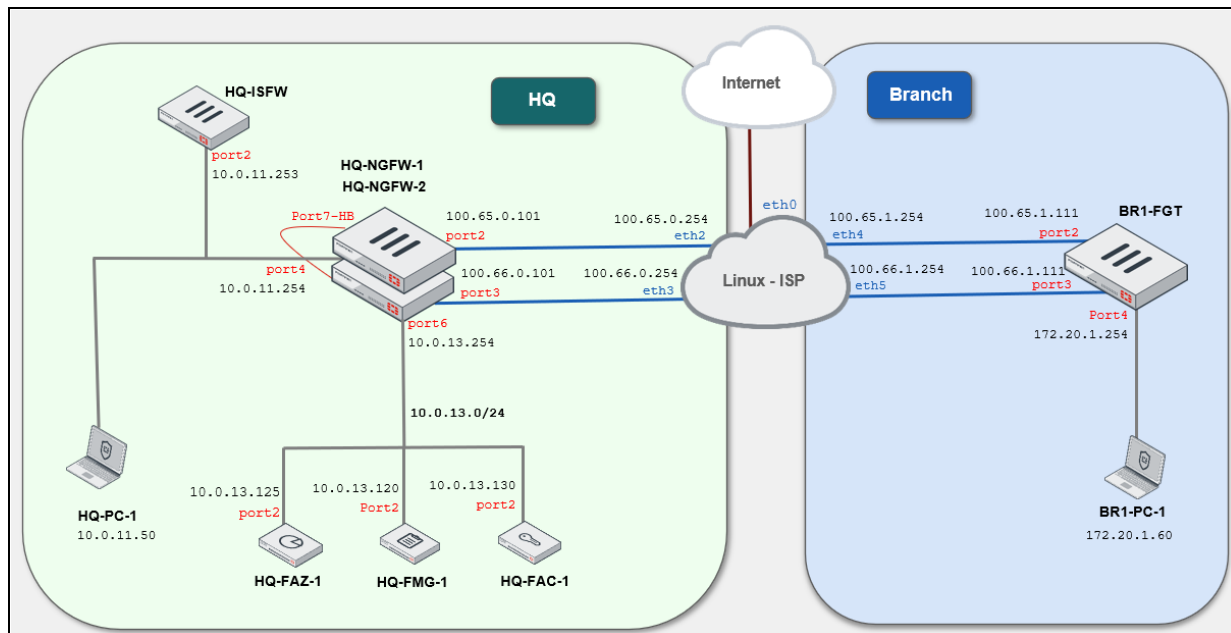
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Change Log

This table includes updates to the *FortiGate Administrator Lab Guide* dated 1/16/2025 to the updated document version dated 6/18/2025.

Change	Location
References to Skype removed	Lab 8 Configuring FortiGuard Web Filtering on page 125

Network Topology



Lab 1: System and Network Settings

In this lab, you will learn about FortiGate basic system and network factory default settings and apply changes and modify the default factory settings. You will also perform administrative tasks through the CLI and GUI and back up and restore a configuration file, as well as create a new administrator account and modify administrator access permissions.

Objectives

- Review and change network settings
- Access the FortiGate CLI
- Back up and restore configuration files
- Locate the FortiGate model and FortiOS firmware build in a configuration file
- Create a new administrator user
- Restrict administrator access

Time to Complete

Estimated: 30 minutes

VM Usernames and Passwords

VM	Username	Password
HQ-PC-1	Administrator	Fortinet1!
BR1-PC-1	Administrator	Fortinet1!
HQ-NGFW-1	admin	Fortinet1!
BR1-FGT	admin	Fortinet1!
HQ-ISFW	admin	Fortinet1!
HQ-FAZ-1	admin	Fortinet1!

Exercise 1: Configuring FortiGate System Settings

In this exercise, you will review the HQ-NGFW-1 system settings and make changes to finish setting up FortiGate on your network. You will enable the internal network DHCP server to allow hosts to receive the IP address when they connect to HQ-NGFW-1.

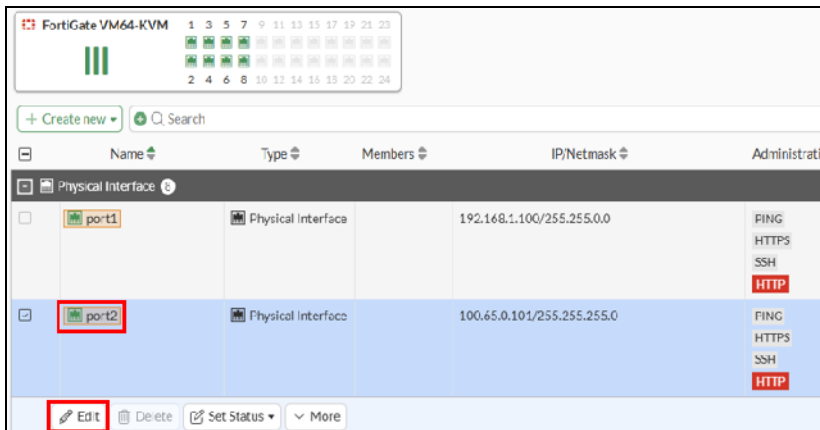
Some of the settings in this lab have been preconfigured and are not the factory default settings of FortiGate.

Review HQ-NGFW-1 Network Settings

First, you will review the port2 network interface on HQ-NGFW-1. Next, you will review the static routes.

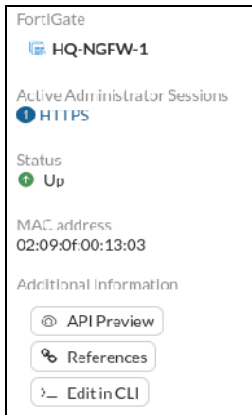
To review the port2 network interface

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Network > Interfaces**.
3. Click **port2**, and then click **Edit**.



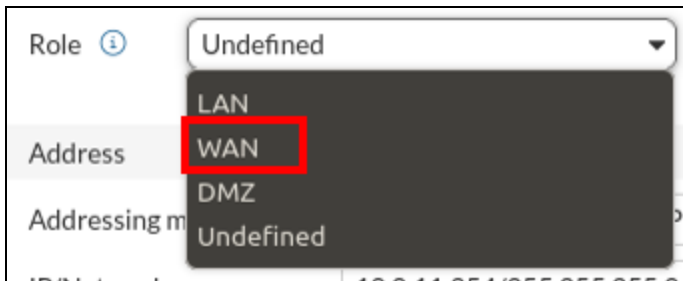
You can double-click an object on the FortiGate GUI to view or edit the content of the object.

4. In the **Edit Interface** window, review the information available on the right.



FortiGate displays its host name and status without the need to navigate away from your current work.

5. In the **Role** field, select **WAN**.



Stop and think!

Why do some of the settings appear or disappear when the role of an interface changes?

Each role reflects the appropriate settings required to configure the interface. The **Undefined** role displays all of the settings you can configure on an interface.



The reason for choosing **WAN** as the role of the interface is to set it up for an external connection. When you select the **WAN** role, some settings, like the DHCP server, will not be available.

6. In the **Estimated bandwidth** fields, review the WAN utilization values.
When the role of the interface is set to **WAN**, you can configure the downstream and upstream maximum bandwidth.
7. Click **Cancel** to clear any changes that you made.

To review the static default gateway on HQ-NGFW-1

1. Continuing on the HQ-NGFW-1 GUI, click **Network > Static Routes**.
2. Click the static route entry, and then click **Edit**.

+ Create new		+ Search			
☒	Destination	Gateway IP	Interface	Status	Comments
☒	0.0.0.0/0	100.65.0.254	port2	Enabled	
☒ Edit ☐ Clone ☐ Disable ☐ Delete More					

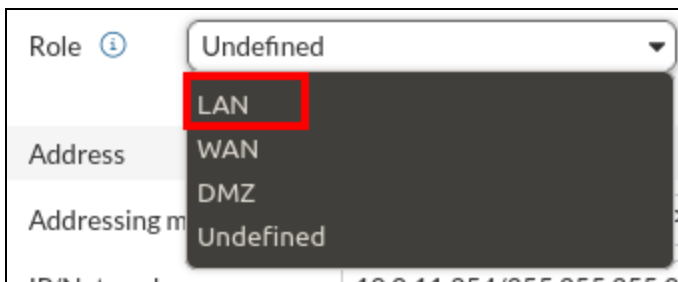
- Expand the **Advanced Options** section.
You can set the priority value of the static route. When two routes have an equal distance, the route with a lower priority number takes precedence.
- Click **Cancel** to clear any changes that you made.

Enable the DHCP Server on HQ-NGFW-1

You will enable the DHCP server on port4.

To enable the DHCP server on port4

- Continuing on the HQ-NGFW-1 GUI, click **Network > Interfaces**.
- Click **port4**, and then click **Edit**.
- In the **Role** field, select **LAN**.



- Enable **DHCP Server**.



Notice that in the **LAN** role, the DHCP server appears on the GUI, unlike when the role is set to **WAN**.

- In the **Address range** field, type 10.0.11.1-10.0.11.250.

DHCP Server

DHCP status: **Enabled** (Up arrow) / Disabled (Down arrow)

Address range: **10.0.11.1-10.0.11.250**

Netmask: 255.255.255.0

Default gateway: **Same as Interface IP** / Specify

DNS server: **Same as System DNS** / Same as Interface IP / Specify

Lease time: **604800** second(s)

Advanced

6. Click **OK** to save the changes.

Exercise 2: Working With the CLI

In this exercise, you will access a FortiGate using the CLI.

1. Open the HQ-NGFW-1 CLI, and then log in with the following credentials:

- Username: `admin`
- Password: `Fortinet1!`

2. Enter the following command:

```
get system status
```

This command displays basic status information about FortiGate. The output includes the FortiGate serial number, operation mode, and so on. When the `More` prompt appears on the CLI, perform one of the following actions:

Action	Command
To continue scrolling	Press the space bar.
To scroll one line at a time	Press <code>Enter</code> .
To exit	Type <code>q</code> .

3. Enter the following command:

```
get ?
```



The `?` character is not displayed on the screen.

This command shows all options that the CLI will accept after the `get` command. Depending on the command, you may need to enter additional words to completely specify a configuration option.

4. Press the up arrow key.

This displays the previous `get system status` command.

5. Try some of the control key sequences that are shown in the following table:

Action	Command
Previous command	Up arrow
Next command	Down arrow
Beginning of line	<code>Ctrl+a</code>
End of line	<code>Ctrl+e</code>
Back one word	<code>Ctrl+b</code>

Action	Command
Forward one word	Ctrl+f
Delete current character	Ctrl+d
Clear screen	Ctrl+l
Abort command and exit	Ctrl+c
Auto repeat history	Ctrl+p

6. Enter the following command:

```
execute ?
```

This command lists all options that the CLI accepts after the `execute` command.

7. Type `exe`, and then press the `Tab` key.

Notice that the CLI completes the current word. The `exe` command with the `Tab` key is recommended.

8. Press the `Tab` key three times.

Each time you press the `Tab` key, the CLI replaces the second word with the next possible option for the `execute` command, in alphabetical order.



You can abbreviate most commands. In lessons and labs, many of the commands that you see are in abbreviated form. For example, instead of typing `execute`, you can type `exe`.

Use this technique to reduce the number of keystrokes that are required to enter a command. Often, experts can configure FortiGate faster using the CLI than using the GUI.

If other commands start with the same characters, your abbreviation must be long enough to be specific, so that FortiGate can distinguish them. Otherwise, the CLI displays an error message about ambiguous commands.

9. On a new line, enter the following command to view the port2 interface configuration (hint: try using the shortcuts you just learned about):

```
show system interface port2
```

10. Enter the following command:

```
show full-configuration system interface port2
```

Stop and think!

Compare both outputs. How are they different?

The `show full-configuration` command displays all the configuration settings for the interface. The `show` command displays only those values that are different from the default values.

Exercise 3: Generating Configuration Backups

In this exercise, you will learn how to generate and restore cleartext and encrypted configuration backups. The configuration files that backups produce enable you to restore FortiGate to an earlier configuration.

Restore a Configuration From a Backup

You will restore a configuration from a backup.

To restore a configuration from a backup

1. Log in to the HQ-PC-1 VM with the following credentials:
 - Username: Administrator
 - Password: Fortinet1!

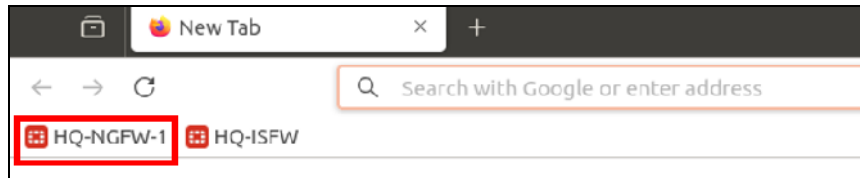


The first time that you log in, you may need to click and drag the screen from the bottom to bring up the login prompt.

2. Open a browser, and then log in to the HQ-NGFW-1 GUI at 10.0.11.254 with the following credentials:
 - Username: admin
 - Password: Fortinet1!

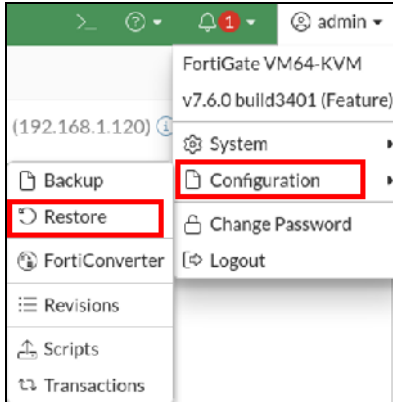


You can also access the HQ-NGFW-1 GUI from the bookmarks bar in the Mozilla Firefox browser.

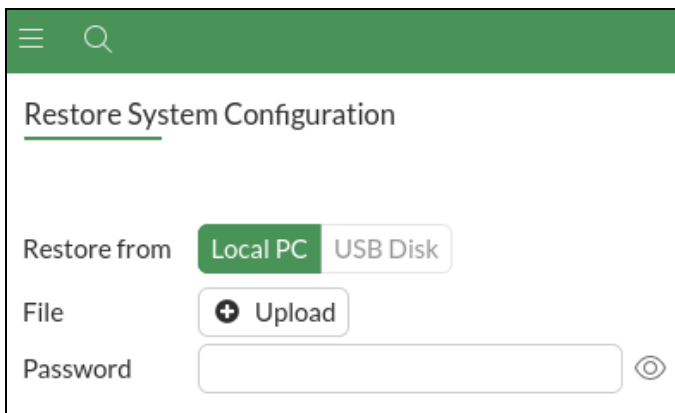


All lab exercises were tested running Firefox on the HQ-PC-1 and BR1-PC-1 VMs. To get consistent results, you should use Firefox to access both the internet and the FortiGate GUIs in this virtual environment.

3. In the upper-right corner, click **admin**, and then click **Configuration > Restore**.



4. Click **Upload** to select the backup configuration file from your local PC.



5. Click **Desktop > Resources > FortiGate-Administrator > System and Network Settings > HQ-NGFW-1-initial.conf**, and then click **Select**.
6. Click **OK**.
7. Click **OK** to reboot.
After your browser uploads the configuration, FortiGate reboots automatically. This takes approximately 30–45 seconds.
8. When the HQ-NGFW-1 GUI login page reappears after reboot, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
9. Click **Network > Interfaces**, and then verify that the network interface settings were restored.

+ Create new		🔍 Search				Group By Type		
☐	Name	Type	Members	IP/Netmask	Administrative access	DHCP clients	DHCP ranges	Ref.
☐	port1	Physical Interface		192.168.1.100/255.255.255.0	PING HTTPS SSH HTTP			0
☐	port2	Physical Interface		100.65.0.101/255.255.255.0	PING HTTPS SSH HTTP			3
☐	port3	Physical Interface		100.66.0.101/255.255.255.0	PING HTTPS SSH HTTP			0
☐	port4	Physical Interface		10.0.11.254/255.255.255.0	PING HTTPS SSH HTTP			2
☐	port5	Physical Interface		10.0.12.254/255.255.255.0	PING HTTPS SSH HTTP			0
☐	port6	Physical Interface		10.0.13.254/255.255.255.0	PING HTTPS SSH HTTP FMC-Access			2
☐	port7	Physical Interface		0.0.0.0/0.0.0.0				0

10. Click **Network > Static Routes**, and then verify that the default route was restored.

+ Create new		Search			
☐	Destination	Gateway IP	Interface	Status	
☐	0.0.0.0/0	100.65.0.254	port2	Enabled	

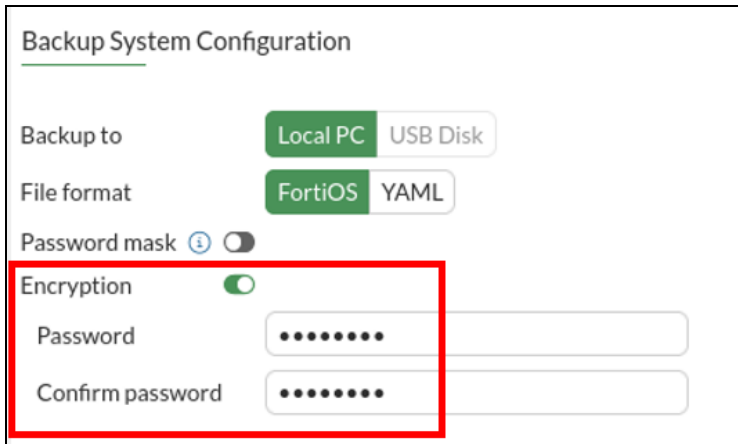
Back Up and Encrypt a Configuration File

Always back up the configuration before making changes to FortiGate (even if the change seems minor or unimportant). There is no *undo*. You should carefully consider the pros and cons of an encrypted backup before you begin encrypting backups. While your configuration, including things like private keys, remains private, an encrypted file hampers troubleshooting because Fortinet Support cannot read the file. Consider saving backups in plaintext, and storing them in a secure place instead.

You will create an encrypted file with the backup of the FortiGate current configuration.

To save an encrypted configuration backup

- On the HQ-PC-1 VM, open a browser, and then log in to the HQ-NGFW-1 GUI at 10.0.11.254 with the following credentials:
 - Username: admin
 - Password: Fortinet1!
- On the HQ-NGFW-1 GUI, in the upper-right corner, click **admin**, and then click **Configuration > Backup**.
- On the **Backup System Configuration** page, enable **Encryption**.
- In the **Password** and **Confirm password** fields, type Fortinet1!.



Backup System Configuration

Backup to: Local PC USB Disk

File format: FortiOS YAML

Password mask: ☐

Encryption: ☒

Password:

Confirm password:

5. Click **OK**.

Firefox saves the encrypted configuration file in the **Downloads** folder, by default. Ensure that you record the password and store it in a secure place.

You can access downloaded files by clicking the download icon in the upper-right corner of the browser.



Restore an Encrypted Configuration Backup

Restoring from a backup enables you to return FortiGate to a previous configuration. As a word of caution, if you cannot recall the password required to decrypt an encrypted backup, you will not be able to restore FortiGate to the backup. Ensure that you record the password and store it in a secure place.

You will restore the configuration backup that you created in the previous procedure.

Take the Expert Challenge!

Restore the configuration from the encrypted backup.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Compare the Headers of Two Configuration Files on page 21](#).

To restore an encrypted configuration backup

1. On the HQ-NGFW-1 GUI, in the upper-right corner, click **admin**, and then click **Configuration > Restore**.
2. On the **Restore System Configuration** page, click **Upload**.
3. Browse to your **Downloads** folder, and then select the configuration file that you created in the previous procedure.
4. In the **Password** field, type `Fortinet1!`, and then click **OK**.
5. Click **OK** to confirm that you want to restore the configuration.
FortiGate reboots.

Compare the Headers of Two Configuration Files

When you troubleshoot issues, or when you restore FortiGate to an earlier OS version or build, it is useful to know where to find the version and build number in a configuration file. This task shows you where to find this information.

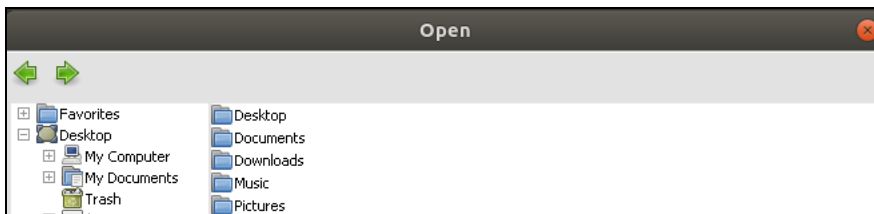
You will open and compare two configuration files using Notepad++.

To compare the headers of two configuration files

1. On the HQ-PC-1 VM, click the Notepad++ icon.



2. Click **File > Open**, and then browse to the **Downloads** folder to open the encrypted configuration file.



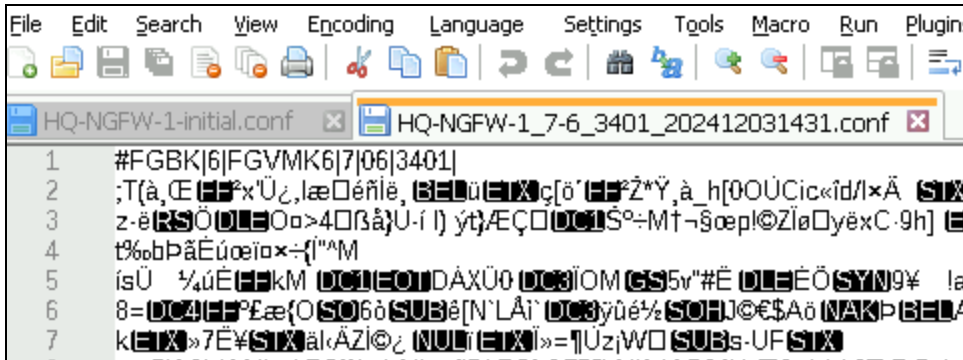
3. Click **File > Open**, and then browse to the initial configuration file:

`Desktop\Resources\FortiGate-Administrator\System and Network Settings\HQ-NGFW-1-initial.conf`

The configuration file opens in a second tab in Notepad++.

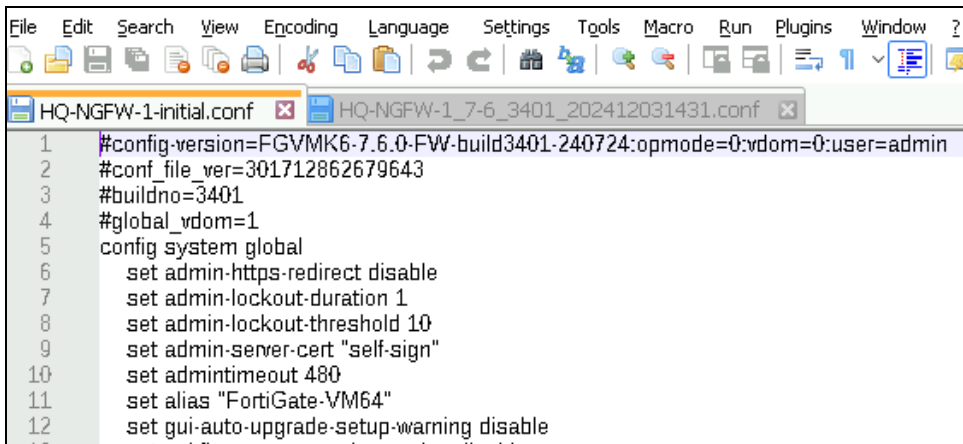
4. Compare the headers in the two files.

The following example is an encrypted file:



```
1 #FGBK[6]FGVMK6[7]06[3401]
2 ;T(ä,CEFFx'Üç,læDéñlë,(BELLÜEIXç[ö'HHZ*Y,ä_h[00UCic«id/I×Ä SIX
3 z-ëRSÖULEOo>4Qßä)U-i I) yq/EÇQUCIS°:Mt-šœpl©ZlæQyëxC-9h] E
4 t%bbPäEúœio×-f{"^M
5 isÜ ¼üEHHKM UCIEOI DAXÜÜ UC3IOM GS5v"#E DLEÖÖ SYN9¥ la
6 8=UC4HFFæ{OSO6ðSUBê[N'LÄi' UC3yüé%SOHJOC$Að NAKP BELA
7 kEIX»7E¥SIXäkÄZl©ç NUL(EIX)»=¶UziWÜ SUBs-UF SIX
```

The following example is a cleartext file:



```
1 #config-version=FGVMK6-7.6.0-FW-build3401-240724:opmode=0:vdom=0:user=admin
2 #conf_file_ver=301712862679643
3 #buildno=3401
4 #global_vdom=1
5 config system global
6     set admin-https-redirect disable
7     set admin-lockout-duration 1
8     set admin-lockout-threshold 10
9     set admin-server-cert "self-sign"
10    set admintimeout 480
11    set alias "FortiGate-VM64"
12    set gui-auto-upgrade-setup-warning disable
```



In both the cleartext and encrypted configuration files, the top line acts as a header, and lists the firmware and model that this configuration belongs to.

5. Close the two tabs in Notepad++, and then close the application.

Exercise 4: Configuring Administrator Accounts

FortiGate offers many options for configuring administrator privileges. For example, you can specify the IP addresses that administrators are allowed to connect from.

In this exercise, you will work with administrator profiles and administrator user accounts. An administrator profile is a role that is assigned to an administrator user that defines what the user is permitted to do on the FortiGate GUI and CLI.

Configure a User Administrator Profile

You will create a new user administrator profile that has read-only access for most of the configuration settings.

To configure a user administrator profile

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **System > Admin Profiles**.
3. Click **Create New**.
4. In the **Name** field, type `Security_Admin_Profile`.
5. In the **Permissions** field, click **Set All**, and then select **Read**.
6. In the permissions table, set **Security Profile** to **Read/Write**.

Access Control	Permissions	Set All
Security Fabric	None Read Read/Write	
FortiView	None Read Read/Write	
User & Device	None Read Read/Write	
Firewall	None Read Read/Write Custom	
Log & Report	None Read Read/Write Custom	
Network	None Read Read/Write Custom	
System	None Read Read/Write Custom	
Security Profile	None Read Read/Write Custom	
VPN	None Read Read/Write	
WAN Opt & Cache	None Read Read/Write	
WiFi & Switch	None Read Read/Write	

7. Click **OK** to save the changes.

Create an Administrator Account

You will create a new administrator account. You will assign the account to the administrator profile you created in the previous procedure. The administrator will have read-only access to most of the configuration settings.

To create an administrator account

1. On the HQ-NGFW-1 GUI, click **System > Administrators**.
2. Click **Create New**, and then click **Administrator** to add a new administrator account.
3. On the **New Administrator** page, configure the following settings:

Field	Value
Username	Security
Type	Local User
Password	Fortinet1!
Confirm Password	Fortinet1!
Administrator Profile	Security_Admin_Profile



Administrator names and passwords are case sensitive. You can't include characters, such as < > () # ", in an administrator account name.

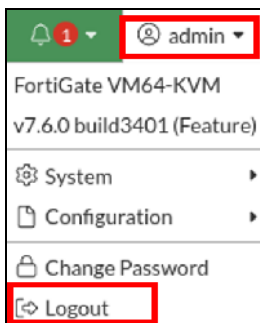
4. Click **OK** to save the changes.

Test the New Administrator Account

You will confirm that the new administrator account has read-write access to only the security profile configuration.

To test the new administrator account

1. Continuing on the HQ-NGFW-1 GUI, click **admin**, and then click **Logout** to log out of the **admin** account GUI session.



2. Log back in to the HQ-NGFW-1 GUI with the following credentials:
 - Username: `Security`
 - Password: `Fortinet1!`
3. In the **FortiGate Setup** window, click **Later**.
4. Enable **Don't show again**, and then click **OK** to close the FortiOS introduction window.
5. Explore the settings that are available on the GUI.
You should see that this account can configure only security profiles.
6. Log out of the GUI.

Restrict Administrator Access

You will restrict access for FortiGate administrators. Only administrators connecting from a trusted subnet are allowed access. This is useful if you must restrict the access points that administrators connect to FortiGate from.

To restrict administrator access

1. On the HQ-PC-1 VM, open a browser, and then log in to the HQ-NGFW-1 GUI with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`

2. Click **System > Administrators**.
3. Edit the **Security** account.
4. Enable **Restrict login to trusted hosts**, and then set **Trusted Host 1** to the `100.65.1.0/24` address.
5. Click **OK** to save the changes.

Test the Restricted Access

You will verify that a **Security** administrator outside the `100.65.1.0/24` subnet can't access FortiGate.

To test the restricted access

1. On the HQ-PC-1 VM, log out of the HQ-NGFW-1 GUI session as the **admin** user.
2. Try to log in to the **Security** account with the following password:
`Fortinet1!`
Authentication will fail.
3. Log in to the BR1-PC-1 VM with the following credentials:
 - Username: `Administrator`
 - Password: `Fortinet1!`
4. On the BR1-PC-1 VM, open a browser, and then log in to the HQ-NGFW-1 GUI at `100.65.0.101` with the following credentials:
 - Username: `Security`
 - Password: `Fortinet1!`

What is the result this time?

Stop and think!

Why were you able to log in using the **admin** account but not the **Security** account with the HQ-PC-1 VM directly connecting to the HQ-NGFW-1 GUI?

This is because the **Trusted Host** setting is configured on the **Security** administrator account but not on the **admin** account.

5. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
6. Enter the following CLI commands to add `10.0.11.0/24` as the second trusted IP subnet (**Trusted Host 2**) to the **Security** administrator account:

```
config system admin
edit Security
set trusthost2 10.0.11.0/24
end
```
7. Return to the HQ-PC-1 VM.
8. Open a browser, and then try to log in to the HQ-NGFW-1 GUI at `10.0.11.254` with the following credentials:

- Username: Security
- Password: Fortinet1!

You should be able to log in.

Lab 2: Firewall Policies and NAT

In this lab, you will configure firewall policies on HQ-NGFW-1, and then perform various tests on the HQ-PC-1 VM to confirm that traffic is matching the appropriate firewall policies based on the configuration.

You will also examine how to configure and test a firewall policy for destination network address translation (DNAT) using a virtual IP (VIP) address, and source network address translation (SNAT) using an IP pool. You will configure and test SNAT using the central SNAT policy, and DNAT using the DNAT policy and VIPs. You can use network address translation (NAT) to perform SNAT and DNAT for the traffic passing through FortiGate.

Objectives

- Configure firewall objects and firewall policies
- Configure source and destination matching in firewall policies
- Apply service objects to a firewall policy
- Configure firewall policy logging options
- Reorder firewall policies
- Read and understand logs
- Configure DNAT settings using a VIP
- Configure SNAT settings using overload IP pools

Time to Complete

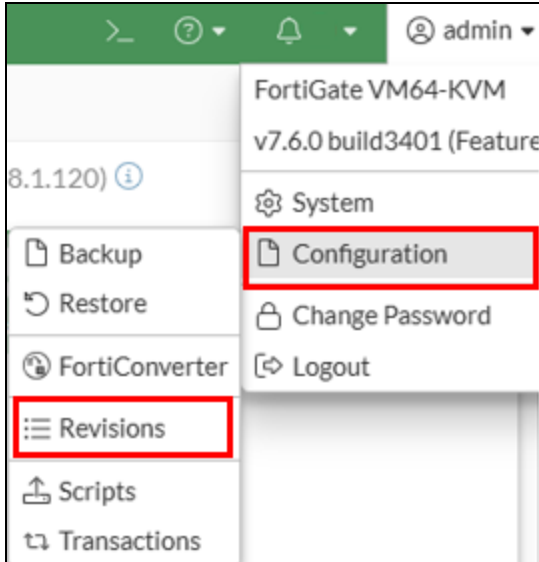
Estimated: 60 minutes

Prerequisites

Before you begin this lab, you must restore a configuration file on HQ-NGFW-1.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click + to expand the list.
4. Select the configuration with the comment **HQ-firewall-policy**, and then click **Revert**.

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Creating Firewall Address Objects and Firewall Policies

In this exercise, you will configure firewall address objects. You will also configure an IPv4 firewall policy that you will apply firewall address objects to, along with a schedule, services, and log options. Then, you will test the firewall policy by passing traffic through it and checking the logs for your traffic.

At its core, FortiGate is a firewall, so almost everything that it does to your traffic is related to your firewall policies.

Create Firewall Address Objects

By default, FortiGate has many preconfigured, well-known address objects in the factory default configuration. However, if those objects don't meet the needs of your organization, you can configure more.

To create a firewall address object

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **Policy & Objects > Addresses**.
3. Click **Create new**.
4. Configure the following settings:

Field	Value
Name	HQ_SUBNET
Interface	any
Type	Subnet
IP/Netmask	10.0.11.0/24

5. Click **OK**.

Create a Firewall Policy

First, you will disable the existing firewall policy. Then, you will create a more specific firewall policy using the firewall address object that you created in the previous task. You will also select specific services and configure log settings.

To disable an existing firewall policy

1. On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.



The FortiGate GUI may ask to use the new policy list layout. Click **Cancel** to continue using the classic layout. The new policy list layout is designed to improve performance when viewing a large list of firewall policies.

- 2. Expand the **port4** → **port2** firewall policy section.
- 3. Right-click the **Internet** firewall policy, and then in the **Set Status** field, select **Disable**.

To create a firewall policy

- 1. Continuing in the **Policy & Objects > Firewall Policy** section, click **Create New** to add a new firewall policy.
- 2. Configure the following settings:

Field	Value
Name	Internet_Access
Schedule	always
Incoming interface	port4
Outgoing interface	port2
Source	HQ_SUBNET
Destination	all
Service	ALL_ICMP, HTTP, HTTPS, DNS, SSH Tip: Type the service name in the search box to quickly find it, and then click the service object to add it to the policy.
Log Allowed Traffic	Select All Sessions .
Generate Logs when Session Starts	Enabled

- 3. Leave all other settings at the default values, and then click **OK** to save the changes.



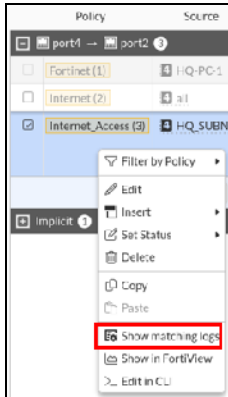
When you create firewall policies, remember that FortiGate is a stateful firewall. As a result, you need to create only one firewall policy that matches the direction of the traffic that initiates the session.

Test the Firewall Policy and View the Generated Logs

Now that you have configured the firewall policy, you will test it by passing traffic through it and viewing the generated logs.

To test and view logs for a firewall policy

1. On the HQ-PC-1 VM, open several browser tabs, and then connect to several external websites, such as:
 - www.google.com
 - www.cnn.com
 - www.bbc.com
2. Return to the browser tab with the HQ-NGFW-1 GUI, and then click **Policy & Objects > Firewall Policy**.
3. Right-click the **Internet_Access** policy, and then click **Show matching logs**.



4. Identify the log entries for your internet browsing traffic.

With the current settings, you should have a few log messages that have **Accept (Start)** in the **Result** column. These are the session start logs.

When sessions close, a separate log entry lists the amount of data that was sent and received.

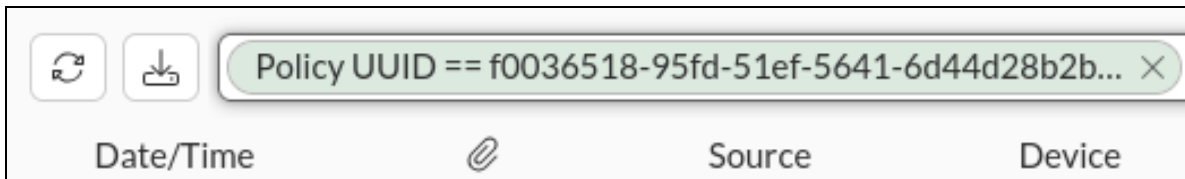


Enabling **Generate Logs when Session Starts** in the firewall policy will generate twice the amount of log messages. You should use this option only when this level of detail is absolutely necessary.



When you click **Show Matching Logs** in the firewall policy, it adds the **Policy UUID** filter in the forward traffic logs.

5. In the **Forward Traffic** logs, click **X** to remove the **Policy UUID** filter.



When you remove the **Policy UUID** filter, the logs are displayed unfiltered. You will use the logs in upcoming labs.

You may filter the DNS log entries by clicking the **Application Name** filter icon, enable **NOT**, click **DNS**, and then click **Apply**.



⌵ Resize to Contents

Filter

Exact Match ☒ NOT

DNS, ?

Suggestions

HTTPS

DNS ✓

udp/443

HTTP

NTP

Apply

Exercise 2: Reordering Firewall Policies and Firewall Policy Actions

In the applicable interface pair section, FortiGate looks for a matching policy, beginning at the top. Usually, you should put more specific policies at the top—otherwise, more general policies will match the traffic first, and more granular policies will never be applied.

In this exercise, you will create a new firewall policy with more specific settings, such as the source, destination, and service, and you will set the action to **DENY**. Then, you will move this firewall policy above the existing firewall policies and observe the behavior that reordering the firewall policies creates.

Create a Firewall Policy

You will create a new firewall policy to match a specific source, destination, and service, and you will set the action to **DENY**.



The **LINUX_ETH2** firewall address with the 100.65.0.254/32 IP/netmask is preconfigured for you, and you will use this address when you create the firewall policy.

Take the Expert Challenge!

Configure a firewall policy on the HQ-NGFW-1 GUI using the following settings:

- Name the firewall policy **Block_Ping**.
- Use port4 as the incoming interface and port2 as the outgoing interface.
- Block all ping traffic from the 10.0.11.0/24 subnet destined for the 100.65.0.254 address. Use the preconfigured address objects **HQ_SUBNET** and **LINUX_ETH2**.
- Enable log violation traffic.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you have performed these steps, see [Test the Reordering of a Firewall Policy on page 35](#).

To create a firewall policy

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!

- Click **Policy & Objects > Firewall Policy**, and then click **Create New**.
- Configure the following settings:

Field	Value
Name	Block_Ping
Incoming interface	port4
Outgoing interface	port2
Source	HQ_SUBNET
Destination	LINUX_ETH2
Service	PING Tip: Type the service name in the search box to quickly find it, and then click the service object to add it to the policy.
Action	DENY
Log Violation Traffic	Enabled

- Click **OK** to save the changes.

Test the Reordering of a Firewall Policy

Now that your configuration is ready, you will test it by moving the **Block_Ping** firewall policy above the **Internet_Access** firewall policy. The objective is to confirm that, after you reorder the firewall policies, the following occurs:

- Traffic is matched to a more specific firewall policy.
- The policy ID remains the same.

To test the reordering of a firewall policy

- On the HQ-PC-1 VM, open a terminal.
- Ping the destination address (**LINUX_ETH2**) that you configured in the **Block_Ping** firewall policy.

```
ping 100.65.0.254
```

Stop and think!

Why are you still able to ping the destination address, even though you just configured a policy to block it?

The ping should still work because it matches the **ACCEPT** policy and not the **DENY** policy that you created. The **Block_Ping** policy was never checked because the traffic matched the policy at the top (**Internet_Access**). This demonstrates the behavior that FortiGate looks for a matching policy, beginning at the top.

- Leave the terminal window open and running.
- On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.

5. Hover over the first column to the left of the **Policy** column until your cursor changes to a hand and you can move the policy.

	Policy	Source	Destination
	port4 → port2 4		
☐	Fortinet (1)	4 HQ-PC-1	FORTINET
☐	Internet (2)	4 all	4 all
☐	Internet_Access (3)	4 HQ_SUBNET	4 all
☐	Block_Ping (4)	4 HQ_SUBNET	4 LINUX_ETH2
☐	Implicit 1		

6. Drag the **Block_Ping** firewall policy up and place it above the **Internet_Access** firewall policy. When you move the **Block_Ping** policy up, the **ID** value remains the same.

	Policy	Source	Destination	Schedule	Service
	port4 → port2 4				
☐	Fortinet (1)	4 HQ-PC-1	FORTINET	always	Web Access
☐	Internet (2)	4 all	4 all	always	ALL
☐	Block_Ping (4)	4 HQ_SUBNET	4 LINUX_ETH2	always	PING
☐	Internet_Access (3)	4 HQ_SUBNET	4 all	always	ALL_ICMP HTTP +3



If the changes that you made are not displayed, refresh the page. Alternatively, you can log out of the FortiGate GUI, and then log back in.

7. On the HQ-PC-1 VM, review the terminal window that is running the continuous ping. You should see that the pings now fail.

Stop and think!

Why are the pings failing?

This demonstrates the outcome of the policy reordering. After moving the more granular policy above the general access policy, the traffic is matched to the more granular policy and, based on the **DENY** action, the traffic stops being processed.

8. Close the terminal window.
9. On the HQ-NGFW-1 GUI, click **Log & Report > Forward Traffic**.

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID
2024/10/29 07:43:23	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)
2024/10/29 07:43:22	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)
2024/10/29 07:43:21	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)
2024/10/29 07:43:20	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)
2024/10/29 07:43:19	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)
2024/10/29 07:43:18	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)
2024/10/29 07:43:17	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)
2024/10/29 07:43:16	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)
2024/10/29 07:43:15	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)
2024/10/29 07:43:14	10.0.11.50		100.65.0.254	PING	Deny	Block_Ping (4)

You should see many policy violation logs reporting the blocked ping.



Clear the log filter that you applied in the previous exercise.

Exercise 3: Configuring DNAT Settings Using a VIP

VIPs are typically used to translate external, or public, IP addresses to internal, or private, IP addresses.

In this exercise, you will examine how to configure a VIP for the HQ-PC-1 VM. Then, you will create an egress-to-ingress firewall policy and apply the VIP. This allows internet connections to the HQ-PC-1 VM. You will also verify the DNAT and SNAT behavior using CLI commands.

Create a VIP

For DNAT on FortiGate, you use a VIP as the destination address of a firewall policy.

You will configure the VIP to map the HQ-PC-1 VM (10.0.11.50) to 100.65.0.200, which is part of the port2 subnet. To refer to the lab diagram, see [Network Topology on page 9](#).

To create a VIP

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **Policy & Objects > Virtual IPs**, and then click **Create New**.
3. Configure the following settings:

Field	Value
Name	VIP-INTERNAL-HOST
Interface	port2 This port is connected to the internet with IP address 100.65.0.101/24.
External IP address/range	100.65.0.200 This IP address is in the same range as the port2 subnet.
Map to IPv4 address/range	10.0.11.50

New Virtual IP

Name: VIP-INTERNAL-HOST

Comments: Write a comment... 0/255

Color: Change

Network

Interface: port2

Type: Static NAT 1:1 QUN

External IP address/range: 100.65.0.200

Map to: 10.0.11.50

Optional Filters

Port Forwarding

4. Click **OK**.

Create a Firewall Policy

You will configure a new firewall policy using the VIP that you just created as the destination address.

To create a firewall policy

1. On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Click **Create New**.
3. Configure the following settings:

Field	Value
Name	Web-Server-Access
Incoming interface	port2
Outgoing interface	port4
Source	all
Destination	VIP-INTERNAL-HOST Tip: This is listed under the VIRTUAL IP/SERVER section.
Service	HTTP, HTTPS Tip: In the section on the right, in the search field, type the service name, and then click the services to add.

4. In the **Firewall/Network Options** section, disable **NAT**.

Create New Policy

Name: Web Server Access

Schedule: always

Action: ACCEPT (selected), DENY

Incoming interface: port2

Outgoing interface: port1

Source & Destination: Show logic

Source: all

User/group: +

Destination: VIP-INTERNAL-HOST

Service: HTTP, HTTPS

Firewall/Network Options

Inspection mode: flow based (selected), proxy based

NAT: disabled

5. In the **Logging Options** section, in the **Log allowed traffic** field, select **All sessions**.

Security Profiles

AntiVirus: disabled

Web filter: disabled

DNS filter: disabled

Application control: disabled

IPS: disabled

File filter: disabled

SSL inspection: SSL no-inspection (selected)

Logging Options

Log allowed traffic: enabled (Security events, All sessions selected)

Generate logs when session starts: disabled

Comments: 0/1023

Enable this policy: enabled

6. Click **OK**.

Test the VIP Firewall Policy

Now that you have configured a firewall policy with the VIP as the destination, you can test your VIP by accessing it from the BR1-PC-1 VM, which is behind the BR1-FGT internal network. A Linux VM acts as a router between the

two FortiGate devices, and routes the traffic from BR1-FGT to HQ-NGFW-1. For more information, see [Network Topology on page 9](#).

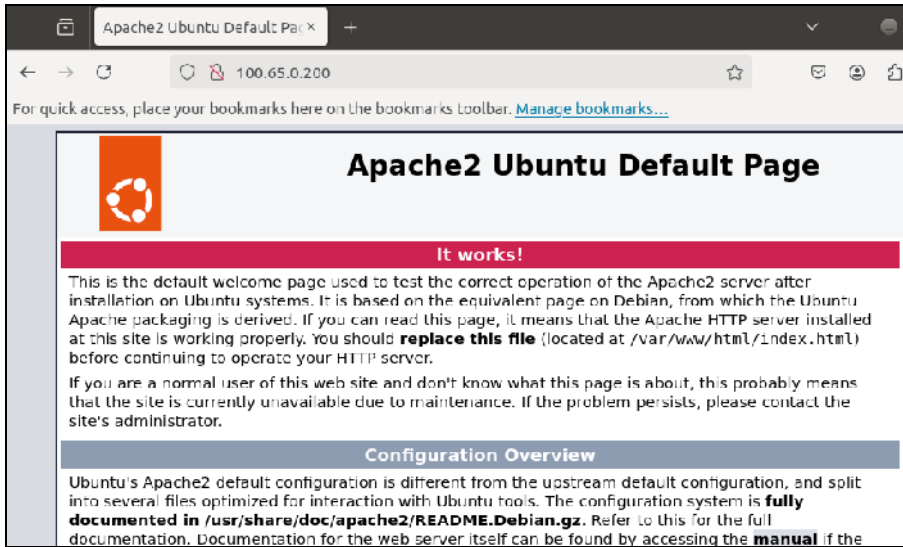
You will also test how the VIP translates the source address when traffic leaves the HQ-PC-1 VM.

To test VIPs (DNAT)

1. On the BR1-PC-1 VM, open a browser, and then browse to the following URL:

<http://100.65.0.200>

If the VIP operation is successful, a simple web page opens.



2. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
3. Enter the following command to check the destination NAT entries in the session table:

```
get system session list
```

The following example shows a sample output:

```
HQ-NGFW-1# get system session list
PROTO EXPIRE SOURCE SOURCE-NAT DESTINATION DESTINATION-NAT
tcp 3594 100.65.1.111:49478 - 100.65.0.200:80 10.0.11.50:80
```

You will notice that the 100.65.0.200 destination address is translated to 10.0.11.50, which is the mapping you configured in the VIP.



The HTTP session may have been deleted by the time you enter the `get system session list` command. You can repeat steps 1–3 to generate a new HTTP connection and, therefore, another HTTP session through HQ-NGFW-1.

Test SNAT

As a result of the VIP (which is a destination NAT), FortiGate uses the VIP external address as the NAT IP address when performing SNAT for the internal-to-external direction of the traffic, as long as the matching outgoing firewall policy has NAT enabled. That is, FortiGate doesn't use the IP address of the egress interface.

To test SNAT

1. Return to the HQ-NGFW-1 CLI session, and then enter the following command to clear any existing sessions:

```
diagnose sys session clear
```



The `diagnose sys session clear` CLI command clears all sessions, including the SSH session you created. This is expected behavior.

This clears the session to HQ-NGFW-1 from the HQ-PC-1 VM.

2. Close the HQ-NGFW-1 CLI window.
3. On the HQ-PC-1 VM, open a few browser tabs, and then connect to a few websites, such as:
 - www.fortinet.com
 - www.yahoo.com
 - www.bbc.com
4. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!

5. Enter the following command to view the session information:

```
get system session list
```

The following example shows a sample output:

HQ-NGFW-1 # get sys session list						
PROTO	EXPIRE	SOURCE	SOURCE-NAT	DESTINATION	DESTINATION-NAT	
tcp	3598	10.0.11.50:44952	100.65.0.200:44952	34.107.221.82:80	-	
tcp	3599	10.0.11.50:44950	100.65.0.200:44950	34.107.221.82:80	-	
tcp	3547	10.0.11.50:43326	100.65.0.200:43326	34.117.188.166:443	-	
tcp	3547	10.0.11.50:51526	100.65.0.200:51526	34.160.144.191:443	-	
tcp	3488	10.0.11.50:51524	100.65.0.200:51524	34.107.243.93:443	-	



The outgoing connections from the HQ-PC-1 VM are now translated with the VIP address 100.65.0.200, instead of the IP address of the firewall egress interface (100.65.0.101).

This is a behavior for SNAT when using a static NAT VIP. That is, when you enable NAT in a policy, the external address of a static NAT VIP takes precedence over the IP address of the destination interface, if the source address of the connections matches the internal address of the VIP.

6. Close the HQ-NGFW-1 CLI window.
7. Close all browser tabs except the HQ-NGFW-1 GUI.

Exercise 4: Using Dynamic NAT With IP Pools

An IP pool is used to translate the source address to an address from that pool, rather than the IP address of the egress interface.

Currently, HQ-NGFW-1 translates the source IP address of all traffic generated from the HQ-PC-1 VM to 100.65.0.200, because the internal address of the VIP matches the address of the HQ-PC-1 VM, and the VIP is a static NAT VIP.

In this exercise, you will examine how to create an IP pool, apply it to the ingress-to-egress firewall policy, and verify the SNAT address using CLI commands.

Create an IP Pool

You will create an IP pool from the range of public IP addresses that are available on the egress port (port2).

To create an IP pool

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Policy & Objects > IP Pools**.
3. Click **Create New**, and then configure the following settings:

Field	Value
Name	INTERNAL-HOST-EXT-IP
External IP Range	100.65.0.100-100.65.0.100

New Dynamic IP Pool

Name: INTERNAL-HOST-EXT-IP

Comments: Write a comment... 0/255

Type: Overload

External IP Range: 100.65.0.100-100.65.0.100

NAT64: ☐

ARP Reply: ☒

4. Click **OK**.

Edit a Firewall Policy to Use the IP Pool

You will apply the IP pool to change the behavior from static NAT to dynamic NAT in the ingress-to-egress firewall policy.

To edit the firewall policy

- 1. On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
- 2. Right-click the **Internet** firewall policy.
- 3. Click **Set Status > Enable**.
- 4. Right-click the firewall policy again, and then click **Edit**.
- 5. In the **Firewall/Network Options** section, configure the following settings:

Field	Value
NAT	Enabled
IP pool configuration	Use Dynamic IP Pool

- 6. Click the **+** sign that appeared when you clicked **Use Dynamic IP Pool**, and then in the section on the right, click **INTERNAL-HOST-EXT-IP**.

Your configuration will look similar to the following example:

Edit Policy

Name

Internet

Schedule

always

Action

ACCEPT

DENY

Incoming interface

port4

Outgoing interface

port2

Source & Destination

Show logic

Source

all

User/group

Destination

all

Service

ALL

Firewall/Network Options

Inspection mode

Flow-based

Proxy-based

NAT

ON

IP pool configuration

Use Outgoing Interface Address

Use Dynamic IP Pool

INTERNAL-HOST-EXT-IP

- 7. Click **OK**.

Test Dynamic NAT With IP Pools

Now that your configuration is ready, you can test dynamic NAT with IP pools by browsing to a few external websites. If successful, you will see that the IP address of the HQ-PC-1 VM (10.0.11.50) is translated to the IP pool address of 100.65.0.100.

To test dynamic NAT with IP pools

1. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Enter the following commands to clear sessions sourced from 10.0.11.50:

```
diagnose sys session filter clear
diagnose sys session filter src 10.0.11.50
diagnose sys session clear
```



You built the filter to match sessions sourced from 10.0.11.50. This way, when you enter the `diagnose sys session clear` CLI command, it clears only the sessions sourced from 10.0.11.50. As a result, your SSH session is not disconnected, unless your SSH session is originating from the HQ-PC-1 VM. This is why it is important to build the session filter before using the `session clear` command.

3. On the HQ-PC-1 VM, open a few browser tabs, and then connect to a few websites, such as:
 - www.fortinet.com
 - www.yahoo.com
 - www.bbc.com
4. On the HQ-NGFW-1 CLI, enter the following command to verify the SNAT address that the sessions are using:

```
get system session list
```

The following image shows a sample output:

HQ-NGFW-1 # get sys session list					
PROTO	EXPIRE	SOURCE	SOURCE-NAT	DESTINATION	DESTINATION-NAT
tcp	3599	10.0.11.50:46162	100.65.0.100:46162	54.197.240.163:443	-
tcp	3593	10.0.11.50:51412	100.65.0.100:51412	151.101.2.217:443	-
tcp	3588	10.0.11.50:35858	100.65.0.100:35858	3.161.213.116:443	-
tcp	3584	10.0.11.50:59688	100.65.0.100:59688	3.161.254.190:443	-
tcp	3598	10.0.11.50:45260	100.65.0.100:45260	3.161.215.181:443	-

5. Notice that the SNAT address is now 100.65.0.100, as configured in the IP pool, and the IP pool has overridden the static NAT VIP.
6. Close the HQ-NGFW-1 CLI window.
7. Close all browser tabs except the HQ-NGFW-1 GUI.

Lab 3: Routing

In this lab, you will configure the router settings and test scenarios to learn how FortiGate makes routing decisions.

Objectives

- Route traffic based on the destination IP address, as well as other criteria
- Balance traffic among multiple paths
- Implement route failover
- Diagnose a routing problem

Time to Complete

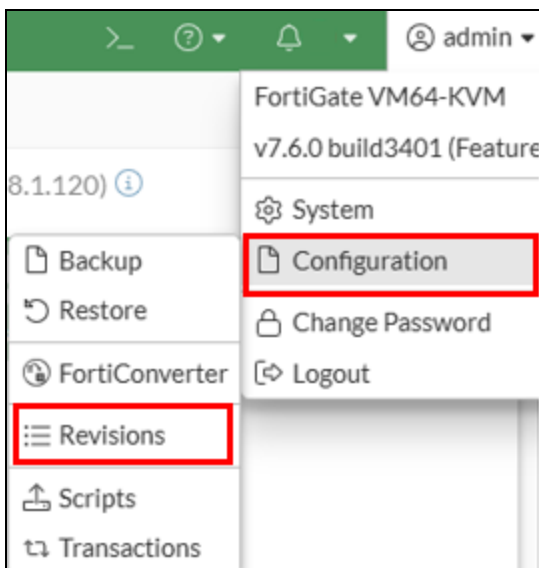
Estimated: 50 minutes

Prerequisites

Before you begin this lab, you must restore a configuration file on HQ-NGFW-1.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **HQ-routing**, and then click **Revert**.

Delete

Details

Diff

Revert

Save Changes

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Configuring Route Failover

In the lab network, HQ-NGFW-1 has two interfaces connected to the internet: port2 and port3. In this exercise, you will configure the port2 connection as the primary internet link and the port3 connection as the backup internet link. HQ-NGFW-1 should use the port3 connection only if the port2 connection is down. To achieve this objective, you will configure two default routes with different administrative distances, and then you will disable the primary default route interface to activate the standby route.

Verify the Routing Configuration

You will verify the existing routing configuration on HQ-NGFW-1.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI (admin/Fortinet1!), complete the following:

- View the existing static route configuration on HQ-NGFW-1.
- Enable the **Distance** and **Priority** columns on the static route configuration page.
- Make a note of the **Distance** and **Priority** values of the existing default route.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

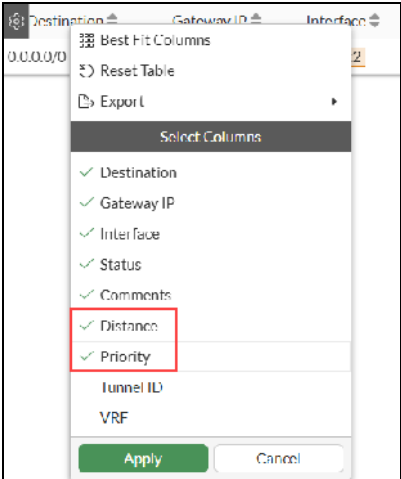
After you complete the challenge, see [Configure a Second Default Route on page 50](#).

To verify the routing configuration

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Network > Static Routes**.
3. Verify the existing default route for **port2**.

+ Create new		+ Q Search			
☐	Destination ▾	Gateway IP ▾	Interface ▾	Status ▾	Comments ▾
☐	0.0.0.0/0	100.65.0.254	 port2	✔ Enabled	

4. Right-click any of the column headers to open the context-sensitive menu.
5. In the **Select Columns** section, select **Distance** and **Priority**, and then click **Apply**.



The **Distance** and **Priority** columns appear on the GUI.

Note that, by default, static routes have a **Distance** value of 10 and a **Priority** value of 1.

Configure a Second Default Route

You will create a second default route using the port3 interface. To make sure this second default route remains the standby route, you will assign it a higher administrative distance than the first default route.

Take the Expert Challenge!

- On the HQ-NGFW-1 GUI, configure a second default route using **port3**.
- Assign it a **Distance** of 20 and a **Priority** of 5.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Configure the Firewall Policies on page 51](#).

To configure a second default route

1. Continuing on the HQ-NGFW-1 GUI, click **Network > Static Routes**.
2. Click **Create New**.
3. Configure the following settings:

Field	Value
Gateway Address	100.66.0.254
Interface	port3
Administrative Distance	20

- Click + to expand the **Advanced Options** section.
- In the **Priority** field, type 5.

New Static Route

Destination Subnet Internet Service

Gateway Address 0.0.0.0/0.0.0.0

Interface port3

Administrative Distance 20

Comments Write a comment...

Status Enabled Disabled

Advanced Options

Priority 5

- Click **OK**.
- FortiGate adds a second default route.

Create new	Search						
☐	Destination	Gateway IP	Interface	Status	Comments	Distance	Priority
☐	0.0.0.0/0	100.65.0.254	port2	Enabled		10	1
☐	0.0.0.0/0	100.66.0.254	port3	Enabled		20	5

Configure the Firewall Policies

You will modify the existing **Internet** firewall policy to log all sessions. You will also create a second firewall policy to allow traffic through the secondary interface.

Take the Expert Challenge!

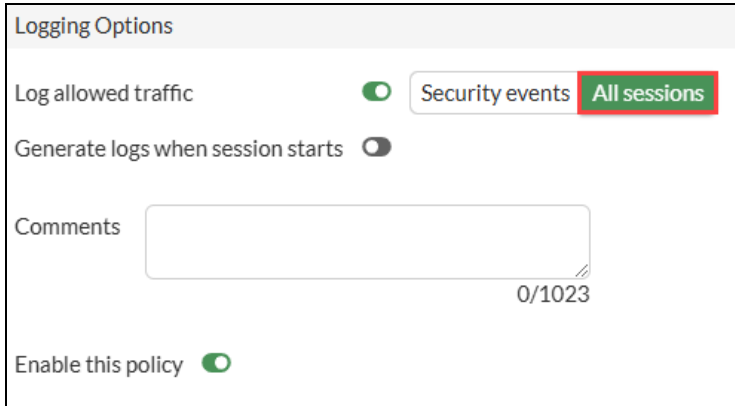
- Continuing on the HQ-NGFW-1 GUI, enable logging for all sessions in the existing **Internet** firewall policy.
- Create a second firewall policy named `Backup_Internet`.
- Configure the **Backup_Internet** policy to allow traffic from **port4** to **port3** with NAT enabled.
- Enable logging on the **Backup_Internet** policy for all sessions.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [View the Routing Table on page 53](#)

To configure the firewall policies

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Double-click the existing **Internet** policy to edit it.
3. Enable **Log allowed traffic**, and then select **All sessions**.



All sessions logging ensures that FortiGate logs all traffic, not only sessions that security profiles inspected. This will assist you in verifying traffic routing using the **Forward Traffic** logs.

4. Click **OK**.
5. Click **Create New**.
6. Configure a second firewall policy with the following settings:

Field	Value
Name	Backup_Internet
Schedule	always
Incoming Interface	port4
Outgoing Interface	port3
Source	HQ_SUBNET
Destination	all
Service	ALL
Log allowed traffic	All sessions

7. Click **OK**.

View the Routing Table

The HQ-NGFW-1 configuration now has two default routes with different distances. You will view the routing table to see which route was installed in the routing table and which route was installed in the routing table database.

To view the routing table

1. On the HQ-NGFW-1 CLI, log in with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Enter the following command to list the routing table entries:

```
get router info routing-table all
```

The second default route is not listed.

3. Enter the following command to list the routing table database entries:

```
get router info routing-table database
```

4. Confirm that the second default route is listed as inactive.

```
HQ-NGFW-1 # get router info routing-table database
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       > - selected route, * - FIB route, p - stale info

Routing table for VRF=0
S    0.0.0.0/0 [20/0] via 100.66.0.254, port3, [5/0]
S    *> 0.0.0.0/0 [10/0] via 100.65.0.254, port2, [1/0]
C    *> 10.0.11.0/24 is directly connected, port4
C    *> 10.0.12.0/24 is directly connected, port5
C    *> 10.0.13.0/24 is directly connected, port6
C    *> 100.65.0.0/24 is directly connected, port2
C    *> 100.66.0.0/24 is directly connected, port3
C    *> 192.168.0.0/16 is directly connected, port1
```



Only active routes show the > symbol, which means they are the selected and active routes. The routing table database contains all active, standby, and inactive routes on FortiGate.

Stop and think!

Why is the port3 default route the standby route?

The port3 default route has a higher administrative distance than the port2 default route. When two or more routes to the same destination have different distances, the higher distance route is not installed in the routing table, but you can still see it in the routing table database. Routes marked as inactive are marked inactive when the corresponding interface is down.

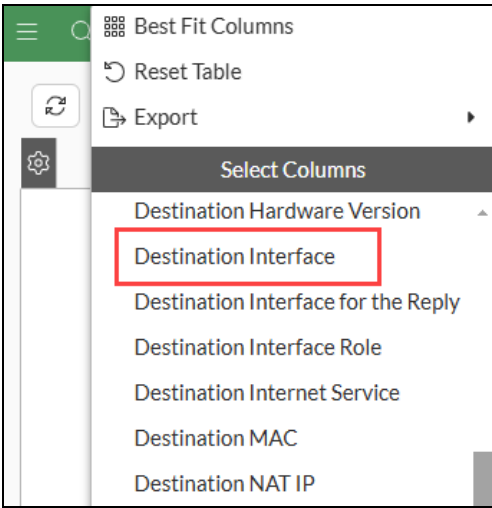
5. Close the HQ-NGFW-1 CLI session.

Test the Route Failover

First, you will access various websites and use the **Forward Traffic** logs to verify that the port2 route is being used. Next, you will force a failover by reconfiguring the port2 interface setting and bringing the interface down. Finally, you will generate some more traffic, and use the **Forward Traffic** logs to verify that the port3 route is being used.

To confirm the port2 route is the primary route

1. Continuing on the HQ-NGFW1 GUI, click **Log & Report > Forward Traffic**.
2. Right-click any of the column headers to open the context-sensitive menu.
3. In the **Select Columns** section, select **Destination Interface**.



4. Scroll down in the context-sensitive menu, and then click **Apply**.
The **Destination Interface** column is displayed.

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID	Destination Interface
2024/10/31 09:02:26	10.0.11.50		142.251.133.170	HTTP	✓ Accept (2.79 kB / 6.76 kB)	Internet (1)	port2
2024/10/31 09:02:17	10.0.11.50		8.8.8.8 (dns.google)	DNS	✓ Accept (320 B / 342 B)	Internet (1)	port2

5. On the HQ-PC-1 VM, in the browser, open a few new tabs, and then visit a few websites, such as:
- <http://neverssl.com>
 - <http://eu.httpbin.org>
6. On the HQ-NGFW1 GUI, click **Log & Report > Forward Traffic**.
7. Click the refresh icon.
8. Locate the relevant log entries for the websites you accessed, and then verify that the **Destination Interface** is **port2**.

Date/Time	@	Source	Device	Destination	Application Name	Result	Policy ID	Destination Interface
2024/10/31 09:24:50		10.0.11.50		52.81.79.32 (multiplatform)	HTTP	✓ Accept (3.6 kB / 410.99 kB)	Internet (1)	port2
2024/10/31 09:24:50		10.0.11.50		52.81.79.32 (multiplatform)	HTTP	✓ Accept (1.8 kB / 203.96 kB)	Internet (1)	port2
2024/10/31 09:24:47		10.0.11.50		52.81.79.32 (multiplatform)	HTTP	✓ Accept (13.94 kB / 1.49 MB)	Internet (1)	port2
2024/10/31 09:24:47		10.0.11.50		52.81.79.32 (multiplatform)	HTTP	✓ Accept (2.49 kB / 19.65 kB)	Internet (1)	port2
2024/10/31 09:24:37		10.0.11.250		96.45.46.46 (dns.google.com)	DNS	✓ Accept (377 B / 3.05 kB)	Internet (1)	port2
2024/10/31 09:23:49		10.0.11.50		142.251.32.67 (fontassets.gstatic.com)	HTTPS	✓ Accept (2.89 kB / 6.46 kB)	Internet (1)	port2
2024/10/31 09:23:48		10.0.11.50		142.251.32.67 (fontassets.gstatic.com)	HTTPS	✓ Accept (2.79 kB / 6.39 kB)	Internet (1)	port2
2024/10/31 09:23:47		10.0.11.50		142.251.41.42 (fontassets.gstatic.com)	HTTPS	✓ Accept (3.74 kB / 10.13 kB)	Internet (1)	port2
2024/10/31 09:23:47		10.0.11.50		142.251.41.42 (fontassets.gstatic.com)	HTTPS	✓ Accept (2.79 kB / 6.82 kB)	Internet (1)	port2
2024/10/31 09:23:39		10.0.11.50		34.293.124.45 (googleadservices.com)	HTTP	✓ Accept (1.65 kB / 2.9 kB)	Internet (1)	port2
2024/10/31 09:23:38		10.0.11.50		34.293.124.45 (googleadservices.com)	HTTP	✓ Accept (608 B / 2.59 kB)	Internet (1)	port2

This verifies that the port2 route is currently the route being used.

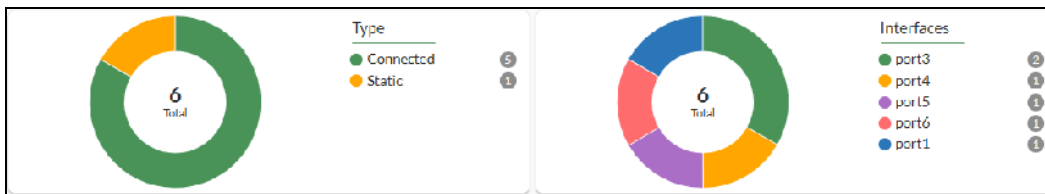
To force the failover

- Continuing on the HQ-NGFW1 GUI, click **Network > Interfaces**.
- Double-click the **port2** interface to edit it.
- In the **Miscellaneous** section, in the **Status** field, select **Disabled**.
- Click **OK**.

The port2 internet connection is now down, and FortiGate removes the corresponding route from the routing table.

To verify the route change

- Continuing on the HQ-NGFW1 GUI, click **Dashboard > Network**, and then click **Static & Dynamic Routing** to expand it to full screen.
- In the routing table, verify that the **port3** route replaced the **port2** route.



Network	Gateway IP	Interfaces	Distance	Type
0.0.0.0/0	100.66.0.254	port3	20	Static
10.0.11.0/24	0.0.0.0	port4	0	Connected
10.0.12.0/24	0.0.0.0	port5	0	Connected
10.0.13.0/24	0.0.0.0	port6	0	Connected
100.66.0.0/24	0.0.0.0	port3	0	Connected
192.168.0.0/16	0.0.0.0	port1	0	Connected

To verify traffic logs

- On the HQ-PC-1 VM, in the browser, open a few new tabs, and then visit a few websites, such as:
 - <http://neverssl.com>
 - <http://eu.httpbin.org>
- Return to the browser tab where you are logged in to the HQ-NGFW1 GUI, and then click **Log & Report > Forward Traffic**.

3. Locate the relevant log entries for the websites you accessed, and then verify that the **Destination Interface** indicates **port3**.

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID	Destination Interface
2024/10/31 09:14:34	10.0.11.253	96.45.65.45 (onefortiguard.net)	DNS	✓ Accept (137 B / 4.1 KB)	Backup Access (2)	port3	
2024/10/31 09:14:19	10.0.11.50	52.87.99.32 (eu.httpbin.org)	HTTP	✓ Accept (4.11 KB / 203.91 KB)	Backup Access (2)	port3	
2024/10/31 09:14:19	10.0.11.50	52.87.99.32 (eu.httpbin.org)	HTTP	✓ Accept (4.22 KB / 401.15 KB)	Backup Access (2)	port3	
2024/10/31 09:14:10	10.0.11.50	52.87.99.32 (eu.httpbin.org)	HTTP	✓ Accept (1.92 KB / 89.65 KB)	Backup Access (2)	port3	
2024/10/31 09:14:17	10.0.11.50	52.87.99.32 (eu.httpbin.org)	HTTP	✓ Accept (6.11 KB / 1.49 MB)	Backup Access (2)	port3	
2024/10/31 09:13:52	10.0.11.253	96.45.65.46 (onefortiguard.net)	DNS	✓ Accept (469 B / 5.3 KB)	Backup Access (2)	port3	
2024/10/31 09:13:22	10.0.11.50	142.251.133.170 (fortisguard.com)	HTTPS	✓ Accept (2.76 KB / 6.87 KB)	Backup Access (2)	port3	
2024/10/31 09:13:16	10.0.11.50	142.251.132.67 (fortisguard.com)	HTTPS	✓ Accept (2.79 KB / 6.41 KB)	Backup Access (2)	port3	
2024/10/31 09:13:16	10.0.11.50	142.251.132.67 (fortisguard.com)	HTTPS	✓ Accept (2.79 KB / 6.46 KB)	Backup Access (2)	port3	
2024/10/31 09:13:17	10.0.11.50	142.251.133.170 (fortisguard.com)	HTTPS	✓ Accept (31.79 KB / 101.14 KB)	Backup Access (2)	port3	
2024/10/31 09:13:09	10.0.11.50	142.251.41.78 (encripted-lookup.com)	HTTPS	✓ Accept (1.77 KB / 11.44 KB)	Backup Access (2)	port3	
2024/10/31 09:13:09	10.0.11.50	142.251.124.45 (uniquemanagerielkissnews.com)	HTTP	✓ Accept (1.63 KB / 2.89 KB)	Backup Access (2)	port3	
2024/10/31 09:13:08	10.0.11.50	142.251.124.45 (uniquemanagerielkissnews.com)	HTTP	✓ Accept (600 B / 2.59 KB)	Backup Access (2)	port3	

This verifies that HQ-NGFW1 is using the port3 default route.

Restore the Routing Table

Before you begin the next exercise, you will restore the port2 interface settings and bring the interface up, which will restore the port2 default route as the best route in the routing table.

To restore the port2 health monitor configuration

1. Continuing on the HQ-NGFW1 GUI, click **Network > Interfaces**.
2. Double-click the **port2** interface to edit it.
3. In the **Miscellaneous** section, in the **Status** field, select **Enabled**.
4. Click **OK**.

To verify the routing table

1. Continuing on the HQ-NGFW1 GUI, click **Dashboard > Network**, and then click **Static & Dynamic Routing** to expand it to full screen.
2. In the routing table, verify that the **port2** route replaced the **port3** route.

7

total

Connected

Static

6

1

Interfaces

port2

port4

port5

port6

port3

port1

2

1

1

1

1

1

Route Lookup

Create Address

Edit

Search

Network	Gateway IP	Interfaces	Distance	Type
0.0.0.0/0	100.65.0.254	port2	10	Static
10.0.11.0/24	0.0.0.0	port4	0	Connected
10.0.12.0/24	0.0.0.0	port5	0	Connected
10.0.13.0/24	0.0.0.0	port6	0	Connected
100.65.0.0/24	0.0.0.0	port2	0	Connected
100.66.0.0/24	0.0.0.0	port3	0	Connected
192.168.0.0/16	0.0.0.0	port1	0	Connected

3. Close the browser.

Exercise 2: Configuring ECMP

In this exercise, you will configure equal-cost multi-path (ECMP) routing on Local-FortiGate to load balance the internet traffic between port2 and port3.

Configure Administrative Distance

To establish ECMP, first, you will configure multiple static routes with the same administrative distance.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI (admin/Fortinet1!), complete the following:

- Change the **port3** static route **Administrative Distance** to 10.
- Verify that both **port2** and **port3** default routes are present in the routing table.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Change the ECMP Load Balancing Algorithm on page 59](#).

To configure administrative distance

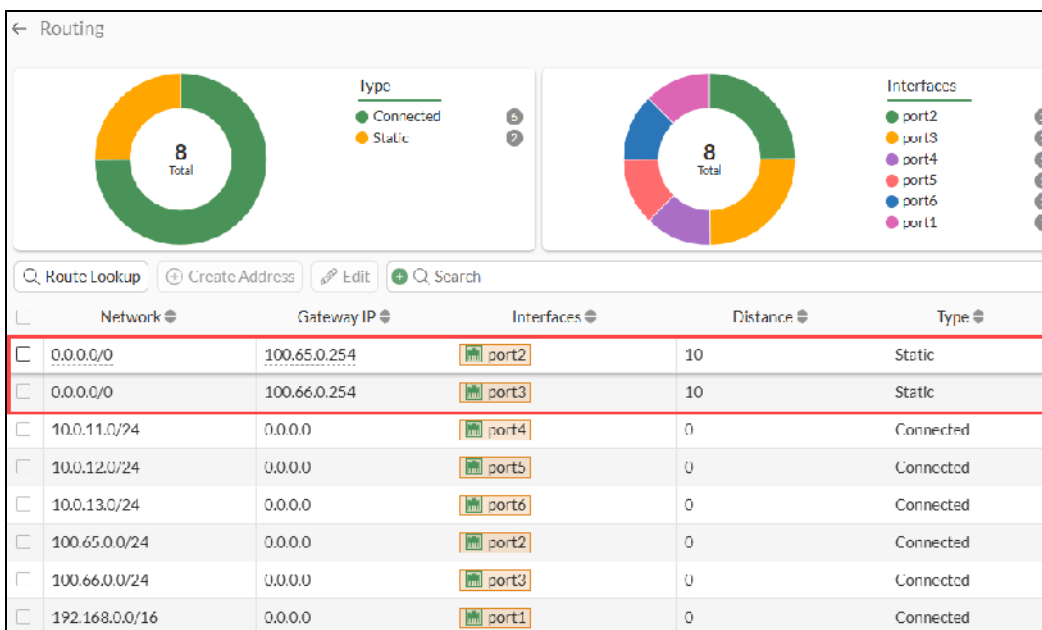
1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Network > Static Routes**.
3. Double-click the **port3** static route to edit it.
4. In the **Administrative Distance** field, change the value to 10.

The screenshot shows the 'Edit Static Route' configuration window in the FortiGate GUI. The 'Destination' is set to 'Subnet' with the address '0.0.0.0/0.0.0.0'. The 'Gateway Address' is '100.66.0.254'. The 'Interface' is 'port3'. The 'Administrative Distance' is highlighted with a red box and set to '10'. The 'Status' is 'Enabled'.

5. Click **OK**.

To verify the routing table

1. Continuing on the HQ-NGFW-1 GUI, click **Dashboard > Network**, and then click **Static & Dynamic Routing** to expand it to full screen.
2. Verify that both default routes are installed in the routing table.



Change the ECMP Load Balancing Algorithm

By default, the ECMP load balancing algorithm is based on the source IP address. This works well when there are multiple clients generating traffic. In the lab network, because you have only one client (the HQ-PC-1 VM), the source IP address method does not balance any traffic to the second route. FortiGate always uses only one route. For this reason, you will change the load balancing method to use both source and destination IP addresses. Using this method, as long as the traffic goes to multiple destination IP addresses, FortiGate load balances the traffic across both routes.

To modify the ECMP load balancing method

1. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Enter the following commands to change the ECMP load balancing method:


```
config system settings
    set v4-ecmp-mode source-dest-ip-based
end
```
3. Leave the HQ-NGFW-1 CLI session open.

Verify Traffic Routing

You will generate some HTTP traffic and verify traffic routing using the **Forward Traffic** logs.

Take the Expert Challenge!

- On the HQ-NGFW-1 VM, open a few new browser tabs, and then generate some HTTP traffic.
- Verify the traffic routing on HQ-NGFW-1, using the **Forward Traffic** logs.
- Identify why all the outgoing packets are still being routed through **port2**.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Configure Priority on page 61](#).

To verify traffic routing

1. On the HQ-PC-1 VM, in the browser, open a few new tabs, and then visit a few websites, such as:
 - <http://neverssl.com>
 - <http://example.com>
 - <http://eu.httpbin.org>
2. On the HQ-NGFW-1 GUI, click **Log & Report > Forward Traffic**.
3. In the relevant log entries for the websites you accessed, identify the **Destination Interface**.

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID	Destination Interface
2024/11/04 00:52:35	10.0.11.50	52.32.164.86 (atom-tracking.demandscience.com)	HTTPS	✓ Accept (3.68 kB / 6.71 kB)	Internet (1)	port2	
2024/11/04 00:52:29	10.0.11.50	96.45.36.159 (site.fortinet.com)	HTTPS	✓ Accept (5.25 kB / 1.74 kB)	Internet (1)	port2	
2024/11/04 00:52:27	10.0.11.50	54.84.141.5 (eu.httpbin.org)	HTTP	✓ Accept (22/111/43/2/3 kB)	Internet (1)	port2	
2024/11/04 00:52:18	10.0.11.50	52.12.117.226 (gsmountain.com)	HTTPS	✓ Accept (2.57 kB / 6.13 kB)	Internet (1)	port2	

Why are all the outgoing packets still being routed through port2?

Stop and think!

The **port3** route is not being used to route internet traffic. Why?

At the beginning of this exercise, you set a distance of 10 on the port3 route but you didn't change its priority. The port3 route priority is still 5, as you configured it in the previous exercise (see [Configure a Second Default Route on page 50](#)). In addition, the port2 route has distance and priority values of 10 and 1, respectively.

When two routes to the same destination have the same distance, both remain in the routing table. However, if the priorities are different, FortiGate uses the route with the lowest priority value—port2 in this case. To achieve ECMP with static routes, the distance and priority values must be the same for all routes.

Configure Priority

You will change the priority value for the **port3** route to match the **port2** route.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI, modify the static routing configuration so both default routes are eligible for ECMP.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Verify ECMP on page 61](#)

To configure priority

1. Continuing on the HQ-NGFW-1 GUI, click **Network > Static Routes**.
2. Double-click the **port3** default route to edit it.
3. Click **+** to expand the **Advanced Options** section.
4. Change the **Priority** value to 1.
5. Click **OK**.

Verify ECMP

Now that both port2 and port3 routes share the same distance and priority values, they are eligible for ECMP. First, you will verify the routing table, and then you will verify traffic routing using the **Forward Traffic** logs.

To verify the routing table

1. Return to the HQ-NGFW-1 CLI session, and then enter the following command on HQ-NGFW-1:

```
get router info routing-table all
```
2. Verify that both default routes are currently active.

```
HQ-NGFW-1 # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
S*    0.0.0.0/0 [10/0] via 100.65.0.254, port2, [1/0]
      [10/0] via 100.66.0.254, port3, [1/0]
C     10.0.11.0/24 is directly connected, port4
C     10.0.12.0/24 is directly connected, port5
C     10.0.13.0/24 is directly connected, port6
C     100.65.0.0/24 is directly connected, port2
C     100.66.0.0/24 is directly connected, port3
C     192.168.0.0/16 is directly connected, port1
```

To configure the CLI sniffer

- Continuing on the HQ-NGFW-1 CLI session, enter the following command:

```
diagnose sniff packet any 'tcp[tcpflags] & tcp-syn !=0 and tcp[tcpflags] & tcp-ack == 0
and port 80' 4
```



The 'tcp[tcpflags] & tcp-syn !=0 and tcp[tcpflags] & tcp-ack == 0' filter matches packets with the SYN flag on, so the output will show only all SYN packets for port 80 (HTTP).

- Leave the HQ-NGFW-1 CLI window open in the background.

To verify ECMP routing

- On the HQ-PC-1 VM, in the browser, open a few new tabs, and then visit a few websites, such as:
 - <http://neverssl.com>
 - <http://example.com>
 - <http://eu.httpbin.org>
- Return to the HQ-NGFW-1 CLI session, and then press **Ctrl+C** to stop the sniffer.
- Analyze the sniffer output.

```
HQ-NGFW-1 # diagnose sniff packet any 'tcp[tcpflags] & tcp-syn !=0 and tcp[tcpflags] & tcp-ack == 0 and port 80' 4
Using Original Sniffing Mode
interfaces=[any]
filters=[tcp[tcpflags] & tcp-syn !=0 and tcp[tcpflags] & tcp-ack == 0 and port 80]
5.693764 port4 in 10.0.11.50.33078 -> 34.223.124.45.80: syn 338166153
5.693810 port3 out 100.66.0.101.33078 -> 34.223.124.45.80: syn 338166153
5.863314 port4 in 10.0.11.50.33080 -> 34.223.124.45.80: syn 1055820813
5.863360 port3 out 100.66.0.101.33080 -> 34.223.124.45.80: syn 1055820813
27.709330 port4 in 10.0.11.50.60874 -> 35.183.176.123.80: syn 3322836919
27.709355 port2 out 100.65.0.101.60874 -> 35.183.176.123.80: syn 3322836919
27.785799 port4 in 10.0.11.50.54304 -> 192.229.211.108.80: syn 2989499725
27.785847 port3 out 100.66.0.101.54304 -> 192.229.211.108.80: syn 2989499725
```

The SYN packets are egressing both port2 and port3. This verifies that HQ-NGFW-1 is now load balancing all internet traffic across both routes.

- On the HQ-NGFW-1 GUI, click **Log & Report > Forward Traffic**.
- Identify the **Destination Interface** in the relevant log entries for the websites you accessed.

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID	Destination Interface
2024/11/04 01:27:03	10.0.11.50		52.0.108.150 (eu.httpbin.org)	HTTP	✓ Accept (226 B / 112 B)	Internet (1)	port2
2024/11/04 01:27:01	10.0.11.50		54.164.141.5 (eu.httpbin.org)	HTTP	✓ Accept (2.94 kB / 6.31 kB)	Backup_Internet (2)	port3
2024/11/04 01:26:59	10.0.11.50		172.217.165.10 (fonts.googleapis.com)	HTTPS	✓ Accept (2.81 kB / 6.77 kB)	Internet (1)	port2
2024/11/04 01:26:25	10.0.11.50		34.223.124.45 (beautifulvondorflluststars.ncvcrss.com)	HTTP	✓ Accept (1.71 kB / 2.91 kB)	Backup_Internet (2)	port3
2024/11/04 01:26:24	10.0.11.50		34.273.124.45 (beautifulvondorflluststars.ncvcrss.com)	HTTP	✓ Accept (608 B / 2.59 kB)	Backup_Internet (2)	port3
2024/11/04 01:26:13	10.0.11.50		142.251.41.78 (encrypted-tls.gstatic.com)	HTTPS	✓ Accept (3.57 kB / 11.44 kB)	Backup_Internet (2)	port3
2024/11/04 01:26:13	10.0.11.50		142.250.69.100 (www.google.com)	HTTPS	✓ Accept (4.01 kB / 22.77 kB)	Internet (1)	port2

Lab 4: Firewall Authentication

In this lab, you will examine how to configure FortiGate to communicate with remote LDAP and RADIUS servers for server-based password authentication.

Objectives

- Configure server-based password authentication with an LDAP server
- Configure server-based password authentication with a RADIUS server

Time to Complete

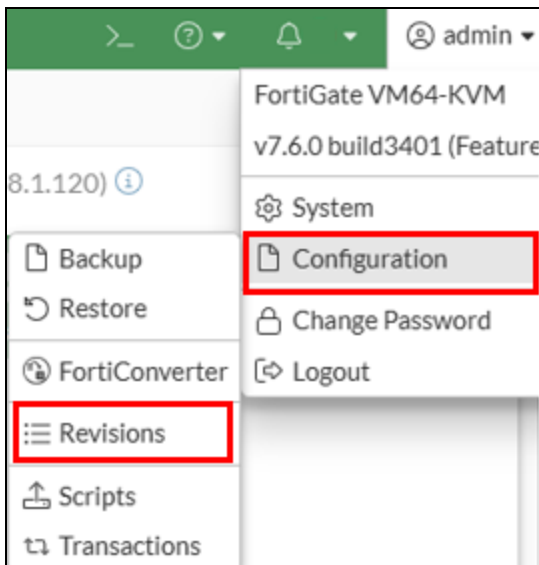
Estimated: 35 minutes

Prerequisites

Before you begin this lab, you must restore a configuration file on HQ-NGFW-1.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **HQ-FW-Auth**, and then click **Revert**.

Delete

Details

Diff

Revert

Save Changes

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Configuring an LDAP Server

In this exercise, you will examine how to configure an LDAP server on FortiGate for remote authentication. Next, you will create a remote authentication group for remote users, and then add that group as a source in a firewall policy. Finally, you will authenticate as one of the remote users, and then monitor the login as the administrator.

Configure an LDAP Server on FortiGate

You will configure FortiGate to point to a preconfigured FortiAuthenticator that is acting as an LDAP server for server-based password authentication.

To configure an LDAP server on FortiGate

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **User & Authentication > LDAP Servers**, and then click **Create New**.
3. Configure the following settings:

Field	Value
Name	External_Server
Server IP/Name	10.0.13.130 This is the IP address of the FortiAuthenticator that is acting as the LDAP server. For more information, see Network Topology on page 9 .
Server Port	389 This is the default port for LDAP.
Common Name Identifier	uid This is the attribute name that is used to find the username on the preconfigured LDAP server.
Distinguished Name	ou=Training,dc=training,dc=lab This is the domain name for the LDAP directory on FortiAuthenticator, with all users located under the Training organizational unit (ou).
Bind Type	Regular
Username	uid=admin,cn=Users,dc=training,dc=lab You are using the credentials of the admin LDAP user to authenticate to the LDAP server.

Field	Value
Password	Fortinet1!
	This is the password that is preconfigured for the adadmin user. You must use it to be able to bind.

4. Click **Test Connectivity**.

You should see a message indicating that the connection was successful.

5. Click **OK**.

Assign an LDAP User Group to a Firewall Group

You will assign an LDAP user group (**AD_users**) that includes two users (**aduser1** and **aduser2**) to a firewall user group, called **Remote-users**, on FortiGate. By doing this, you will be able to configure firewall policies to act on the firewall user group.

Usually, groups are used to more effectively manage individuals who have a shared relationship.



The **Remote-users** firewall group is preconfigured for you. However, you must modify it to add the users from the remote LDAP server you configured in the previous procedure.

Take the Expert Challenge!

On HQ-NGFW-1 (10.0.11.254), assign the Active Directory user group named **AD_users** to the FortiGate firewall user group named **Remote-users**.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you have completed this exercise, see [Configuring an LDAP Server on page 66](#).

To assign a user to a user group

1. On the HQ-NGFW-1 GUI, click **User & Authentication > User Groups**, and then edit the **Remote-users** group.
2. Notice that this group is currently configured as a firewall group.
3. In the **Remote Groups** table, click **Add** to add users from the remote LDAP server.

Edit User Group

Name: Remote-users

Type: Firewall

Members: +

Remote Groups

+ Add Edit Delete

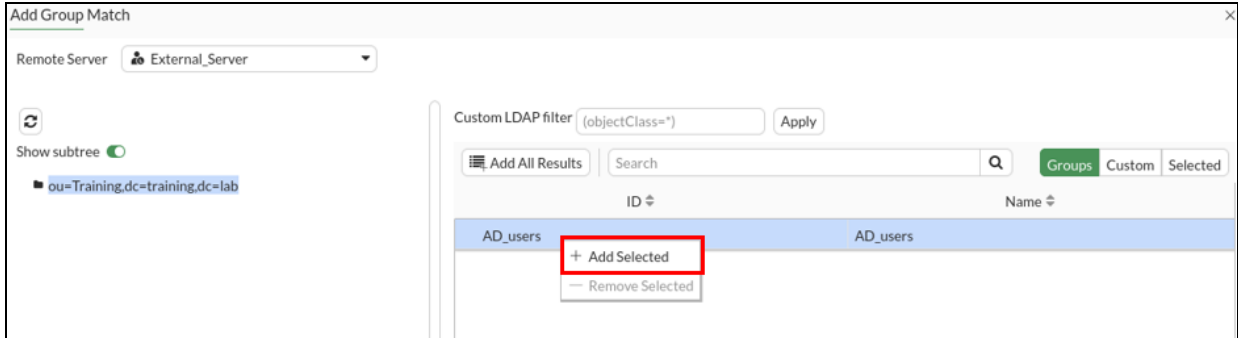
Remote Server	Group Name
No results	

The **Add Group Match** window opens.

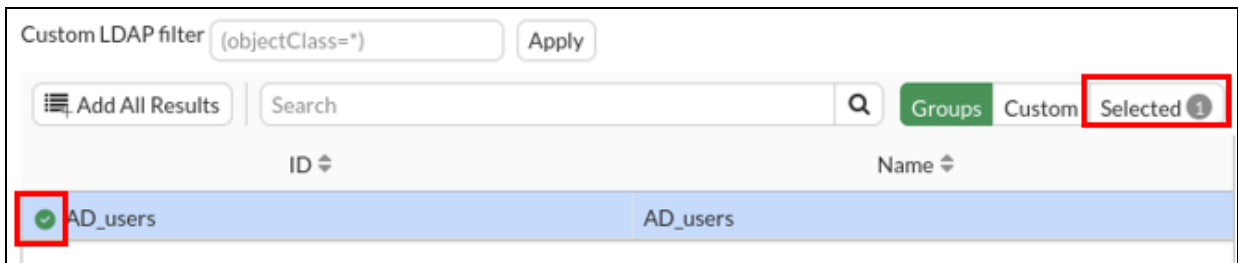
Add Group Match

Remote Server: [dropdown]

4. In the **Remote Server** field, select **External_Server**.
5. On the **Groups** tab, right-click **AD_users**, and then select **Add Selected**.

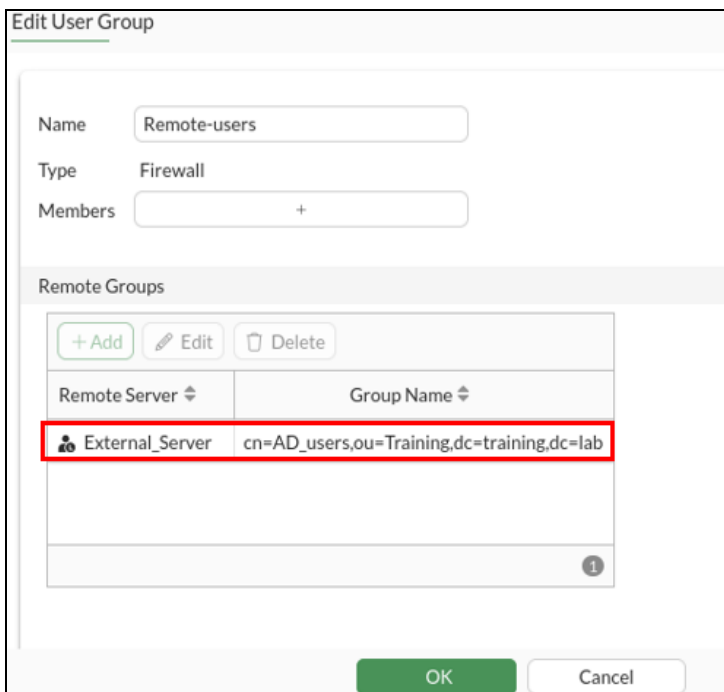


AD_users has a green check mark beside it, which indicates that it was added.



6. Click **OK**.

The users in this Active Directory group are now included in the FortiGate **Remote-users** firewall user group. Only users from the remote LDAP server that match this user group entry can authenticate.



7. Click **OK**.

Add the Remote User Group to the Firewall Policy

Now that you have added the LDAP server to the **Remote-users** firewall user group, you can add the group to a firewall policy. This allows you to control access to network resources, because policy decisions are made for the group as a whole.

To add the remote user group to the firewall policy

1. On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**, and then double-click the existing port4 to port2 firewall policy.

Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT	Type	Security Profiles	Log
port4 → port2	Internet (1)	HQ_SUBNET	all	always	ALL	✓ ACCEPT	✓ NAT	Standard	SSL certificate-inspection	UTM

2. Configure the following setting:

Field	Value
User/group	Click +, and then select Remote-users (located under User).

3. In the **Security Profiles** section, enable **Web Filter**, and then select **Category_Monitor**.
This web filter was preconfigured and is set to block the following categories: **Potentially Liable**, **Adult/Mature Content**, and **Security Risk**.
4. In the **Logging Options** section, ensure **Log Allowed Traffic** is enabled, and then select **All Sessions**.
5. Click **OK**.

Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT	Type	Security Profiles	Log
port4 → port2	Internet (1)	HQ SUBNET	all	always	ALL	✓ ACCEPT	✓ NAT	Standard	WEB Category_Monitor SSL certificate-inspection	✓ All

To test whether aduser1 can successfully authenticate

1. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Enter the following command:


```
diagnose test authserver ldap <LDAP server name> <LDAP user name> <password>
```

Where:

- <LDAP server name> is External_Server (case sensitive)
- <LDAP user name> is aduser1
- <password> is Fortinet1!

A message like the following example should appear to indicate that authentication was successful:

```
HQ-NGFW-1 # diagnose test authserver ldap External_Server aduser1 Fortinet1!  
authenticate 'aduser1' against 'External_Server' succeeded!  
Group membership(s) - cn=AD_users,ou=Training,dc=training,dc=lab
```

3. Close the HQ-NGFW-1 CLI.

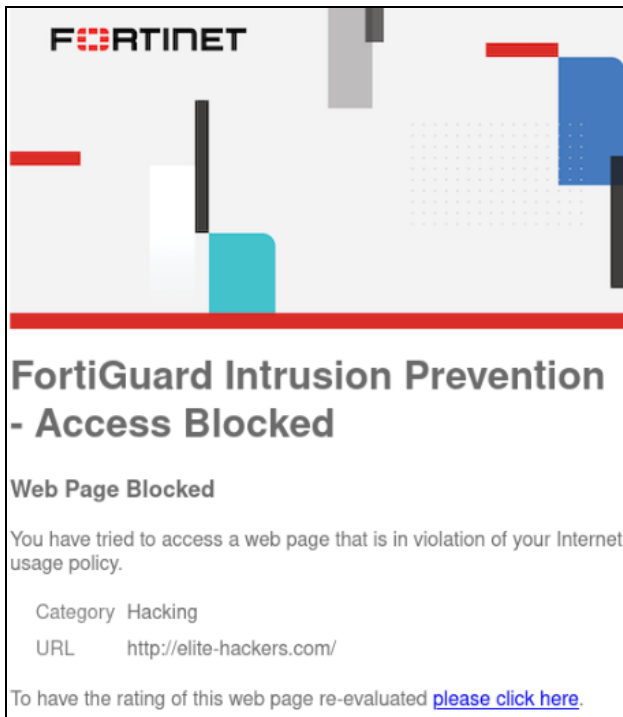
Authenticate and Monitor the Authentication

You will authenticate through the firewall policy as `aduser1`. This user is a member of the **Remote-users** group on FortiGate. Then, you will monitor the authentication.

To authenticate as a remote user

1. On the HQ-PC-1 VM, open a new browser tab, and then go to `elite-hackers.com`.
You are asked to log in to the network.
2. Log in with the following credentials:
 - Username: `aduser1`
 - Password: `Fortinet1!`

This URL is set to be blocked by the web filter security profile you enabled in the firewall policy.



The blocked page displays a replacement message that includes useful information, such as the **URL** and **Category**.

To monitor active authenticated users

1. Return to the browser tab where you are logged in to the HQ-NGFW-1 GUI as `admin`.
2. Click **Dashboard > Assets&Identities**, and then click **Firewall Users** to expand it to full screen to view this login authentication and monitor the firewall authenticated user.

User Name	IP Address	User Group	Duration	Traffic Volume	Method
aduser1	10.0.11.50	Remote users	1m 23s	1.05 MB	Firewall

aduser1 is listed along with other information, such as **User Group** and **IP Address**.

3. Click **aduser1**, and then click **Deauthenticate**.

User Name	IP Address	User Group	Duration	Traffic Volume	Method
aduser1	10.0.11.50	Remote users	1m 23s	1.05 MB	Firewall



The `config user setting` CLI command determines how long a user can remain authenticated. However, you can choose to manually revoke a user authentication by selecting the user in the **Firewall User Monitor** list, and then clicking **Deauthenticate**. After the user is deauthenticated, the user disappears from the list, because it is reserved for active users only.

4. In the **Confirm** window, click **OK**.

Confirm

⚠ Are you sure you want to deauthenticate the selected user(s)?

OK Cancel

This deauthenticates the user. The user must log in again to access the resources that the firewall policy protects.

Remove the User Group From the Firewall Policy

You will remove the user group that is assigned to the firewall policy for authentication.

To remove the remote user group from the firewall policy

1. On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**, and then double-click the existing port4 to port2 firewall policy.

Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT	Type	Security Profiles	Log
Internet (1)	HQ_SUBNET Remote users	all	always	ALL	ACCEPT		NAT	Standard	WCD Category_Monitor SQL certificate inspection	All
Edit Insert Disable Delete More										

2. In the **User/group** field, remove the **Remote-users** user group.

Edit Policy

NameInternet

Schedulealways

ActionACCEPTDENY

Incoming interfaceport4

Outgoing interfaceport2

Source & DestinationShow logic

SourceHQ_SUBNET

User/groupRemote-users

Destinationall

ServiceALL

3. Click **Close**, and then click **OK** to save the changes.

Exercise 2: Configuring a RADIUS Server on FortiGate

In this exercise, you will examine how to configure a RADIUS server on FortiGate for remote authentication. Next, you will create a remote authentication group for remote users, and then add that group as a source in a firewall policy. Finally, you will authenticate as one of the remote users, and then monitor the login as the administrator.

Configure a RADIUS Server on FortiGate

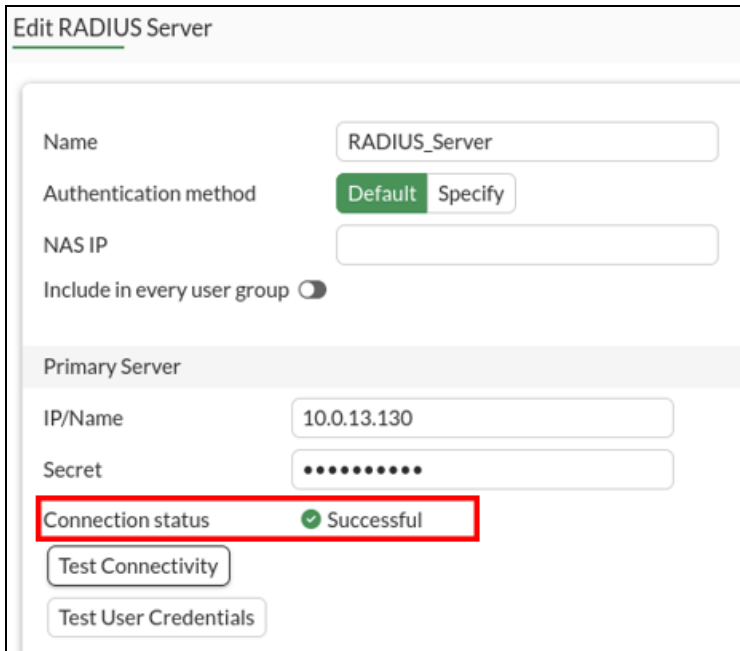
You can configure FortiGate to point to a preconfigured FortiAuthenticator that is acting as a RADIUS server for server-based password authentication.

To configure a RADIUS server on FortiGate

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **User & Authentication > RADIUS Servers**, and then click **Create New**.
3. Configure the following settings:

Field	Value
Name	RADIUS_Server
Authentication method	Default
Primary Server IP/Name	10.0.13.130 This is the IP address of the FortiAuthenticator that is acting as the RADIUS server. For more information, see Network Topology on page 9 .
Secret	Fortinet1!

4. Click **Test Connectivity**.



You should see a message indicating that the connection was successful.

5. Click **OK**.

Assign a RADIUS User Group to a Firewall Group

You will assign a RADIUS user group (**Training**) that includes a user (**radius1**) to a firewall user group, called **Training**, on FortiGate. By doing this, you will be able to configure firewall policies to act on the firewall user group.

Usually, groups are used to more effectively manage individuals who have a shared relationship.



The **Training** firewall group is preconfigured for you. However, you must modify it to add the users from the remote RADIUS server you configured in the previous procedure.

Take the Expert Challenge!

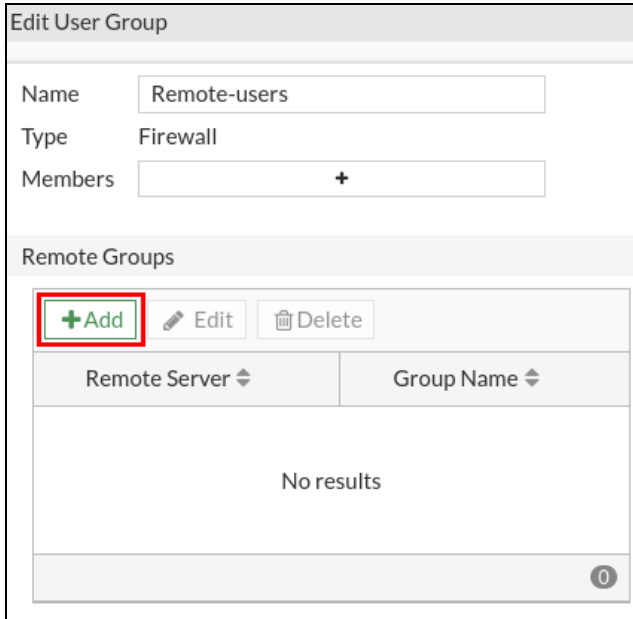
On HQ-NGFW-1 (10.0.11.254), assign the RADIUS user group named **Training** to the FortiGate firewall user group named **Training**.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you have completed this exercise, see [Configuring a RADIUS Server on FortiGate on page 74](#).

To assign a user to a user group

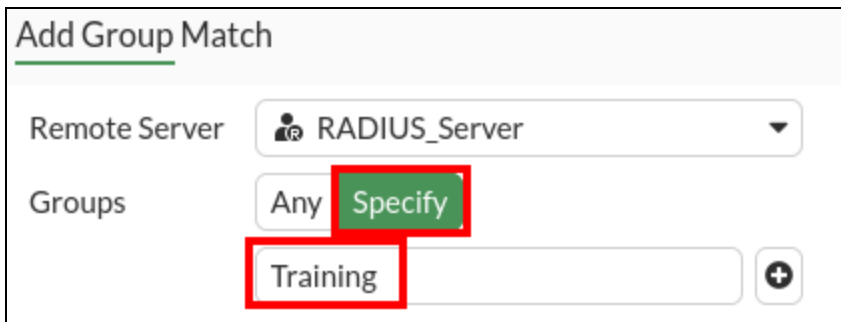
1. On the HQ-NGFW-1 GUI, click **User & Authentication > User Groups**, and then edit the **Training** group.
2. Notice that this group is currently configured as a firewall group.
3. In the **Training** table, click **Add** to add users from the remote RADIUS server.



The **Add Group Match** window opens.



4. In the **Remote Server** field, select **RADIUS_Server**.
5. In the **Groups** field, select **Specify**, and then type the group name **Training**.



6. Click **OK**.

The user in this RADIUS server group is now included in the FortiGate **Training** firewall user group. Only users from the remote RADIUS server that match this user group entry can authenticate.



The remote RADIUS server is configured with using the RADIUS attribute value pair (AVP) 26, known as a vendor-specific attribute (VSA). This attribute allows the Fortinet-Group-Name VSA to be included in the RADIUS response. In FortiOS, the user group must be configured to specifically match this group.

Name	Training
Type	Firewall
Members	+
Remote Groups	
+ Add Edit Delete	
Remote Server	Group Name
RADIUS_Server	Training

7. Click **OK**.

Add the Training User Group to the Firewall Policy

Now that you have added the RADIUS server to the **Training** firewall user group, you can add the group to a firewall policy. This allows you to control access to network resources, because policy decisions are made for the group as a whole.

To add the Training user group to the firewall policy

- 1. On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**, and then double-click the existing port4 to port2 firewall policy.

Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT	Type	Security Profiles	Log
port4 -> port2	Internet (1)	10.0.0.0/24	always	ALL	ACCEPT		NAT	Standard	Web, SQL, Category_Monitor, Certificate-Inspection	All
Edit Insert Disable Delete More										

- 2. Configure the following setting:

Field	Value
User/group	Click +, and then select Training (located under User).

- 3. Click **OK**.

Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT	Type	Security Profiles	Log
port4 → port2	Internet (1)	HQ_SUBNET	all	always	ALL	✓ ACCEPT	✓ NAT	Standard	WEB Category_Monitor SSL certificate-inspection	UTM
		Training								

To test whether the radius1 user can successfully authenticate

1. On the HQ-NGFW-1 CLI, log in with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Enter the following command:

```
diagnose test authserver radius <RADIUS server name> mschap2 <RADIUS user name>  
<password>
```

Where:

- <RADIUS server name> is RADIUS_Server (case sensitive)
- <RADIUS user name> is radius1
- <password> is Fortinet1!

A message like the following example should appear to indicate that authentication was successful:

```
HQ-NGFW-1 # diagnose test authserver radius RADIUS_Server mschap2 radius1 Fortinet1!  
authenticate 'radius1' against 'mschap2' succeeded, server=primary assigned_rad_session_id=16359576514562  
session_timeout=0 secs idle_timeout=0 secs!  
Group membership(s) - Training
```

3. Close the HQ-NGFW-1 CLI.

Authenticate and Monitor the Authentication

You will authenticate through the firewall policy as `radius1`. This user is a member of the **Training** group on FortiGate. Then, you will monitor the authentication.

To authenticate as a remote RADIUS user

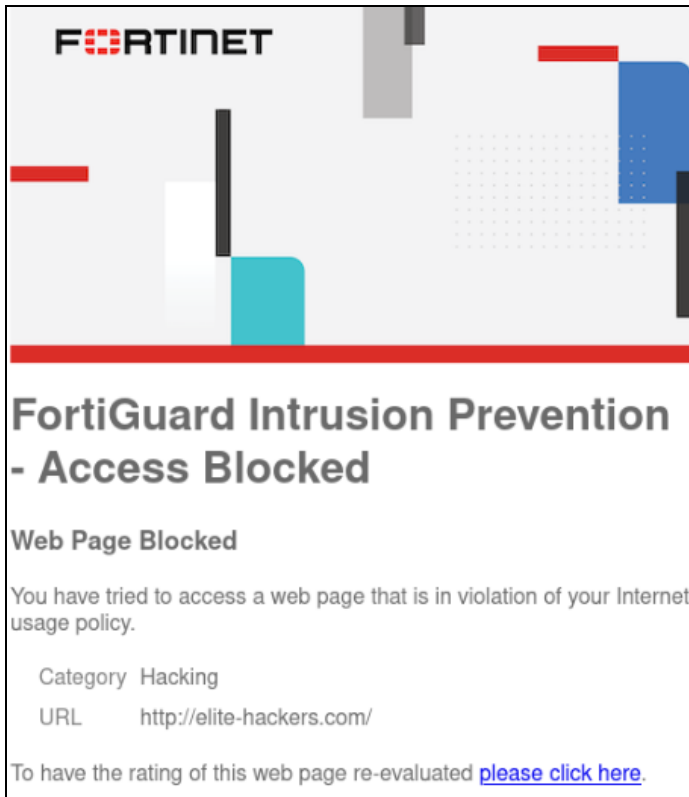
1. On the HQ-PC-1 VM, open a new browser tab, and then go to `elite-hackers.com`.

You are asked to log in to the network.

2. Log in with the following credentials:

- Username: radius1
- Password: Fortinet1!

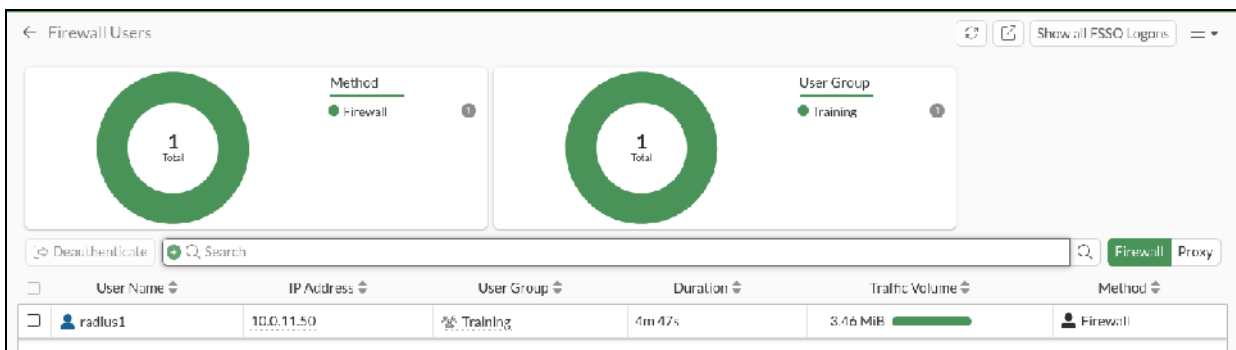
This URL is set to be blocked by the web filter security profile you enabled in the firewall policy.



The blocked page displays a replacement message that includes useful information, such as the **URL** and **Category**.

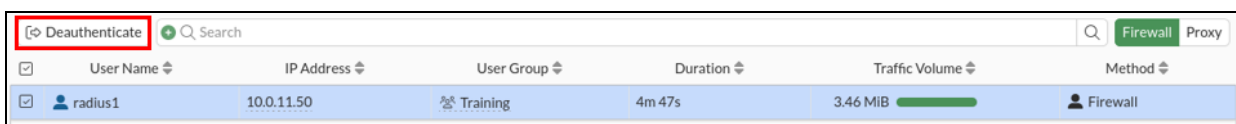
To monitor active authenticated users

1. Return to the browser tab where you are logged in to the HQ-NGFW-1 GUI as `admin`.
2. Click **Dashboard > Assets & Identities**, and then click **Firewall Users** to expand it to full screen to view this login authentication and monitor the firewall authenticated user.



You will see the user **radius1** listed along with other information, such as **User Group** and **IP Address**.

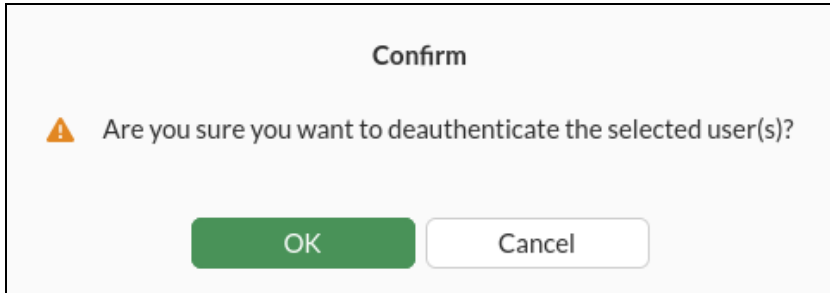
3. Click **radius1**, and then click **Deauthenticate**.





The `config user setting` CLI command determines how long a user can remain authenticated. However, you can choose to manually revoke a user authentication by selecting the user in the **Firewall User Monitor** list, and then clicking **Deauthenticate**. After the user is deauthenticated, the user disappears from the list, because it is reserved for active users only.

4. In the **Confirm** window, click **OK**.



This deauthenticates the user. The user must log in again to access the resources that the firewall policy protects.

Remove the User Group From the Firewall Policy

You will remove the user group that is assigned to the firewall policy for authentication.

To remove the remote user group from the firewall policy

1. On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**, and then double-click the existing port4 to port2 firewall policy.

Policy	Source	Destination	Schedule	Service	Action	IP Pool
[-] port4 → port2 1						
☒	Internet (1) HQ_SUBNET Training	all	always	ALL	✓ ACCEPT	
[Edit] [Insert ▼] [Disable] [Delete] [More ▼]						

2. In the **Source** field, remove the **Training** user group.

Edit Policy

Name ⁱ Internet

Schedule always

Action ☒ ACCEPT ☐ DENY

Incoming interface port4

Outgoing interface port2

Source & Destination Show logic

Source HQ_SUBNET

User/group Training

Destination all

Service ALL

3. Click **Close**, and then click **OK** to save the changes.

Lab 5: Fortinet Single Sign-On Configuration

In this lab, you will test user authentication using Fortinet Single Sign-On (FSSO).

Objectives

- Review the FSSO configuration on FortiGate
- Test the transparent or automatic user identification by generating user logon events
- Monitor the FSSO status and operation

Time to Complete

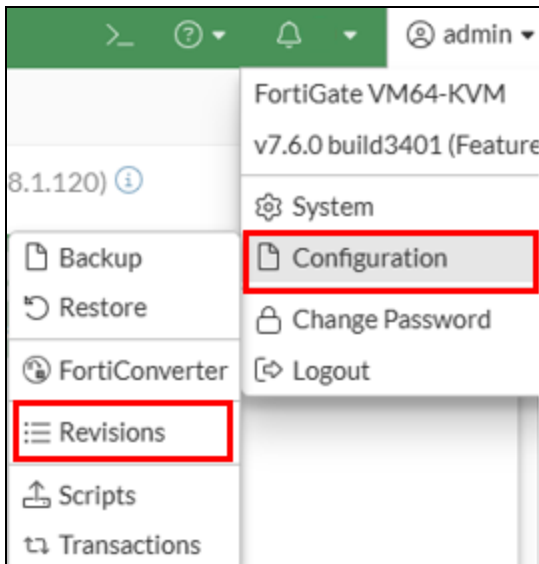
Estimated: 35 minutes

Prerequisites

Before you begin this lab, you must restore a configuration file on HQ-NGFW-1.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **HQ-FSSO**, and then click **Revert**.

Delete

Details

Diff

Revert

Save Changes

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Configuring FortiGate for FSSO Authentication

In this exercise, you will configure FortiGate for FSSO and test user authentication. The lab uses a demo environment to emulate the behavior of an active FSSO DC agent from the HQ-PC-1 VM using a Python script. Therefore, you will not configure a DC agent to send logon events from the HQ-PC-1 VM.



In a real-world environment, you must configure FortiGate to identify users by polling their logon events using an FSSO agent, and you must install and configure a collector agent. FSSO agents are available on the Fortinet Support website (<http://support.fortinet.com>).

For FortiGate to communicate and poll information from the FSSO collector agent, you must assign the polled user to a firewall user group, and then add the user group as a source on a firewall policy.

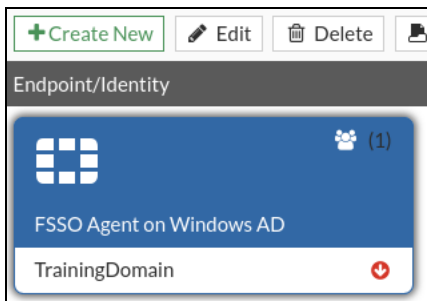
Finally, you can verify the user logon event that FortiGate collects. This event is generated after a user logs in to the Windows Active Directory domain. Therefore, no firewall authentication is required.

Review the FSSO Configuration on FortiGate

You will review the FSSO configuration and FSSO user groups on FortiGate. FSSO allows FortiGate to automatically identify the users who connect using SSO. Then, you will add FSSO user groups to the firewall policies.

To review the FSSO server and FSSO user group configuration on FortiGate

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Security Fabric > External Connectors**.
3. Select **TrainingDomain**, and then click **Edit**.



4. In the upper-right corner, review the **Endpoint/Identity** status, and see that the status is **Disconnected**.
5. Leave the browser window open.

To run a script to simulate a user logon event

1. On the HQ-PC-1 VM, open a terminal window, and then enter the following commands to simulate a user logon event:

```
cd Desktop/FSSO/  
python2 fssoreplay.py -l 8000 -f sample.log
```

2. Keep the terminal window open.

The script continues to run in the background.

To review the FSSO connection and FSSO user groups

1. Continuing in the **TrainingDomain** window, click **Apply & Refresh**.
2. Select **TrainingDomain**, and then click **Edit**.
3. In the **Users/Groups** field, click **View**.

Endpoint/Identity

FSSO Agent on Windows AD

Connector Settings

Name: TrainingDomain

Primary FSSO agent: 10.0.11.50

Trusted SSL certificate: ☐

User group source: Collector Agent Local

Users/Groups: 1 View

Apply & Refresh OK Cancel

The **training.lab/Users** monitored group is displayed.

Collector Agent Group Filters

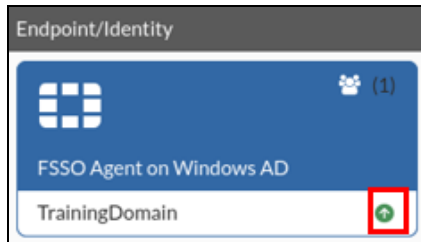
Delete Search

AD Group

training.lab/Users

4. Click **X** to close the **Collector Agent Group Filters** window.
5. Click **OK**.

A green up arrow confirms that the communication with the FSSO collector agent is up.



To assign the FSSO user to an FSSO user group

1. Continuing on the HQ-NGFW-1 GUI, click **User & Authentication > User Groups**.
2. Click **Create New**, and then configure the following settings:

Field	Value
Name	Training
Type	Fortinet Single Sign-On (FSSO)
Members	training.lab/Users



The FSSO user is automatically listed because of the selected group type—FSSO.

3. Click **OK**.

Assign FSSO Users to a Firewall Policy

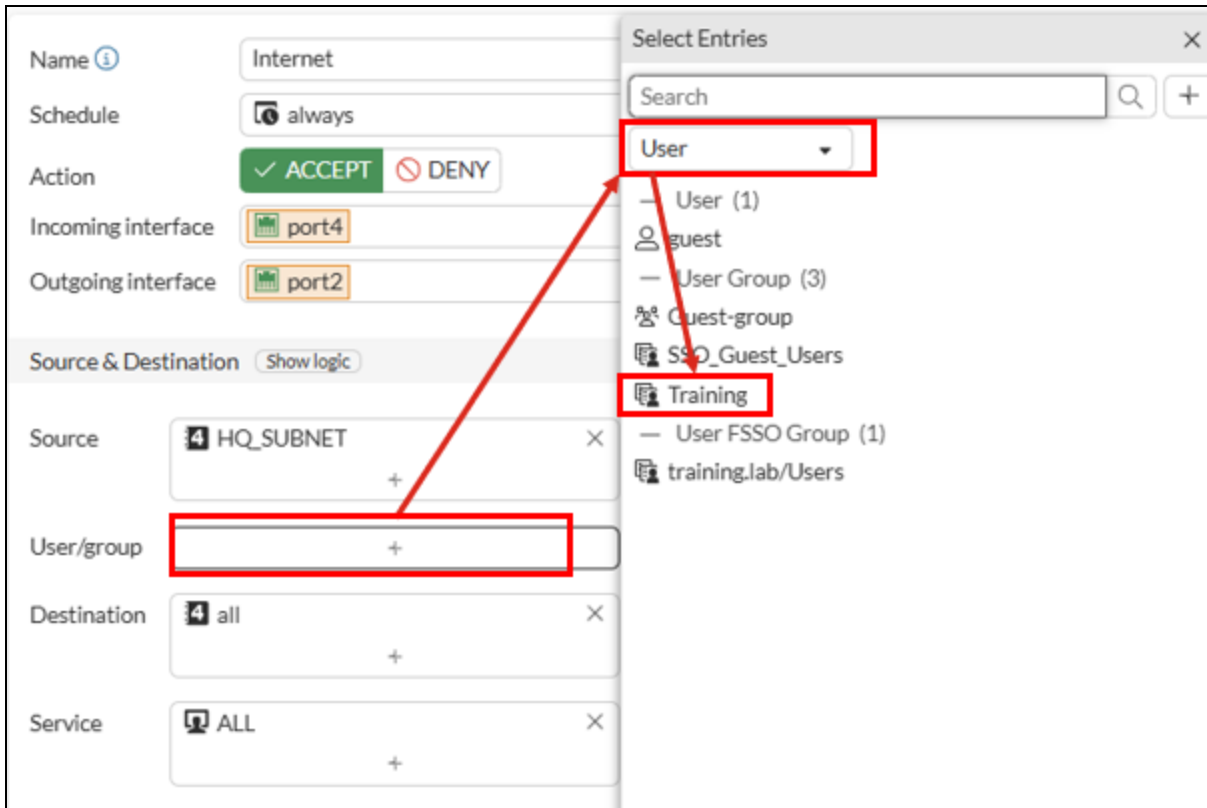
You will assign your FSSO user group as a source in a firewall policy. This allows you to control access to network resources based on user identity.

To test the connection without assigning the FSSO user group to a firewall policy

1. On the HQ-PC-1 VM, open a new browser, and then go to <https://www.fortinet.com>.
You can see that all users can access the Fortinet website.

To add the FSSO user group to your firewall policy

1. Return to the browser where you are logged in to the HQ-NGFW-1 GUI, and then click **Policy & Objects > Firewall Policy**.
2. Edit the **Internet** firewall policy.
3. In the **User/group** field, click **+**, select **User**, and then select **Training** to add the **Training** group.



4. Click **Close**, and then click **OK**.

Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT	Type	Security Profiles	Log
Internet (1)	HQ_SUBNET Training	all	always	ALL	ACCEPT		NAT	Standard	no-inspection	UTM

Test FSSO

After a user logs in, they are automatically identified based on their IP address. As a result, FortiGate allows the user to access network resources as policy decisions are made. You will test FSSO.

To test the connection after assigning the FSSO user to the firewall policy

- On the HQ-PC-1 VM, open a new browser tab, and then go to <http://support.fortinet.com>.



The Python script that is running on the HQ-PC-1 VM is already sending user logon events with the following information:

- user:** aduser1
- IP:** 10.0.11.50

In this case, the website loads successfully because aduser1 belongs to the configured user group on a firewall policy.

To review the connection status between the FSSO collector agent and FortiGate

1. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Enter the following commands to show the connection status between FortiGate and each collector agent:

```
diagnose debug enable
diagnose debug authd fsso server-status
```

3. Observe the CLI output.

Your FortiGate is connected to the FSSO collector agent.

```
Server Name Connection Status Version Address
-----
TrainingDomain connected FSAE server 1.1 10.0.11.50
```

To monitor communication between the FSSO collector agent and FortiGate

1. Continuing on the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Enter the following commands:

```
diagnose debug enable
diagnose debug application authd 8256
```
3. On the HQ-PC-1 VM, in a terminal window, press **Ctrl+C** to stop the script, and then enter the following command again to simulate a user logon event:

```
python2 fssoreplay.py -l 8000 -f sample.log
```
4. View the output of the **diagnose** command.

```
[_process_logon: 1079]: ADUSER1(10.0.11.50, 0) logged on from TrainingDomain.
[_process_logon:1122di]: ADUSER1 (10.0.11.50, 0) from TrainingDomain exists
fsae_io_ctx_process_msg[TrainingDomain]: received heartbeat 100004
fsae_io_ctx_process_msg[TrainingDomain]: received heartbeat 100005
```



You generated a logon event on the HQ-PC-1 VM using the script, and it was forwarded to FortiGate.

5. Enter the following command to stop the debug process:

```
diagnose debug reset
```

To display the FSSO logon events

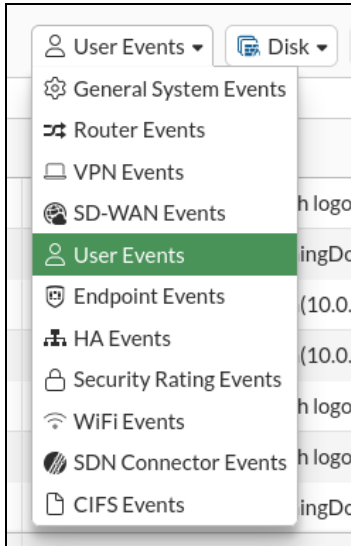
1. Continuing on the HQ-NGFW-1 VM, enter the following command:

```
diagnose debug authd fsso list
```
2. Review the output, which shows the FSSO logon events.

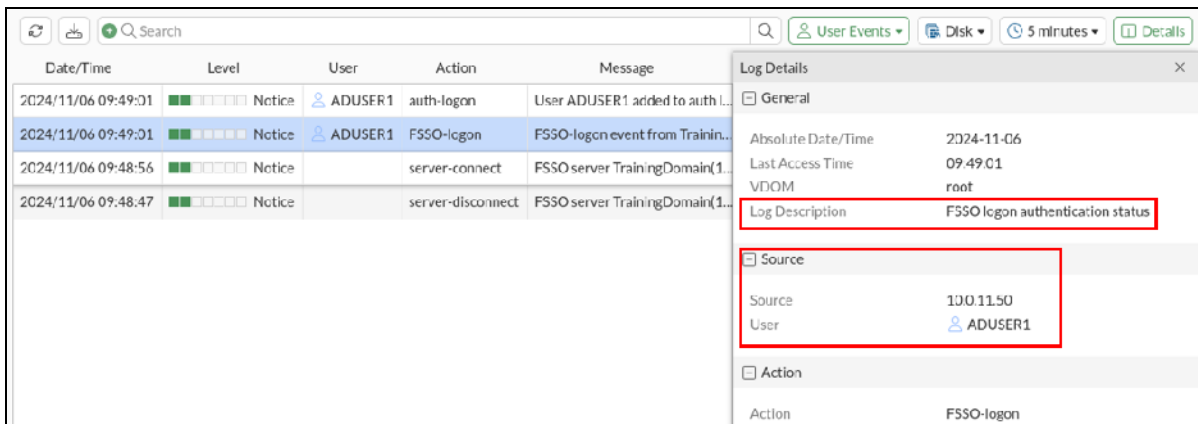
```
----FSSO logons----
IP:10.0.11.50 User: ADUSER1 Groups: training.lab/Users Workstation
C7280677811.training.lab MemberOf: Training training.lab/Users
Total number of logons listed: 1, filtered: 0
----end of FSSO logons----
```

To review the user event logs

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Log & Report > System Events**, and then in the **User Events** widget, click the **View Logs** arrow.

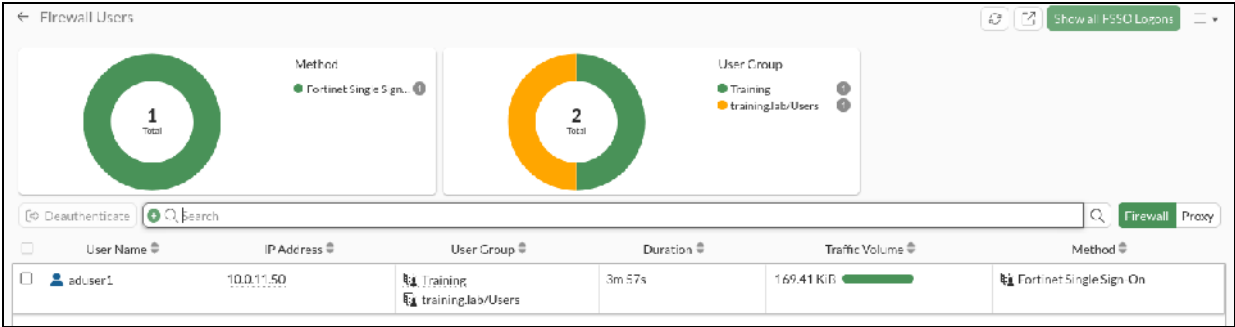


3. Select a log, and then click **Details** to view more information about it.



To monitor FSSO logon events

1. Continuing on the HQ-NGFW-1 GUI, click **Dashboard > Assets & Identities**, and then click **Firewall Users** to expand it to full screen.
2. Click **Show all FSSO Logons**, and then click **Refresh** if the user's details don't appear.



Lab 6: Certificate Operations

In this lab, you will configure full SSL inspection using a self-signed SSL certificate on FortiGate to inspect outbound traffic. Next, you will review some situations that prevent full SSL inspection, and implement workarounds. Finally, you will learn how to handle some certificate anomalies.

Objectives

- Configure and enable full SSL inspection on outbound traffic
- Handle certificate anomalies

Time to Complete

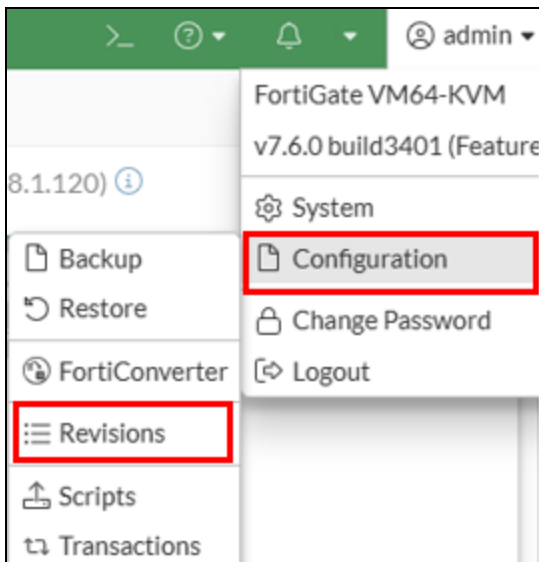
Estimated: 40 minutes

Prerequisites

Before you begin this lab, you must restore a configuration file on HQ-NGFW-1.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **HQ-Certificate**, and then click **Revert**.

Delete

Details

Diff

Revert

Save Changes

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Configuring Full SSL Inspection on Outbound Traffic

Full SSL inspection on outbound traffic allows FortiGate to inspect encrypted internet traffic and apply security profiles to that traffic. It protects your network and end users from potential malware that could come from encrypted websites, like HTTPS websites, that internal users visit. FortiGate employs a man-in-the-middle (MITM) technique to inspect the traffic and apply security profiles, such as antivirus, web filter, and application control.

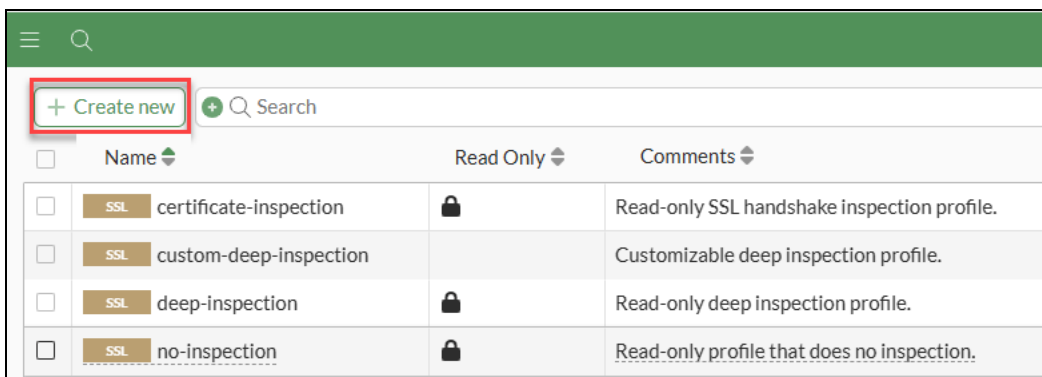
In this exercise, you will configure and enable full SSL inspection on all outbound traffic.

Configure SSL Inspection

By default, FortiGate includes four security profiles for SSL/SSH inspection: **certificate-inspection**, **custom-deep-inspection**, **deep-inspection**, and **no-inspection**. You can modify the settings for the **custom-deep-inspection** profile only or create a personalized profile. The other profiles are read-only. Because this exercise involves configuring full SSL inspection on FortiGate, you will configure a new SSL/SSH inspection profile for this purpose.

To configure SSL inspection

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Security Profiles > SSL/SSH Inspection**.
3. Click **Create new** to create a new profile.



4. In the **Name** field, type `Custom_Full_Inspection`.
5. In the **SSL Inspection Options** section, verify that the following settings are configured (default values):

Field	Value
Enable SSL inspection of	Multiple Clients Connecting to Multiple Servers

Field	Value
Inspection method	Full SSL Inspection
CA certificate	Fortinet_CA_SSL

SSL Inspection Options

Enable SSL inspection of	Multiple Clients Connecting to Multiple Servers
Inspection method	SSL Certificate Inspection Full SSL Inspection
CA certificate	Fortinet_CA_SSL
Blocked certificates	Allow Block View Blocked Certificates
Untrusted SSL certificates	Allow Block Ignore View Trusted CAs List
Server certificate SNI check	Enable Strict Disable
Enforce SSL cipher compliance	
Enforce SSL negotiation compliance	
RPC over HTTPS	
MAPI over HTTPS	

6. Scroll down to the bottom of the page, and then in the **Common Options** section, confirm the following:
- The **Invalid SSL certificates** field is set to **Custom**.
 - Confirm that the other settings are configured as shown in the following image (default values):

Common Options

Invalid SSL certificates	Allow Block Custom
Expired certificates	Keep Untrusted & Allow Block Trust & Allow
Revoked certificates	Keep Untrusted & Allow Block Trust & Allow
Validation timed-out certificates	Keep Untrusted & Allow Block Trust & Allow
Validation failed certificates	Keep Untrusted & Allow Block Trust & Allow
Log SSL anomalies	

7. Click **OK**.

Enable SSL Inspection in a Firewall Policy

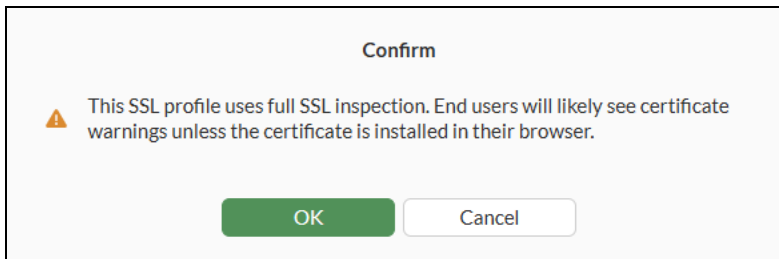
You must enable SSL inspection in a firewall policy to start inspecting SSL traffic. In this policy, you will use SSL inspection associated with web filtering. For the purposes of this lab, you will enable the default web filter security profile.

To enable SSL inspection in a firewall policy

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Edit the **Internet** firewall policy.
3. In the **Security Profiles** section, enable the following security profiles:

Security Profile	Value
Web Filter	default
SSL Inspection	Custom_Full_Inspection This is the profile you created previously.

4. In the **Logging Options** section, enable **Log Allowed Traffic**, and then select **All Sessions**.
5. Click **OK**.
FortiGate displays a warning message to highlight that full SSL inspection is activated and might trigger warnings in users' browsers.



6. Read the warning message, and then click **OK**.

Install the Fortinet_CA_SSL Certificate

FortiGate includes an SSL certificate, named Fortinet_CA_SSL, that you can use for full SSL inspection. The SSL inspection profile you created in the previous step uses it. This certificate is signed by a certificate authority (CA) named FortiGate CA, which is not public. Because the CA is not public, each time a user connects to a secure website, the browser displays a certificate warning. This is because the browser receives traffic that was encrypted by certificates that were signed by FortiGate, using a CA it does not know and trust.

You can avoid this warning by downloading the Fortinet_CA_SSL certificate and installing it on all workstations as a public authority.

You will first test access to a secure website *without* the Fortinet_CA_SSL certificate installed in the browser. Then, you will install the Fortinet_CA_SSL certificate in the browser and test access to the secure website again.

To test full SSL inspection without a trusted CA

1. Connect to the HQ-PC-1 VM, and then log in with the following credentials:
 - Username: Administrator
 - Password: Fortinet1!
2. Open a browser, and then go to an HTTPS site, such as:
`https://www.goto.com`

3. Notice the certificate warning.



Warning: Potential Security Risk Ahead

Firefox detected a potential security threat and did not continue to **www.goto.com**. If you visit this site, attackers could try to steal information like your passwords, emails, or credit card details.

What can you do about it?

The issue is most likely with the website, and there is nothing you can do to resolve it.

If you are on a corporate network or using antivirus software, you can reach out to the support teams for assistance. You can also notify the website's administrator about the problem.

[Learn more...](#)

Go Back (Recommended)Advanced...

This warning appears because the browser receives certificates that were signed by the FortiGate CA private key, and the corresponding CA certificate is not in the certificate store of the HQ-PC-1 VM.

4. Click **Advanced**, and then click **View Certificate**.
- You can see that the certificate is issued by Fortinet (Issuer Name), and that it is valid. The subject alternative names list includes a reference to the website you visited (`goto.com`, in our example).

Issuer Name

Country	US
State/Province	California
Locality	Sunnyvale
Organization	Fortinet
Organizational Unit	Certificate Authority
Common Name	FGVMC
Email Address	support@fortinet.com

Validity

Not Before	Fri, 28 Jun 2024 00:00:00 GMT
Not After	Sat, 28 Jun 2025 23:59:59 GMT

Subject Alt Names

DNS Name	gotomeeting.com
DNS Name	*.goto.com
DNS Name	*.goto.de
DNS Name	*.gotomeeting.com
DNS Name	*.gotomypc.com

5. Do not click **Accept the Risk and Continue**.



If the certificate warning indicates **HTTP Strict Transport Security (HSTS)**, the option **Accept the Risk and Continue** does not appear.

6. Leave the browser tab open, and then continue to the next procedure.

To install the Fortinet_CA_SSL certificate in the browser

1. On the HQ-PC-1 VM, open a new browser tab, and then connect to HQ-NGFW-1 at 10.0.11.254.
This time, you might see a warning because the FortiGate GUI presented a certificate that was signed by a CA that your browser doesn't trust.



Warning: Potential Security Risk Ahead

Firefox detected a potential security threat and did not continue to **10.0.11.254**. If you visit this site, attackers could try to steal information like your passwords, emails, or credit card details.

[Learn more...](#)

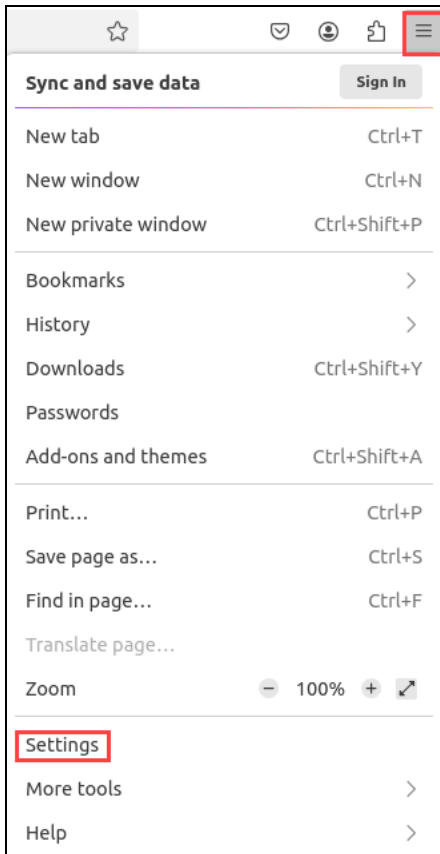
Go Back (Recommended)Advanced...

2. If you receive the warning message, click **Advanced**, and then click **Accept the Risk and Continue**.
3. Log in to the HQ-NGFW-1 GUI at 10.0.11.254 with the following credentials:
 - Username: admin
 - Password: Fortinet1!
4. Click **System > Certificates**.
5. In the **Local CA Certificate** section, click **Fortinet_CA_SSL**, and then click **Download**.

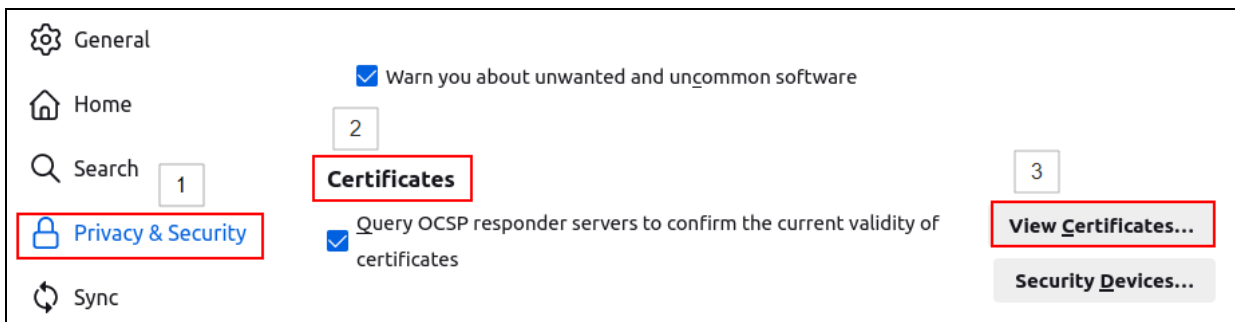
+ Create/Import	Edit	Delete	View Details	Download	Search	Q
Name	Subject	Comments	Issuer	Expires	Status	
Local CA Certificate 2						
Fortinet_CA_SSL	C = US, ST = California, L = Sunnyvale, O = Fortinet, ...	This is the default CA ce...	Fortinet	2034/11/06 07:45:17	Valid	
Fortinet_CA_Untrusted	C = US, ST = California, L = Sunnyvale, O = Fortinet, ...	This is the default CA ce...	Fortinet	2034/11/06 07:45:17	Valid	

The browser downloads the certificate to the **Downloads** folder on your HQ-PC-1 VM.

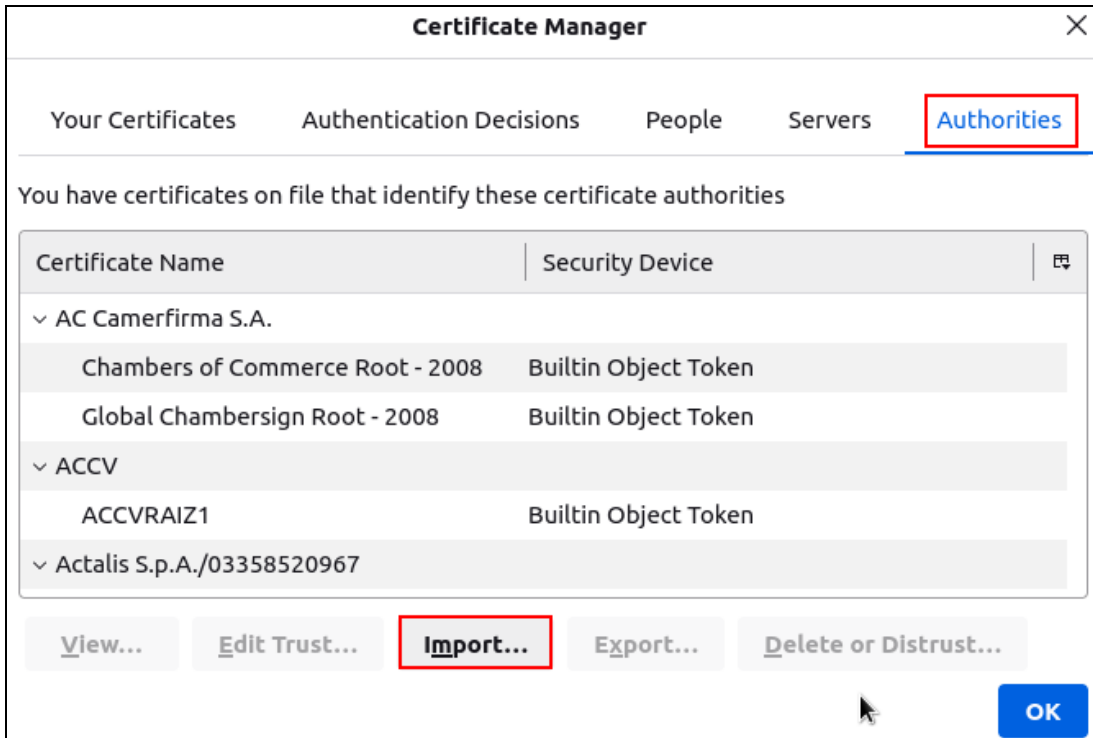
6. Continuing in Firefox, in the upper-right corner, click the **Open menu** icon, and then click **Settings**.



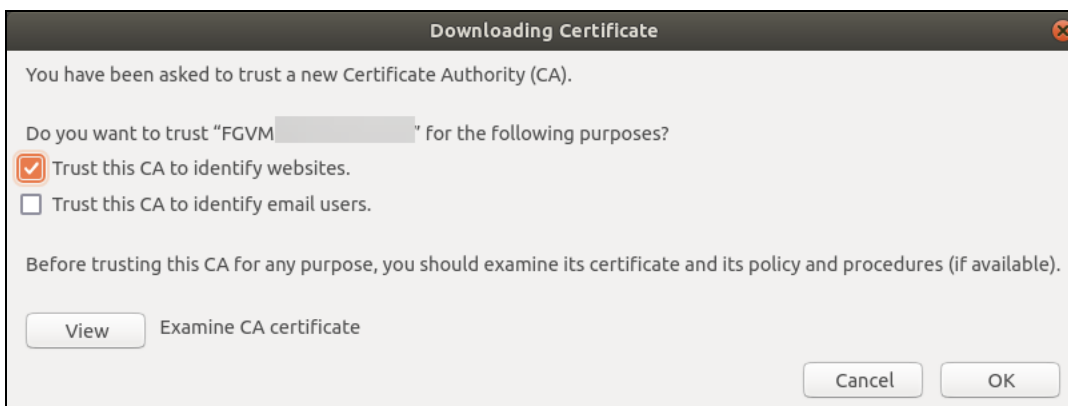
7. Click **Privacy & Security**.
8. In the **Certificates** section, click **View Certificates**.



9. In the **Certificate Manager** window, click the **Authorities** tab.
You can see a list of certificates from the public CA. They are loaded by default in your browser.



10. Click **Import**.
11. In the **Downloads** folder, click `Fortinet_CA_SSL.cer`, and then click **Select**.
12. In the **Downloading Certificate** window, select the **Trust this CA to identify websites** checkbox, and then click **OK**.



The **Fortinet_CA_SSL** certificate is added to the Firefox **Authorities** certificate store.

You can scroll down to see it in the list of authority certificates.

13. Click **OK** to exit the **Certificate Manager**.
14. Restart Firefox.

Test Full SSL Inspection

Now that you have imported the Fortinet_CA_SSL certificate into your browser, you will not receive certificate warnings when you access a secure website. The CA that signed this certificate is not public, but your browser trusts it, because you added it as a trusted authority in the previous procedure.

You will test to make sure that you do not receive a certificate warning.

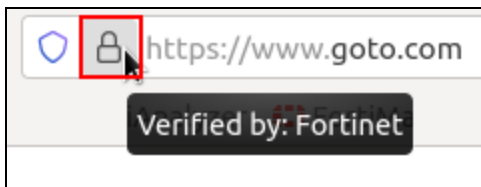
To test full SSL inspection

1. Continuing on the HQ-PC-1 VM, open a new browser session, and then go to a secure website, such as:

`https://www.goto.com`

This time, your browser opens the website without certificate warnings.

2. In the browser navigation bar, hover over the lock icon to see details.



You can see that the certificate is signed by Fortinet CA, and that the browser considers it valid.

3. Close the browser.

Exercise 2: Dealing With Anomalies

When you work with certificates, you might face some issues due to invalid or revoked certificates. You might also have to deal with restrictions that prevent the use of full SSL inspection.

In this exercise, you will learn how to import a certificate revocation list (CRL) on the FortiGate GUI. Next, you will explore how FortiGate responds when it receives invalid certificates for traffic that match a deep inspection SSL profile. Finally, you will configure an exception to exclude a website from full SSL inspection.

Manage Invalid Certificates

A certificate can be invalid because it expired or because the CA that issued it revoked it. A company might want to revoke a certificate because, for example, it was compromised, the key was lost, or it was assigned to a user who left the company. To inform others of revoked certificates, CA administrators periodically publish CRLs. You will import a CRL.

To import a CRL

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **System > Certificates**.
3. In the **Remote CA Certificate** section, right-click the **Fortinet_Wifi_CA** certificate, and then select **View Details**.
4. In the **Certificate Details** window, scroll down to the **Extensions** section, and then find **X509v3 CRL Distribution Points**.
5. Highlight one of the URIs, and then press **Ctrl+C** to copy it for the distribution point.

The screenshot shows the 'Certificate Details' window for the 'Fortinet_Wifi_CA' certificate. The 'Extensions' section is expanded, displaying a table of X509v3 extensions. The 'X509v3 CRL Distribution Points' extension is highlighted, showing a list of URIs. One URI is highlighted in red: <http://crl4.digicert.com/DigiCertGlobalRootCA.crl>.

Extension	Value
X509v3 Subject Key Identifier	B7:6B:A2:E4:A8:AA:84:8C:79:EA:B4:DA:0F:98:B2:C5:95:76:B9:F4
X509v3 Authority Key Identifier	03:DE:50:25:56:D1:4C:BB:66:F0:A3:F2:1B:1B:C3:97:B2:3D:D1:55
X509v3 Key Usage	Digital Signature, Certificate Sign, CRL Sign
X509v3 Extended Key Usage	TLS Web Server Authentication, TLS Web Client Authentication
X509v3 Basic Constraints	CA:TRUE, pathlen:0
Authority Information Access	OCSP - URI: http://ocsp.digicert.com/CAIssuers - URI: http://crls.digicert.com/DigiCertGlobalRootCA.crl
X509v3 CRL Distribution Points	Full Name: URI: http://crl4.digicert.com/DigiCertGlobalRootCA.crl Full Name: URI: http://crl4.digicert.com/DigiCertGlobalRootCA.crl
X509v3 Certificate Policies	Policy: 2.23.140.1.1 Policy: 2.23.140.1.2.1 Policy: 2.23.140.1.2.2 Policy: 2.23.140.1.2.3

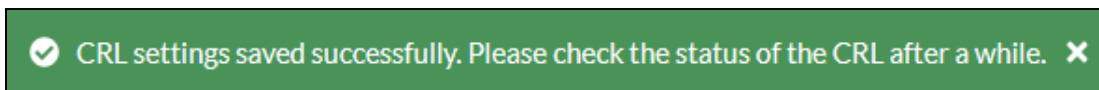
6. Click **Close** to exit the **Certificate Details** window.
7. Click **Create/Import**, and then select **CRL**.



8. Enable **HTTP**, and then in the **URL of the HTTP server** field, paste the URI of the CRL HTTP server that you just copied.

The screenshot shows the 'Import CRL' configuration page. The 'Import Method' section has 'File Based' and 'Online Updating' buttons. The 'HTTP' option is selected with a green circle. The 'URL of the HTTP server' field contains the text 'digicert.com/DigiCertGlobalRootCA.crl'. The 'LDAP' and 'SCEP' options are unselected.

9. Click **OK**.
The FortiGate GUI briefly displays an acknowledgment message similar to the following example:



10. Wait a few seconds, and then click **System > Certificates** again to refresh the page.
The CRL section now includes the CRL you just added.

Name	Subject
CRL 1	
CRL_1	
Local CA Certificate 2	
Fortinet_CA_SSL	C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = Certificate Aut...
Fortinet_CA_Untrusted	C = US, ST = California, L = Sunnyvale, O = Fortinet, OU = Certificate Aut...



Note that you can load CRLs on FortiGate only for a CA that FortiGate trusts. If you want to load the CRL that corresponds to your company CA, you must first load your company CA certificate on FortiGate.



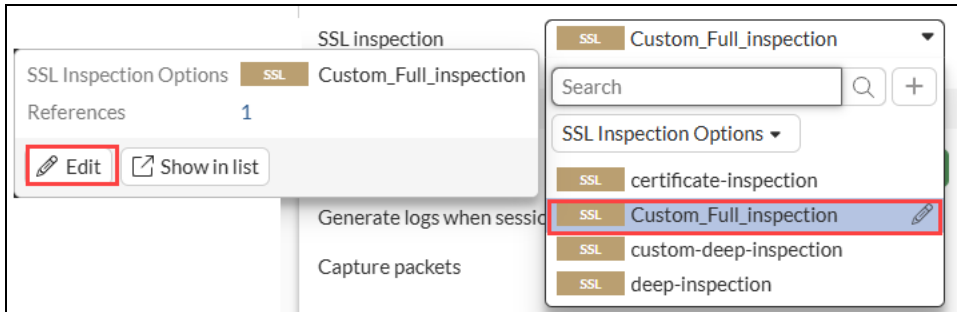
The Online Certificate Status Protocol (OCSP) is used for obtaining the revocation status of an X.509 digital certificate. It can be used as an alternative to CRLs. OCSP is disabled by default on FortiGate.

In this lab, we activated OCSP using the following CLI commands to receive certificate validation from well-known CAs that support OCSP:

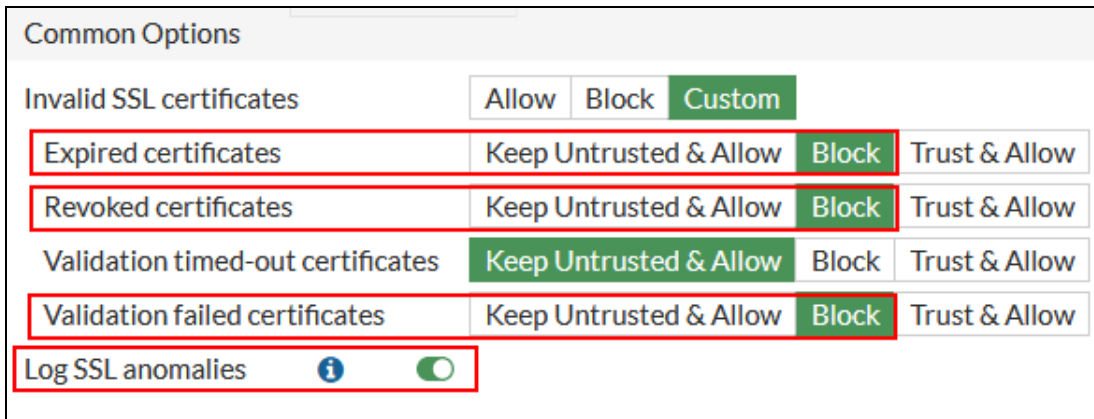
```
config vpn certificate setting
    set ocsf-option certificate
    set ocsf-status enable
    set strict-ocsf-check enable
end
```

To block an invalid certificate with full SSL inspection

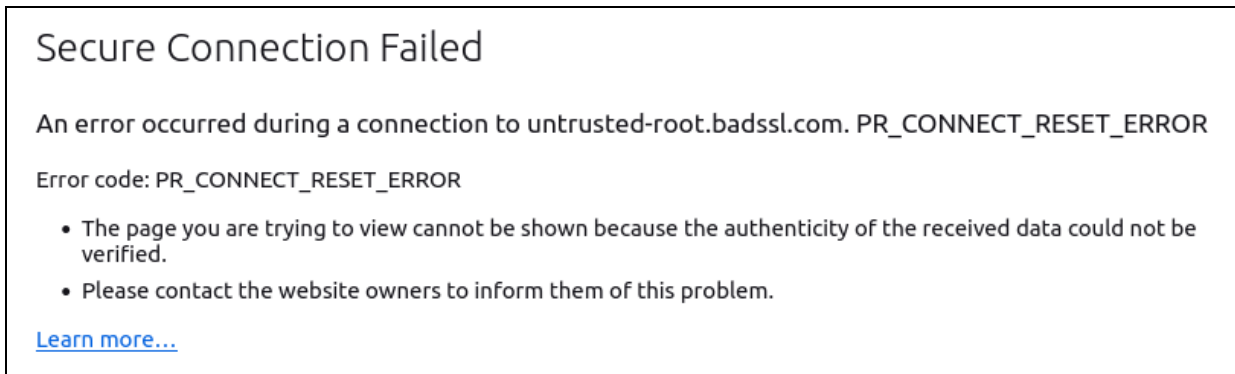
- Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
- Select the **Internet** policy, and then click **Edit**.
- Set the **Inspection mode** field to **Flow-based**.
- Scroll down to the **Security Profiles** section, and then click the pen icon to edit the **SSL Inspection** profile **Custom_Full_inspection**.



- Confirm that the settings are the same as what is shown in the following image:



- Do not make any changes, and then click **Cancel** to exit the SSL inspection profile menu.
- Click **OK** to exit the policy configuration menu.
- Connect to the HQ-PC-1 VM, and then log in with the following credentials:
 - Username: Administrator
 - Password: Fortinet1!
- In another browser tab, visit <https://expired.badssl.com/>.
FortiGate blocks access to the website and the browser displays a warning message similar to the following image:



To review SSL log messages

- Continuing on the HQ-NGFW-1 GUI, click **Log & Report > Security Events**.
- Expand the **SSL** widget to display the log list.

You can see that FortiGate blocked access to the website.

Date/Time	Action	Service	Source	Source Interface	Destination	Destination Interface
2021/12/04 09:05:23	Blocked	SSL	10.0.11.50	port1	104.154.89.105 (expired.badssl.com)	port2
2024/12/04 09:05:23	Blocked	SSL	10.0.11.50	port4	104.154.89.105 (expired.badssl.com)	port2
2021/12/04 09:05:23	Blocked	SSL	10.0.11.50	port1	104.154.89.105 (expired.badssl.com)	port2
2024/12/04 09:05:22	Blocked	SSL	10.0.11.50	port4	104.154.89.105 (expired.badssl.com)	port2

- Double-click a log message to review the details.

Log Details	
Source	
Source	10.0.11.50
Source Port	44732
Source Country/Region	Reserved
Source Interface	port4
Source UUID	703e6ff6-791a-51e7-daa0-9859ce6c1d02
Destination	
Destination	104.154.89.105
Destination Port	443
Destination Country/Region	United States
Destination Interface	port2
Destination UUID	7bc87d34-7916-51e7-3d5b-71812a61b98e
Hostname	expired.badssl.com
Application Control	
Protocol	6
Service	SSL
Data	
Message	SSL connection is blocked, certificate-status: revoked expired
Action	
Action	Blocked
Policy ID	Internet (1)
Policy UUID	b11ac58c-791b-51e7-4600-12f829a689d9
Policy Type	Firewall

You can see that the log message is similar for expired and revoked certificates.

Allow Exceptions to Full SSL Inspection

When replacing a certificate prevents users from accessing some websites, you can define exceptions and exclude some websites from full SSL inspection. You can also exclude some websites or website categories from full SSL inspection for legal reasons. For example, in some countries, it is forbidden to perform deep inspection on traffic between users and financial institution servers.

You will add an exception to the SSL/SSH deep inspection profile that you have already configured.

To configure a site exception to full SSL inspection

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Security Profiles > SSL/SSH Inspection**.
3. Edit the **Custom_Full_Inspection** profile.
4. In the **Exempt from SSL Inspection** section, In the **Addresses** field, click +

5. Click **+ Create**, and then click **+ Address** to create new **Addresses**.
6. Configure the following settings to create a new address object:

New Address	Value
Name	badssl
Type	Subnet
IP/Netmask	104.154.89.105/32

7. Click **OK**.
8. In the **SSL/SSH Inspection Profile** window, in the **Addresses** field, select the new **badssl** address object, and then click **Close**.
9. Click **OK** to save the configuration change to the SSL/SSH inspection profile.
10. Click **Policy & Objects > Firewall Policy**.
11. Edit the **Internet** policy, and then in the **SSL Inspection** field, make sure that **Custom_Full_Inspection** is selected.
12. Click **OK**.

To check the full SSL inspection exception

1. On the HQ-PC-1 VM, navigate to one of the websites you tested previously, such as <https://expired.badssl.com/>.
2. Click **Advanced**, and then click **Accept the Risk and Continue** to accept the browser warning.
Now, you can visit the website.



Usually, you will configure full SSL inspection exceptions only for websites that do not support MITM and that your company trusts. Those websites should have a valid certificate and therefore do not trigger a browser warning.

Lab 7: Antivirus

In this lab, you will examine how to configure, use, and monitor antivirus scanning on HQ-NGFW-1 in both flow-based and proxy-based inspection modes.

Objectives

- Configure antivirus scanning in both flow-based and proxy-based inspection modes
- Understand FortiGate antivirus scanning behavior
- Scan multiple protocols
- Read and understand antivirus logs

Time to Complete

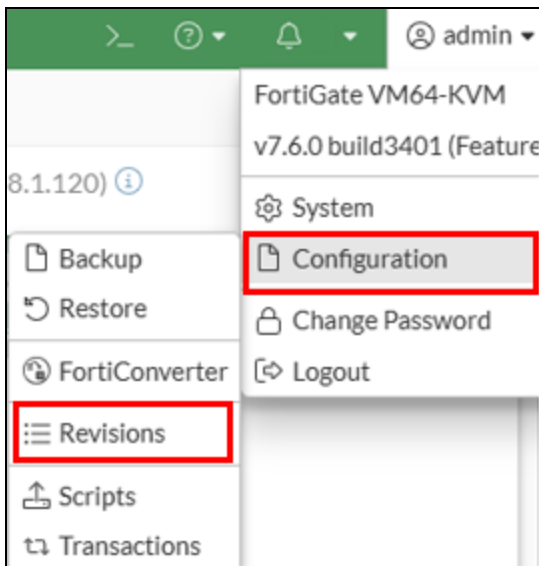
Estimated: 30 minutes

Prerequisites

Before you begin this lab, you must restore a configuration file to HQ-NGFW-1.

To restore the FortiGate configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **HQ-AV**, and then click **Revert**.

Delete

Details

Diff

Revert

Save Changes

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Configuring Flow-Based Antivirus Scanning

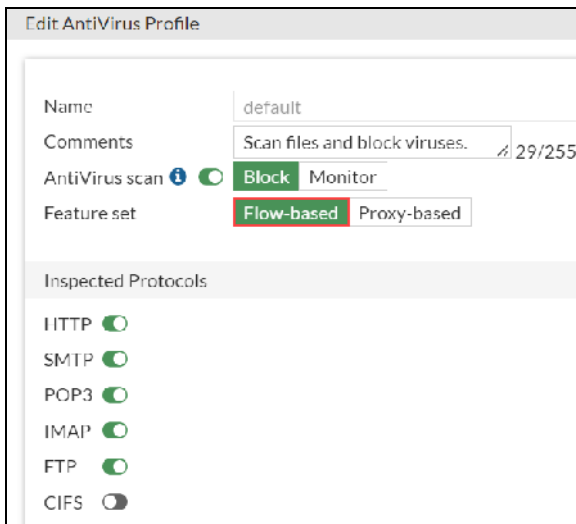
In this exercise, you will configure a firewall policy with an antivirus profile in flow-based inspection mode. Next, you will perform a test to download a file located on an FTP server. Finally, you will view the logs and summary information related to the antivirus scanning.

Configure the Antivirus Profile Inspection Mode

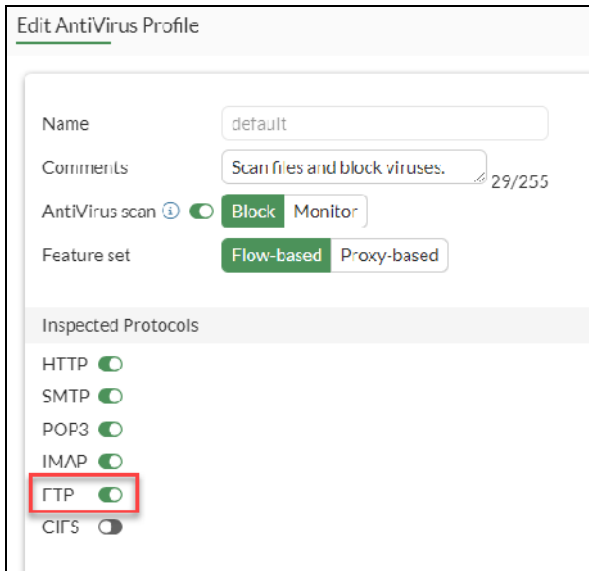
You will verify that the antivirus profile is configured with an inspected protocol of FTP and that flow-based is selected.

To verify the antivirus profile

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Security Profiles > AntiVirus**.
3. Right-click the **default** antivirus profile, and then click **Edit**.
4. In the **Feature set** field, verify that **Flow-based** is selected.



5. In the **Inspected Protocols** section, verify that **FTP** is enabled.



6. Click **Cancel**.

Enable the Antivirus Profile in a Firewall Policy

By default, flow-based inspection mode is enabled in the FortiGate firewall policy. You will configure the antivirus profile in the firewall policy.

To configure the firewall policy with the antivirus profile

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Double-click the **Internet** policy to edit it.
3. In the **Inspection mode** field, verify that **Flow-based** is selected.



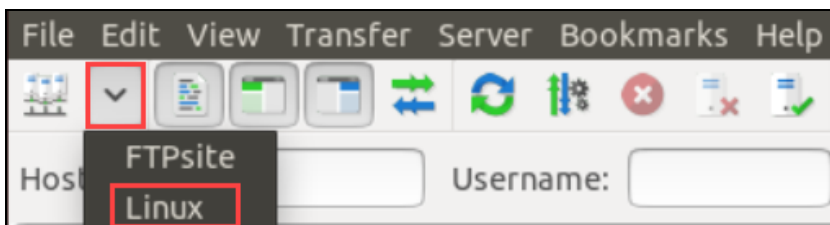
4. In the **Security Profiles** section, enable **AntiVirus**, and then select **default**.
5. Keep the default values for the remaining settings, and then click **OK** to save the changes.

Test the Flow-Based Antivirus Profile

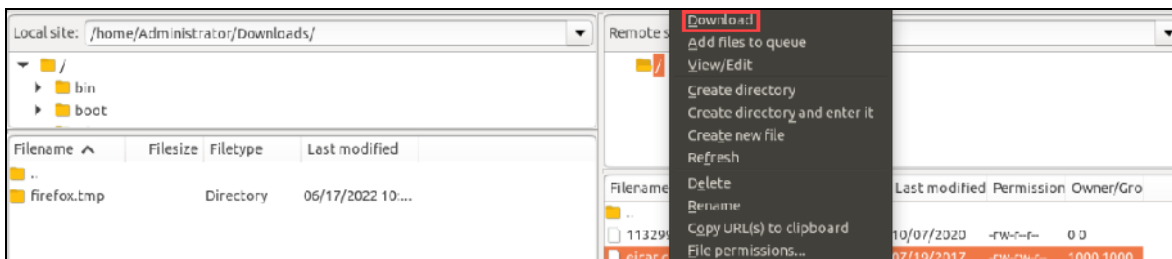
You will test the flow-based antivirus profile using FTP.

To test the antivirus configuration

1. On the HQ-PC-1 VM, on the desktop, open the FileZilla FTP client software.
2. In the upper-left corner, click the **Site Manager** icon, and then select **Linux**.



3. In the **Remote site** section, right-click the `eicar.com` file, and then select **Download**.



The client should display an error message that the server terminated the connection. FortiGate sends the replacement message as a server response.

```
Command: RETR eicar.com|
Response: 150 Opening BINARY mode data connection for eicar.com (68 bytes).
Error: Could not read from transfer socket: ECONNRESET - Connection reset by peer
Response: 226 Transfer complete.
Error: File transfer failed
```



In flow-based inspection mode, FortiGate does not buffer traffic flowing through the policy. If FortiGate detects a violation in the traffic, it sends a reset packet to the receiver, which terminates the connection, and prevents the payload from being sent successfully.

4. Close the FileZilla FTP client.

View the Antivirus Logs

The purpose of logs is to help you monitor your network traffic, locate problems, establish baselines, and make adjustments to network security, if necessary. You will view the antivirus logs.

To view the forward logs

1. Continuing on the HQ-NGFW-1 GUI, click **Log & Report > Forward Traffic**.
2. Locate the antivirus log message from when you tried to access the file using FTP, and then double-click the log entry to view the details.

The **Details** tab shows forward traffic log information, along with the action taken.

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID	Log Details
2024/11/26 09:31:26	10.0.11.50		100.65.0.254	FTP	✓ Accept (1.82 kB / 1.73 kB)	Internet (1)	Details Security
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/45809	✗ Deny (Deny: UTM Blocked)	Internet (1)	Action
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/6447	✗ Deny (Deny: UTM Blocked)	Internet (1)	Security Action
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/50796	✗ Deny (Deny: UTM Blocked)	Internet (1)	Threat

3. Click **Security**.

The **Security** tab shows virus information.

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID	Log Details
2024/11/26 09:31:26	10.0.11.50		100.65.0.254	FTP	✓ Accept (1.82 kB / 1.73 kB)	Internet (1)	Details Security
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/45809	✗ Deny (Deny: UTM Blocked)	Internet (1)	Antivirus
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/6447	✗ Deny (Deny: UTM Blocked)	Internet (1)	Log Details
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/50796	✗ Deny (Deny: UTM Blocked)	Internet (1)	Log Details

To view the security logs

1. Continuing on the HQ-NGFW-1 GUI, click **Log & Report > Security Events > AntiVirus**.

Date/Time	Source	Service	File Name	Virus/Threat	User	Details	Action	Infection Type
2024/11/26 09:31:03		FTP	eficar.com	HECAB-1151-100		Host: 100.65.0.254	Blocked	Malicious
2024/11/26 09:31:03		FTP	eficar.com	HECAB-1151-100		Host: 100.65.0.254	Blocked	Malicious
2024/11/26 09:31:03		FTP	eficar.com	HECAB-1151-100		Host: 100.65.0.254	Blocked	Malicious



To view the logs, you may need to clear the filters in the search bar and increase the time frame to 1 hour.

2. Locate the antivirus log message from when you tried to access the file using FTP, and then double-click the log entry to view the security details.

View the Antivirus Logs

Summary Logs

🔍 Search

Antivirus Disk 1 hour Details

Date/Time	@	Service	Source	File Name	Verus/Threat	User	Details	Action	Infection Type
2024/11/26 09:31:53		FTP	100.11.50	efcar.com	EICAR TEST FILE	Host: 100.65.0.254		Blocked	Malicious
2024/11/26 09:31:53		FTP	100.11.50	efcar.com	EICAR TEST FILE	Host: 100.65.0.254		Blocked	Malicious
2024/11/26 09:31:53		FTP	100.11.50	efcar.com	EICAR TEST FILE	Host: 100.65.0.254		Blocked	Malicious

Log Details

[-] Action

Action: Blocked

Threat: 2

Policy ID: Internet(1)

Policy UUID: b11ac58c-791b-51e7-4000-12f829a689d9

Policy type: Firewall

[-] Security

Level: Warning

Threat Level: Critical

Threat Score: 50

Exercise 2: Using Antivirus Scanning in Proxy-Based Inspection Mode

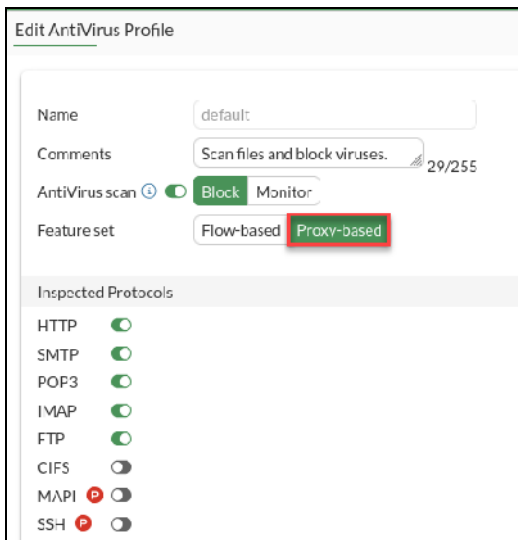
In this exercise, you will examine how to use antivirus in proxy-based inspection mode to understand how FortiGate performs antivirus scanning. You will observe the behavior of antivirus scanning, with and without deep inspection, to understand the importance of performing full-content inspection.

Change the Inspection Mode in an Antivirus Profile

You will change the inspection mode in the default antivirus profile, which is applied to the firewall policy, to inspect traffic.

To change the inspection mode in an antivirus profile

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Security Profiles > AntiVirus**.
3. Right-click the **default** antivirus profile, and then click **Edit**.
4. In the **Feature set** field, select **Proxy-based**.



5. Click **OK**.

Change the Inspection Mode in a Firewall Policy

Inspection mode is configured on a per-policy basis on FortiGate. You will change the inspection mode from flow-based to proxy-based.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI, complete the following:

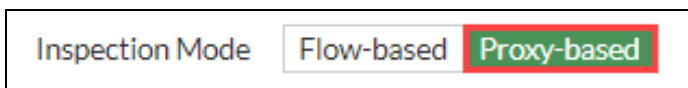
- Edit the **Internet** firewall policy, and change the **Inspection Mode** to **Proxy-based**.
- Enable the **default** antivirus profile.
- Use the **certificate-inspection** profile for SSL inspection.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Test the Antivirus Configuration on page 115](#).

To change the inspection mode in a firewall policy

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Double-click the **Internet** policy to edit it.
3. In the **Inspection Mode** field, select **Proxy-based**.



4. In the **Protocol Options** field, verify that the **default** profile is selected.
5. In the **Security Profiles** section, in the **AntiVirus** field, verify that the **default** profile is selected.
6. In the **SSL Inspection** field, keep the default **certificate-inspection** profile.



The **Protocol Options** profile provides the required settings to hold traffic in proxy while the inspection process is carried out. The default profile is preconfigured to follow the standardized parameters for the common protocols used in networking.

SSL Inspection selects the **certificate-inspection** profile by default. You can select any preconfigured SSL inspection profile in the associated field.

7. Keep the default values for the remaining settings, and then click **OK** to save the changes.

Test the Antivirus Configuration

You will download the EICAR test file to your HQ-PC-1 VM. The EICAR test file is an industry-standard virus used to test antivirus detection without causing damage. The file contains the following characters:

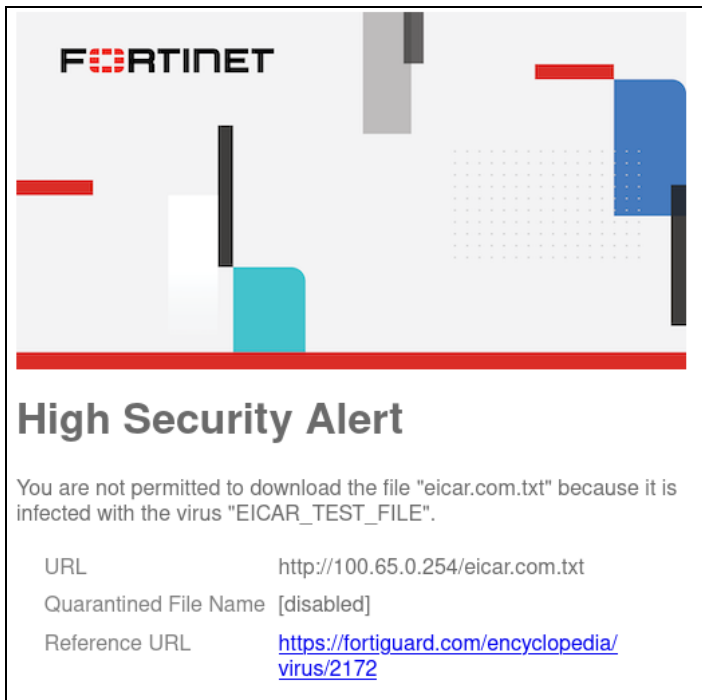
```
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
```

To test the antivirus configuration

1. On the HQ-PC-1 VM, open a browser, and then access the following website:
`http://100.65.0.254/test_av.html`
2. In the **Download area** section, download any EICAR sample file.

Download area using the standard protocol HTTP or secure, SSL enabled protocol HTTPS			
eicar.com	eicar.com.txt	eicar_com.zip	eicarcom2.zip

FortiGate should block the download attempt, and insert a replacement message similar to the following example:



Stop and think!

Why can FortiGate display a replacement message?

In proxy-based inspection mode, the file is buffered. If a virus is detected, FortiGate can then replace the file by a message, which provides security information.

Test an Alternate Download Method

You will test the proxy-based antivirus configuration using the **Save Link As** method to download the EICAR text file.

To test the antivirus configuration

1. On the HQ-PC-1 VM, open a new browser tab, and then go to the following website:
`http://100.65.0.254/test_av.html`

2. In the **Download area** section, right-click **eicar.com.txt**, and then select **Save Link As**.



3. Change the download location to **Desktop**, and then click **Save**.
You should see the file you downloaded on the desktop. Why was the download allowed?
4. On your desktop, right-click the `eicar.com.txt` downloaded file, click **Open With Other Application**, click **Notepad++**, click **Select** to open the file you downloaded, and then scroll to read the bottom of the text file.
Is the content of the file what it is supposed to be?

Stop and think!

Remember, you are using proxy-based inspection mode. When a firewall policy inspection mode is set to proxy, traffic flowing through the policy is buffered by FortiGate for inspection. This means that FortiGate holds the packets for a file, email, or web page until the entire payload is inspected for violations (virus, spam, or malicious web links). After FortiOS has finished the inspection, FortiGate either releases the payload to the destination (if traffic is clean) or drops and replaces it with a message (if the traffic contains violations). FortiGate injects the block message into the partially downloaded file. The client can use Notepad++ to open and view the file.

5. Close **Notepad++**.
6. Delete the downloaded `eicar.com.txt` file from the desktop.

View the Antivirus Logs

You will check and confirm the logs for the tests you just performed.

To view the antivirus logs

1. Continuing on the HQ-NGFW-1 GUI, click **Log & Report > Forward Traffic**.
You may need to remove any log filters in the search bar and increase the time frame.
2. Locate the antivirus log message, and then double-click it.
The **Details** tab shows forward traffic log information, along with the action taken.

Date/Time	Source	Device	Destination	Application Name	Result	Log Details
2024/11/26 10:26:51	10.0.11.50		100.65.0.254	HTTP	Deny (Deny: UTM Blocked)	Details
2024/11/26 10:27:11	10.0.11.50		100.65.0.254	HTTP	Deny (Deny: UTM Blocked)	Details
2024/11/26 10:23:42	100.65.0.101		34.117.188.166 (spocs.getpocket.com)	HTTPS	Accept (1.06 kB / 3.38 kB)	Details
2024/11/26 10:23:42	100.65.0.101		34.110.138.217 (merino.services.mozilla.com)	HTTPS	Accept (1.06 kB / 3.42 kB)	Details
2024/11/26 09:31:26	10.0.11.50		100.65.0.254	FTP	Accept (1.82 kB / 1.73 kB)	Details
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/45809	Deny (Deny: UTM Blocked)	Details
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/6467	Deny (Deny: UTM Blocked)	Details
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/50796	Deny (Deny: UTM Blocked)	Details

3. Select the **Security** tab to view security logs.

Security logs provide information that is more specific to security events, such as file name, virus or botnet, and reference.

Date/Time	Source	Device	Destination	Application Name	Result	Log Details
2024/11/26 10:26:51	10.0.11.50		100.65.0.254	HTTP	Deny (Deny: UTM Blocked)	Details
2024/11/26 10:24:41	10.0.11.50		100.65.0.254	HTTP	Deny (Deny: UTM Blocked)	Details
2024/11/26 10:23:42	100.65.0.101		34.117.188.166 (spocs.getpocket.com)	HTTPS	Accept (1.06 kB / 3.38 kB)	Details
2024/11/26 10:23:42	100.65.0.101		34.110.138.217 (merino.services.mozilla.com)	HTTPS	Accept (1.06 kB / 3.42 kB)	Details
2024/11/26 09:31:26	10.0.11.50		100.65.0.254	FTP	Accept (1.82 kB / 1.73 kB)	Details
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/45809	Deny (Deny: UTM Blocked)	Details
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/6467	Deny (Deny: UTM Blocked)	Details
2024/11/26 09:31:08	10.0.11.50		100.65.0.254	tcp/50796	Deny (Deny: UTM Blocked)	Details

4. Click **Log & Report > Security Events > AntiVirus** to view antivirus security logs.

The logs should be similar to the following example:

Date/Time	Service	Source	File Name	Virus/Botnet	User	Details	Action	Infection Type
2024/11/26 10:26:50	HTTP	10.0.11.50	eicar.com.txt	EICAR_Test_File		URL: http://100.65.0.254/eicar.com.txt	Blocked	Malicious
2024/11/26 10:24:40	HTTP	10.0.11.50	eicar.com.txt	EICAR_Test_File		URL: http://100.65.0.254/eicar.com.txt	Blocked	Malicious



To view the logs, you may need to clear the filters in the search bar and increase the time frame to 1 hour.

You can click **Details** to view more virus information related to an antivirus log entry.

Enable SSL Inspection in a Firewall Policy

So far, you have tested unencrypted traffic for antivirus scanning. In order for FortiGate to inspect the encrypted traffic, you must enable deep inspection in the firewall policy. After you enable this feature, FortiGate can inspect SSL traffic using a technique similar to a man-in-the-middle (MITM) attack.

Take the Expert Challenge!

- On the HQ-PC-1 VM, test the configuration by downloading the `eicar.com` file using HTTPS, without enabling the **deep-inspection** profile in the **Full Access** firewall policy.
- Configure HQ-NGFW-1 to scan secure protocols by enabling **SSL Inspection**, using the **deep-inspection** profile in the **Full Access** firewall policy.
- Test the configuration by downloading the `eicar.com` file using HTTPS.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

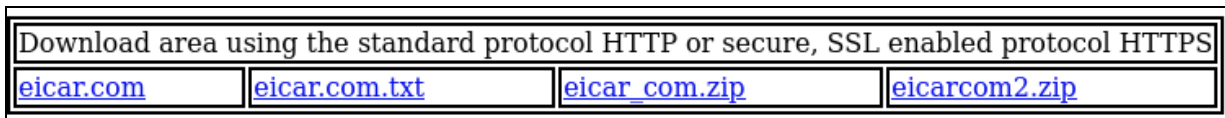
After you complete the challenge, see [Review the Antivirus History on page 120](#).

To test antivirus scanning without SSL inspection enabled in the firewall policy

1. On the HQ-PC-1 VM, open a browser, and then go to the following website:

`https://100.65.0.254/test_av.html`

2. Click **Advanced**.
3. Click **Accept the Risk and Continue**.
4. In the **Download area** section, download the `eicar.com` sample file.



FortiGate should not block the file, because you did not enable full SSL inspection.

5. On the HQ-PC-1 VM, close the browser.

To enable and test the SSL inspection profile in a firewall policy

1. Return to the HQ-NGFW-1 GUI, and then click **Policy & Objects > Firewall Policy**.
2. Double-click the **Internet** firewall policy to edit it.
3. In the **Security Profiles** section, in the **SSL Inspection** field, select **deep-inspection**.
4. Keep the remaining default settings, and then click **OK** to save the changes.
5. Click **OK** to confirm.
6. On the HQ-PC-1 VM, open a browser, and then go to the same website.
`https://100.65.0.254/test_av.html`
7. In the **Download area** section, try to download the same `eicar.com` file again.



If the FortiGate self-signed full-inspection certificate is not installed in the browser, end users will see a certificate warning message. In this environment, the FortiGate self-signed SSL inspection certificate is installed in the browser. If the block page does not appear after 2 minutes, close all browser tabs, and then restart the browser.

You may also need to clear your cache. In Firefox, click **Settings > Privacy & Security**. Scroll to **History**, click **Clear History**, and then ensure the time range to clear is set to **Everything**. Click **OK**.

FortiGate should block the download and replace it with a message.

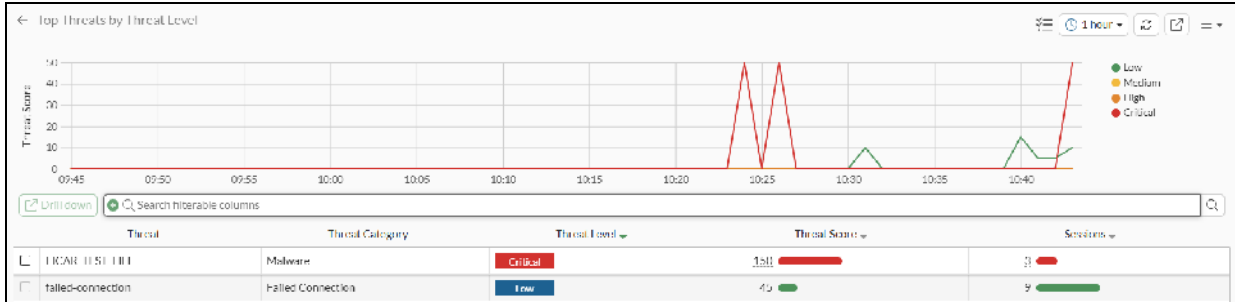
Review the Antivirus History

You will check the security history and the virus definition status.

To view the security history

1. Continuing on the HQ-NGFW-1 GUI, click **Dashboard > Security > Top Threats by Threat Level**.

The graph should be similar to the following example:



You can click **Drill down** to view more details about a specific threat.

To verify the antivirus definitions status

1. Connect over SSH to HQ-NGFW-1.
2. Log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
3. Enter the following commands:

```
diagnose debug application update -1
diagnose debug enable
execute update-av
```

After a few seconds, the output should include information similar to the following example:

```
level(6) alert(0)
upd_status_extract_alci_info[1427]-Finished reading account contracts
installUpd0bRest[1008]-Step 9:Delete backup /tmp/update.backup
installUpd0bRest[1048]-Step 10:Tell parent to respawn
upd_install_pkg[1432]-AVEN028 is up-to-date
upd_install_pkg[1432]-AVDB002 is up-to-date
upd_install_pkg[1432]-AVDB007 is up-to-date
upd_install_pkg[1432]-AVDB019 is up-to-date
upd_install_pkg[1458]-FCNI000(fcn) installed successfully
upd_install_pkg[1458]-FDNI000(fdslist) installed successfully
upd_install_pkg[1458]-FSCI000(contract) installed successfully
upd_install_pkg[1432]-FLDB002 is up-to-date
upd_install_pkg[1432]-MMDB001 is up-to-date
upd_install_pkg[1432]-DBDB001 is up-to-date
upd_install_pkg[1432]-SFAS000 is up-to-date
upd_install_pkg[1458]-ALCI000(alci) installed successfully
upd_status_save_status[135]-try to save on status file
upd_status_save_status[201]-Wrote status file
  upd act update[319]-Package installed successfully
upd_comm_disconnect_fds[500]-Disconnecting FMG 192.168.1.120:8890
[207]  ssl data ctx free: Done
[1115] ssl free: Done
[199]  __ssl_cert_ctx_free: Done
[1125] ssl ctx free: Done
[1106] ssl_disconnect: Shutdown
do_update[714]-UPDATE successful
```



The output confirms that the device has the latest antivirus packages installed.

4. Enter the following commands:

```
diagnose debug disable
diagnose debug application update 0
```

5. Log out of the SSH session and the HQ-NGFW-1 GUI.

Lab 8: Web Filtering

In this lab, you will configure one of the most used security profiles on FortiGate: web filter. This includes configuring FortiGuard category-based and static URL filters, applying the web filter profile in a firewall policy, testing the configuration, and performing basic troubleshooting.

Objectives

- Configure web filtering on FortiGate
- Apply the FortiGuard category-based option for web filtering
- Apply the static URL option for web filtering
- Troubleshoot the web filter
- Read and interpret web filter log entries

Time to Complete

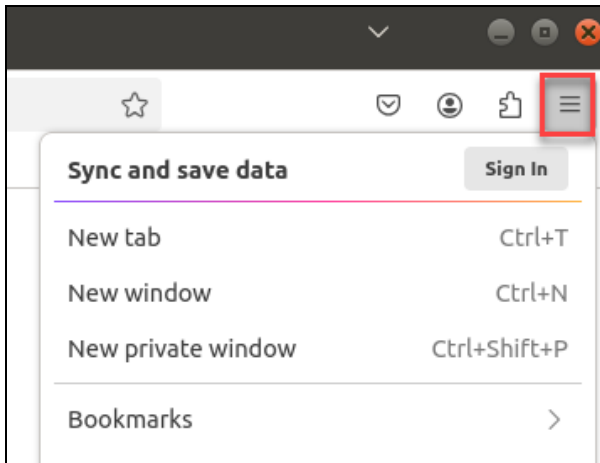
Estimated: 30 minutes

Prerequisites

Before you begin this lab, you must clear the browser history, and then restore a configuration file to HQ-NGFW-1.

To clear the browser history

1. On the HQ-PC-1 VM, open the browser, and then in the upper-right corner, click the menu icon.



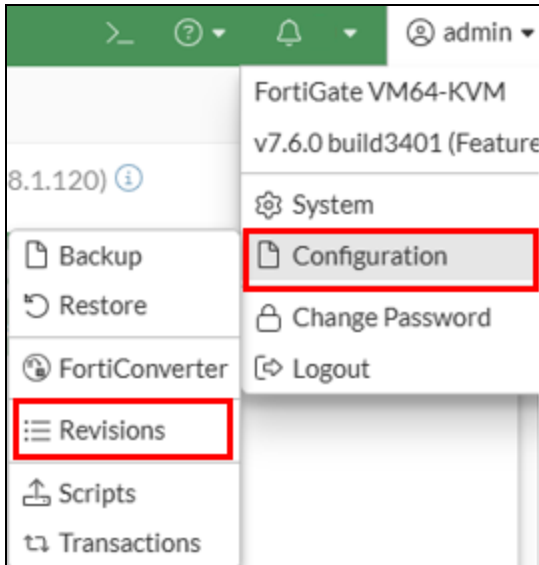
2. Click **Settings > Privacy & Security**.
3. Scroll to **History**, click **Clear History**, and then ensure the time range to clear is set to **Everything**.
4. Click **Clear**.

To restore the FortiGate configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:

- Username: admin
- Password: Fortinet1!

2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.

4. Select the configuration with the comment **HQ-web-filtering**, and then click **Revert**.

✕ Delete

i Details

Diff

↶ Revert

Save Changes ⚠

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Configuring FortiGuard Web Filtering

To configure FortiGate for web filtering based on FortiGuard categories, you must make sure that FortiGate has a valid FortiGuard security subscription license. The license provides the web filtering capabilities necessary to protect against inappropriate websites.

Next, you must configure a category-based web filter security profile on FortiGate, and apply the security profile in a firewall policy to inspect the HTTP traffic.

Finally, you can test different actions that FortiGate has taken, according to the website rating.

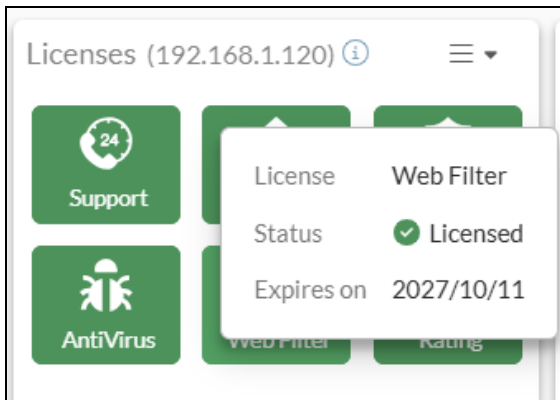
Review the FortiGate Settings

You will review the inspection mode and license status according to the uploaded settings. You will also list the FortiGuard Distribution Servers (FDS) that FortiGate uses to send the web filtering requests.

To review the restored settings on FortiGate

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. On the **Dashboard**, locate the **Licenses** widget, and then hover over **Web Filter** to confirm that the service is licensed and active.

You should see information similar to the following example:





Because of the reboot following the restoration of the configuration file, the web filter license status may be **Unavailable**. In this case, navigate to **System > FortiGuard**. In the **Filtering** section, click **Test Connectivity** to force an update, and then click **OK** to confirm. You can confirm, at the same time, that **Web Filter cache** is enabled.

3. Click **Policy & Objects > Firewall Policy**.
4. Double-click the **Internet** policy to edit it.
5. In the **Inspection Mode** field, select **Proxy-based**.

Notice that the default inspection mode is set to **Flow-based**.

6. Click **OK**.

Determine Web Filter Categories

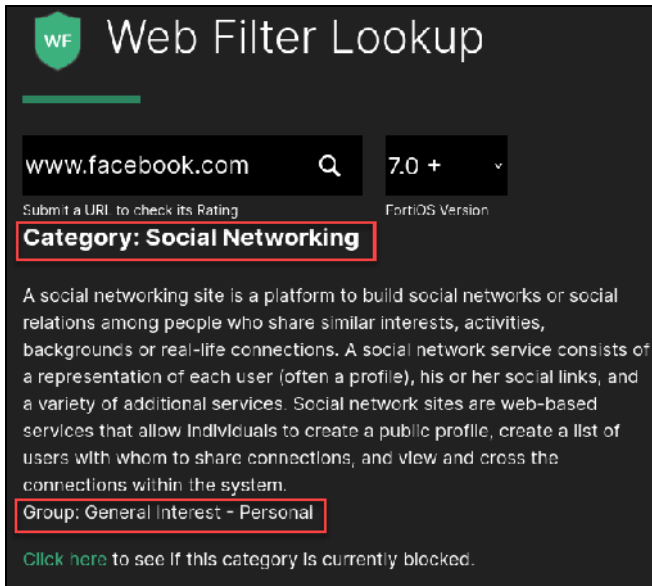
To configure web filter categories, you must first identify how FortiGuard Web Filtering categorizes specific websites.

To determine web filter categories

1. On the HQ-PC-1 VM, open a new browser tab, and then go to <https://www.fortiguards.com/webfilter>.

2. Use the **Web Filter Lookup** tool to search for the following URL:

www.facebook.com



This is one of the websites you will use later to test your web filter.

As you can see, Facebook is listed in the **Social Networking** category.

3. Use the **Web Filter Lookup** tool again to find the web filter category for the following websites:

- `www.ask.com`
- `www.bing.com`

You will test your web filter using these websites also.

The following table shows the category assigned to each URL, as well as the action you will configure FortiGate to take based on your web filter security profile:

Website	Category	Action
www.facebook.com	Social Networking	Block
www.bing.com	Search Engines and Portals	Allow
www.ask.com	Search Engines and Portals	Allow

Configure a FortiGuard Category-Based Web Filter

You will review the default web filtering profile, and then configure the FortiGuard category-based filter.

To configure the web filter security profile

1. Return to the HQ-NGFW-1 GUI, and then click **Security Profiles > Web Filter**.
2. Double-click the **default** web filter profile to edit it.

+ Create new		+ Q Search	
☐	Name	Comments	Ref.
☐	WEB default	Default web filtering.	0
☐	WEB monitor-all	Monitor and log all visited URLs, flow-based.	0
☐	WEB wifi-default	Default configuration for offloading WiFi traffic.	1

3. Verify that **FortiGuard Category Based Filter** is enabled.

☒

FortiGuard Category Based Filter



Allow



Monitor



Block



Warning



Authenticate

Name	Action
 Local Categories 2	
 Potentially Liable 12	
 Adult/Mature Content 15	
 Bandwidth Consuming 6	
 Security Risk 6	
 General Interest - Personal 35	
 General Interest - Business 18	
 Unrated 1	

95



You can click **+** to expand a category or **-** to collapse a category.

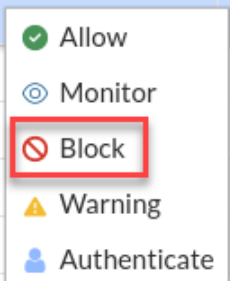
- 4. Review the default actions for each category.**

Category	Action
Local Categories	Disable
Potentially Liable	Block: Extremist Group Allow: all other subcategories Tip: Expand Potentially Liable to view the subcategories.
Adult/Mature Content	Block
Bandwidth Consuming	Allow

Category	Action
Security Risk	Block
General Interest - Personal	Allow
General Interest - Business	Allow
Unrated	Block

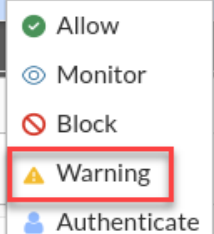
- Expand **General Interest - Personal** to view the subcategories.
- Right-click **Social Networking**, and then select **Block**.

Job Search	✓ Allow
Medicine	✓ Allow
News and Media	✓ Allow
Social Networking	✓ Allow
Political Organizations	✓ Allow
Reference	✓ Allow
Global Religion	✓ Allow
Shopping	✓ Allow



- Expand **Bandwidth Consuming** to view the subcategories.
- Right-click **Internet Telephony**, and then select **Warning**.

Bandwidth Consuming 6	
Freeware and Software Downloads	✓ Allow
File Sharing and Storage	✓ Allow
Streaming Media and Download	✓ Allow
Peer-to-peer File Sharing	✓ Allow
Internet Radio and TV	✓ Allow
Internet Telephony	✓ Allow
Security Risk 6	
Category Usage Quota 53% 95	
+ Create New Edit	



The **Edit Filter** window opens, which allows you to modify the warning interval.

9. Keep the default setting of 5 minutes, and then click **OK**.
10. Click **OK**.

Apply the Web Filter Profile to a Firewall Policy

Now that you have configured the web filter profile, you must apply this security profile to a firewall policy in order to start inspecting web traffic.

You will also enable the logs to store and analyze the security events that the web traffic generates.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI, apply the web filter profile to the existing **Internet** firewall policy. Make sure that logging is also enabled and set to **Security Events**.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

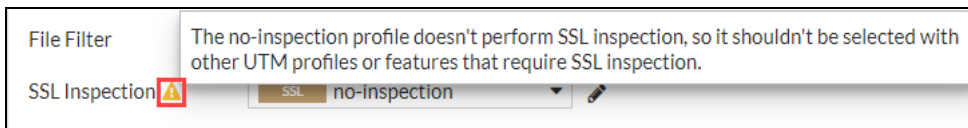
After you complete the challenge, see [Test the Web Filter on page 131](#).

To apply a security profile in a firewall policy

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Double-click the **Internet** policy to edit it.
3. In the **Security Profiles** section, enable **Web Filter**, and then select **default**.



4. Hover over the warning sign that appears beside the **SSL Inspection** field. The message should be similar to the following example:



5. In the **SSL Inspection** field, select **certification-inspection**.



Because web filtering requires URL information and does not inspect the full payload, you can select **certification-inspection** instead of **deep-inspection**.

6. Under **Log allowed traffic**, make sure that **Security events** is selected.
7. Keep all other default settings, and then click **OK**.

Test the Web Filter

You will test the web filter security profile you configured for each category.

To test the web filter

1. On the HQ-NGFW-1 CLI, log in with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Enter the following command to verify the web filter status:

```
get webfilter status
```

The `get webfilter status` and `diagnose debug rating` commands show the list of FDS that FortiGate uses to send web filtering requests. In normal operations, FortiGate sends the rating requests only to the server at the top of the list. Each server is probed for round-trip time (RTT) every 2 minutes.

Stop and think!

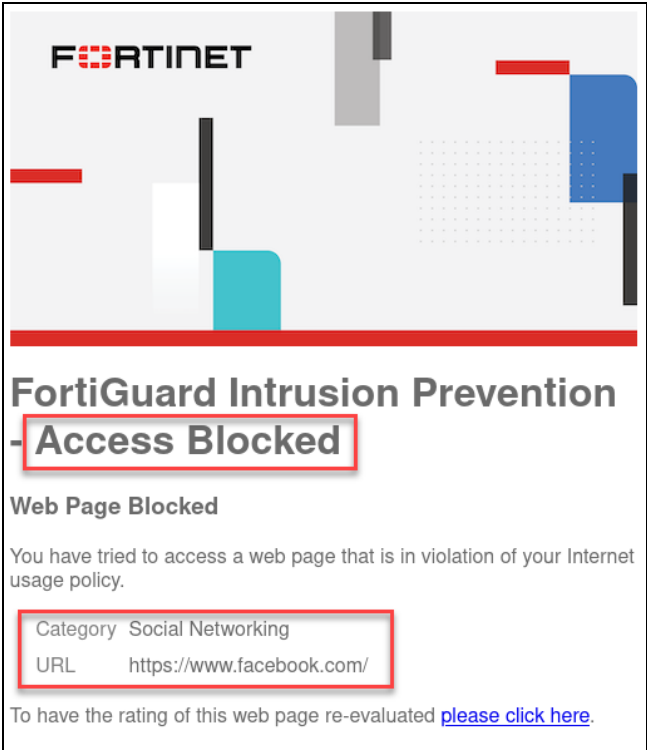
Why does only one IP address from your network appear in the server list?

Your lab environment uses a FortiManager management IP address `192.168.1.120`, which is configured as a local FDS. It contains a local copy of the FDS web rating database.

FortiGate sends the rating requests to FortiManager instead of to the public FDS. For this reason, the output of the command lists the FortiManager IP address only.

3. On the HQ-PC-1 VM, open a new browser tab, and then go to `www.facebook.com`.

A **Page Blocked** notification appears, according to the predefined action for this website category.



- 4. Open a new browser tab, and then go to `www.bing.com`.
This website appears because it belongs to the **Search Engines and Portals** category, which is set to **Allow**.
- 5. Close the HQ-PC-1 VM browser tabs.

Create a Web Rating Override

You will override the category for `www.bing.com`.

To create a web rating override

- 1. Return to the HQ-NGFW-1 GUI, and then click **Security Profiles > Web Rating Overrides**.
- 2. Click **Create New**, and then configure the following settings:

Field	Value
URL	www.bing.com
Category	Security Risk
Sub-Category	Malicious Websites

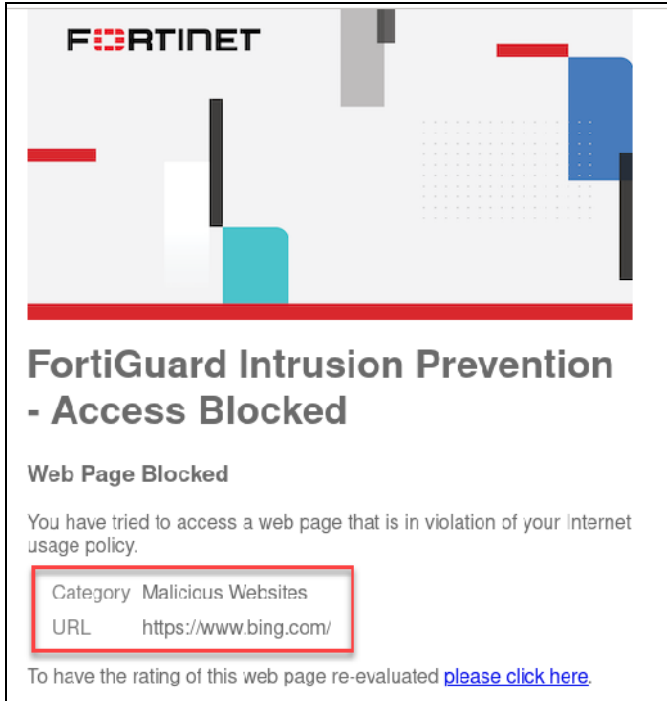
- 3. Click **OK**.

Test the Web Rating Override

You will test the web rating override you created in the previous procedure.

To test the web rating override

1. On the HQ-PC-1 VM, open a new browser tab, and then try to access the `www.bing.com` website again. The website is blocked, and it matches a local rating instead of a FortiGuard rating.



If the website is not blocked, you may close and reopen the browser to start a new session.

Stop and think!

Why is the website `www.bing.com` blocked?

The web rating override changes the category. In the default web profile applied in the firewall policy, the **Malicious Websites** category is set to **Block**. As a consequence, the website `www.bing.com` is now blocked.

Configure an Authenticate Action

You will set the action for the **Malicious Websites** FortiGuard category to **Authenticate**. You will then define a user in order to test the authenticate action.

To set up the authenticate action

1. Continuing on the HQ-NGFW-1 GUI, click **Security Profiles > Web Filter**.
2. Double-click the **default** web filter profile to edit it.
3. Under **FortiGuard Category Based Filter**, expand **Security Risk**, right-click **Malicious Websites**, and then select **Authenticate**.
The **Edit Filter** window opens, which allows you to modify the warning interval and select the user groups.
4. Configure the following settings:

Field	Value
Warning Interval	5 minutes
Selected User Groups	Override_Permissions

5. Click **Close**.
6. Click **OK**.
7. Click **OK**.



For the purpose of this lab, **Override_Permissions** is a predefined user group. To review the user groups, click **User & Authentication > User Groups**.

To create a user

1. Continuing on the HQ-NGFW-1 GUI, click **User & Authentication > User Definition**.
2. Click **Create New**.
3. In the **User Type** field, select **Local User**.
4. Click **Next**, and then configure the following settings:

Field	Value
Username	user1
Password	Fortinet1!

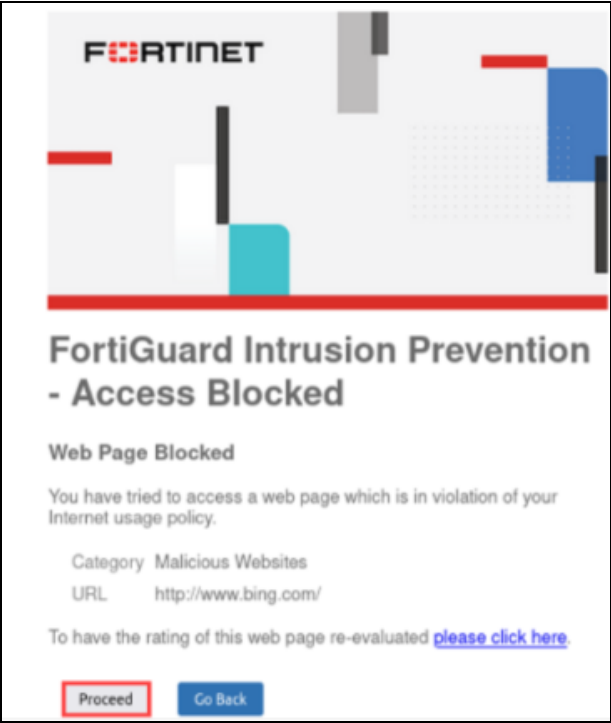
5. Click **Next**.
6. Click **Next**.
7. Enable **User Group**, and then select **Override_Permissions**.
8. Click **Close**.
9. Click **Submit**.

The **user1** user is created.

	Name	Type	Two-factor Authentication	Status	Ref.
☐	guest	Local	✖ Disabled	✔ Enabled	1
☐	user1	Local	✖ Disabled	✔ Enabled	1

To test the web rating override

- 1. On the HQ-PC-1 VM, open a new browser tab, and then try to access www.bing.com.
A warning appears. Notice that it is a different message from the one that appeared before.



- 2. Click **Proceed**.
- 3. Enter the following credentials:

Field	Value
Username	user1
Password	Fortinet1!

- 4. Click **Continue**.
The www.bing.com website now displays correctly.
- 5. Close the HQ-PC-1 VM browser tabs.

Exercise 2: Configuring Static URL Filtering

In this exercise, you will configure a static URL filter and apply the security profile to a firewall policy in flow-based inspection mode. You will then review the web filter logs.

Set Up the Static URL Filter in Flow-Based Inspection Mode

You will create a static URL filter entry and change the inspection mode to flow-based.

To create a static URL filter

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Security Profiles > Web Filter**.
3. Double-click the **default** web filter profile to edit it.
4. In the **Static URL Filter** section, enable **URL Filter**.
5. Click **Create New**, and then configure the following settings:

Field	Value
URL	www.bing.com
Type	Simple
Action	Block
Status	Enable

6. Click **OK**.
Your configuration should match the following example:

Static URL Filter

Block invalid URLs ☐

URL Filter ☒

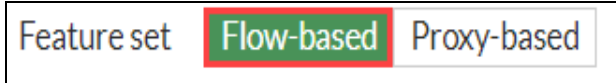
+ Create New Edit Delete Search

URL	Type	Action	Status
www.bing.com	Simple	Block	Enable

7. Click **OK**.

To change the inspection mode to flow-based

1. Continuing on the Local-FortiGate GUI, click **Security Profiles > Web Filter**.
2. Double-click the **default** web filter profile to edit it.
3. In the **Feature set** field, select **Flow-based**.



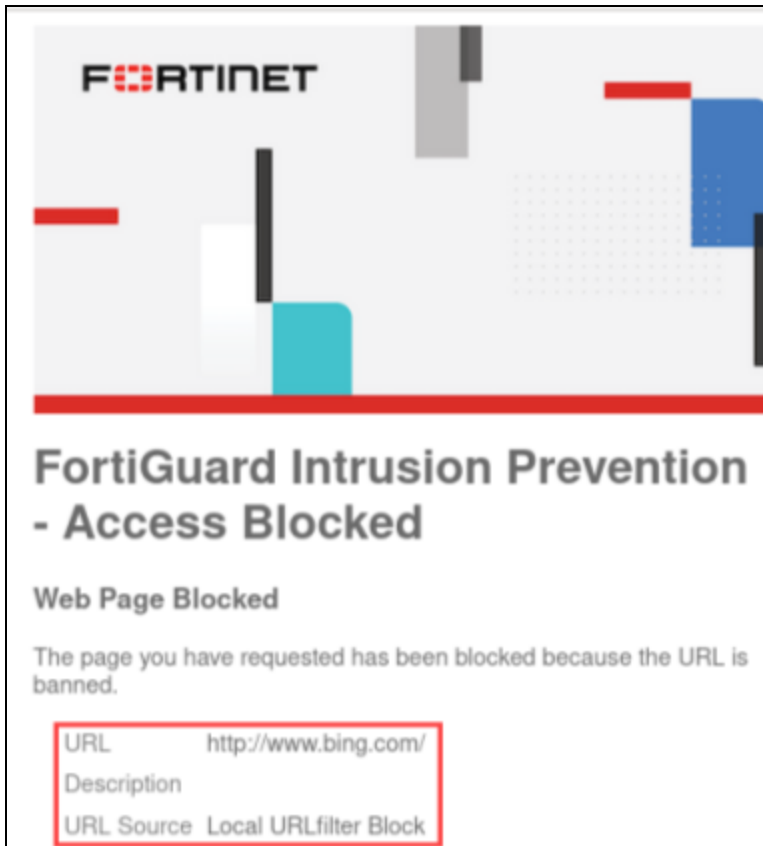
4. Click **OK**.
5. Click **Policy & Objects > Firewall Policy**.
6. Double-click the **Internet** policy to edit it.
7. In the **Inspection Mode** field, select **Flow-based**.



8. Click **OK**.

To test the static URL filter

1. On the HQ-PC-1 VM, open a new browser tab, and then try to access www.bing.com.
A warning appears. Notice that it is a different message from the one that appeared before.



Stop and think!

Why is the replacement message different?

FortiGate applies the static URL filter before the FortiGuard category filter. The `www.bing.com` URL matches the URL filter pattern and therefore is now blocked, and FortiGate displays the corresponding URL filter message.

To review the web filter logs

- 1. Return to your browser tab where you are logged in to the HQ-NGFW-1 GUI, and then click **Log & Report > Security Events**.
- 2. Under **Summary**, click **Web Filter**.

You should see information similar to the following example:

SummaryLogs

🔄📄🔍 Search

🔍

Web Filter

Disk

1 hour

Details

Date/Time	User	Source	Action	URL	Category
2024/11/15 09:01:41		10.0.11.50	Blocked	https://www.bing.com/	
2024/11/15 09:01:41		10.0.11.50	Blocked	https://www.bing.com/	
2024/11/15 09:01:41		10.0.11.50	Blocked	https://www.bing.com/	
2024/11/15 09:00:37		10.0.11.50	Blocked	https://www.bing.com/	
2024/11/15 09:00:37		10.0.11.50	Blocked	https://www.bing.com/	
2024/11/15 08:57:43		10.0.11.50	Passthrough	https://www.bing.com/	Malicious Web
2024/11/15 08:57:43		10.0.11.50	Passthrough	https://www.bing.com/	Malicious Web
2024/11/15 08:57:43		10.0.11.50	Passthrough	https://www.bing.com/	Malicious Web
2024/11/15 08:55:33		10.0.11.50	Passthrough	https://www.bing.com/	Malicious Web
2024/11/15 08:55:32		10.0.11.50	Passthrough	https://www.bing.com/	Malicious Web
2024/11/15 08:54:11		10.0.11.50	Blocked	https://www.bing.com/	Malicious Web
2024/11/15 08:54:11		10.0.11.50	Blocked	https://www.bing.com/	Malicious Web
2024/11/15 08:45:27		10.0.11.50	Blocked	https://www.bing.com/	Malicious Web

Log Details

Policy Type

Firewall

Security

Level

Threat Level

Threat Score

Warning

High

30

Web Filter

Profile

Request Type

Direction

Category ID

Category

Message

default

direct

outgoing

26

Malicious Websites

URL belongs to a category with warnings enabled

Other



To view the logs, you may need to clear the filters in the search bar and increase the time frame to 1 hour.

Stop and think!

Why is the first log entry for the `www.bing.com` website defined as blocked?

Initially, the `www.bing.com` website has the category **Search Engines and Portals**, which was set to **Allow** and does not generate a security log.

To allow a website and generate a security log at the same time, you must set the category to **Monitor**.

Then, according to the logs, `http://www.bing.com` is blocked, but after you clicked **Proceed** and authenticated, the logs show a different action: **passthrough**.

Remember that you overrode the **Search Engines and Portals** category to **Malicious Websites**, which was set to **Block**, and then to **Authenticate**.

3. Double-click a log entry with an empty category.

You should see information similar to the following example:

Summary Logs									
Date/Time	User	Source	Action	URI	Category	Initiator	Sent / Received	Log Details	
2024/11/26 11:36:41	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/			660 B / 0 B	Log Details Policy UID: 012a3b4c-7f1b-51e7-4e00-12f029a609d9 Policy Type: Firewall Security Level: High Threat Level: High Threat Score: 30 Web Filter Threat: default Request Type: direct Direction: outgoing URL Filter Index: 1 URL Filter List: Auto-webfilter-urlfilter_vw/4ex23w Message: URL was blocked because it is in the URL filter list	
2024/11/26 11:36:41	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/			1.09 kB / 0 B		
2024/11/26 11:36:40	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/			1.09 kB / 0 B		
2024/11/26 11:36:40	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/			660 B / 0 B		
2024/11/26 11:36:40	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/			1.89 kB / 0 B		
2024/11/26 11:25:28	10.0.11.50	10.0.11.50	Passthrough	https://www.bing.com/	Malicious Websites		1.25 kB / 0 B		
2024/11/26 11:25:28	10.0.11.50	10.0.11.50	Passthrough	https://www.bing.com/	Malicious Websites		1.89 kB / 0 B		
2024/11/26 11:24:53	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/	Malicious Websites		1.09 kB / 0 B		
2024/11/26 11:24:52	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/	Malicious Websites		1.09 kB / 0 B		
2024/11/26 11:24:52	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/	Malicious Websites		1.09 kB / 0 B		
2024/11/26 11:19:14	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/	Malicious Websites		1.09 kB / 0 B		
2024/11/26 11:19:13	10.0.11.50	10.0.11.50	Blocked	https://www.bing.com/	Malicious Websites		1.89 kB / 0 B		
2024/11/26 11:12:28	10.0.11.50	10.0.11.50	Passthrough	https://edge.atkpc.com/	Internet Telephony		1.9 kB / 0 B		

Stop and think!

Why is the category field empty?

Because the website is blocked by the static URL filter, FortiGuard does not apply the FortiGuard web rating, and does not provide the category.

Lab 9: IPS and Application Control

In this lab, first, you will set up and monitor intrusion prevention system (IPS) profiles. Next, you will configure and use application control in profile-based mode to apply an appropriate action to specific application traffic. Finally, you will analyze the generated logs.

Objectives

- Protect your network against known attacks using IPS signatures
- Configure and test application control on HQ-NGFW-1
- Read and understand application control logs

Time to Complete

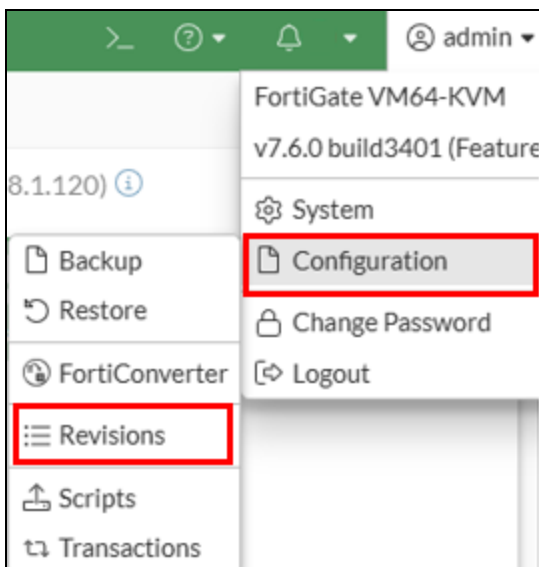
Estimated: 40 minutes

Prerequisites

Before you begin this lab, you must restore a configuration file to HQ-NGFW-1.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **HQ-app-control**, and then click **Revert**.

Delete

Details

Diff

Revert

Save Changes

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Blocking Known Exploits

In this exercise, you will configure and monitor IPS inspection on HQ-NGFW-1.

Configure IPS Inspection

You will configure an IPS sensor that includes the signatures for known attacks based on different severity levels.

To configure IPS inspection

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Security Profiles > Intrusion Prevention**.
3. Click **Create New**.
4. In the **Name** field, type `WEBSERVER`.
5. In the **IPS Signatures and Filters** section, click **Create New**.

New IPS Sensor

Name: **WEBSERVER**

Comments: Write a comment... 0/255

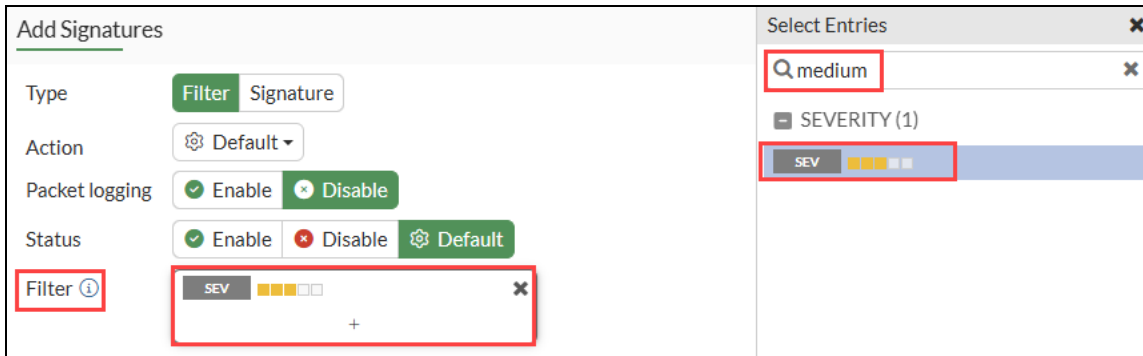
Block malicious URLs: ☐

IPS Signatures and Filters

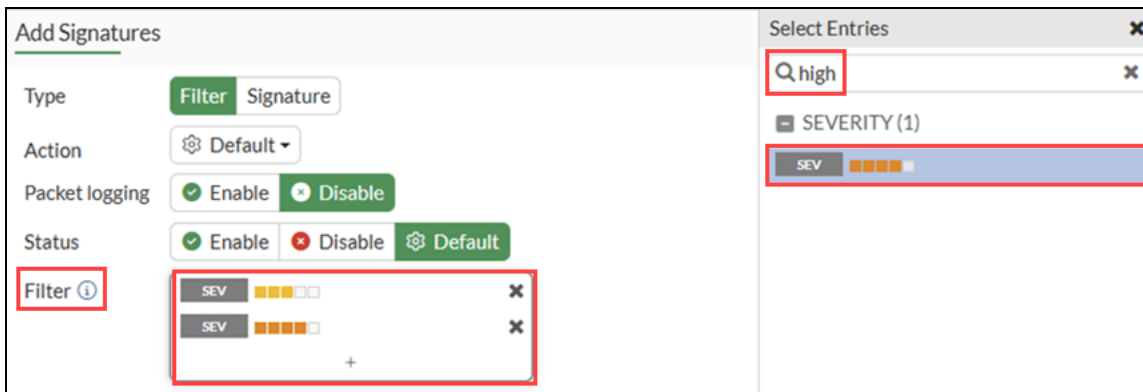
+ Create New Edit Delete

Details	Exempt IPs	Action	Packet Logging
No results			

6. In the **Add Signatures** window, click **+** to add a **Filter**.
7. In the search field, type `medium`, and then click **SEV** to select the medium-severity filter.



8. In the search field, delete `medium`, and then type `high`.
9. Click **SEV** to select the high-severity filter.



10. In the search field, delete `high`, and then type `critical`.
11. Click **SEV** to select the critical-severity filter.



12. In the search field, delete `critical`, and then type `Server`.
13. Click **TGT** to select the server-target filter.

The screenshot shows the 'Add Signatures' configuration window. The 'Filter' tab is active. Under 'Action', 'Default' is selected. 'Packet logging' is set to 'Enable'. Under 'Status', 'Enable' is selected. In the 'Filter' list, 'TGT Server' is highlighted. The 'Select Entries' panel on the right shows a search for 'server', resulting in 'TARGET (1)' with 'TGT Server' selected.

14. In the search field, delete `Server`, and then type `Apache`.
15. Click **App** to select the Apache application filter.

The screenshot shows the 'Add Signatures' configuration window. The 'Filter' tab is active. Under 'Action', 'Default' is selected. 'Packet logging' is set to 'Enable'. Under 'Status', 'Enable' is selected. In the 'Filter' list, 'APP Apache' is highlighted. The 'Select Entries' panel on the right shows a search for 'Apache', resulting in 'APPLICATION (1)' with 'APP Apache' selected.

16. Click **OK** to add the selected filters.



Because FortiGate adds all signatures that match the filters to the IPS sensor, you must configure the filters as specifically as possible.

In this exercise, FortiGate protects an Apache server, and takes the default action for the corresponding signatures.

17. Click **OK**.

Apply an IPS Sensor to a VIP Firewall Policy

You will apply the new IPS sensor to a firewall policy that allows external access to the web server running on the HQ-PC-1 VM.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI, do the following:

- Configure a new virtual IP to map the external IP address 100.65.0.200 to the internal IP address 10.0.11.50, using port2 as the external interface. Name the virtual IP `VIP-WEB-SERVER`.
- Create a new firewall policy to allow all inbound traffic to the virtual IP, and enable the **WEBSERVER** IPS sensor. Name the firewall policy `Web_Server_Access_IPS`.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Generate Attacks From the Linux Server on page 147](#)

To create a virtual IP

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Virtual IPs**.
2. Click **Create New**.
3. Configure the following settings:

Field	Value
Name	VIP-WEB-SERVER
Interface	port2
External IP address/range	100.65.0.200
Map to IPv4 address/range	10.0.11.50

4. Click **OK**.

To configure a firewall policy

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Click **Create New**, and then create a new firewall policy using the following settings:

Field	Value
Name	Web_Server_Access_IPS
Schedule	always
Action	ACCEPT
Incoming interface	port2
Outgoing interface	port4
Source	all
Destination	VIP-WEB-SERVER

Field	Value
Service	ALL
Inspection mode	Flow-based
NAT	Disabled

3. In the **Security Profiles** section, enable **IPS**, and then select **WEBSERVER**.
4. In the **SSL inspection** field, select **certificate-inspection**.

The policy should look like the following example:

Create New Policy

Name: Web_Server_Access_IPS

Schedule: always

Action: ☒ ACCEPT ☐ DENY

Incoming interface: port2

Outgoing interface: port4

Source & Destination (Show logic)

Source: all

User/group:

Destination: VIP-WEB-SERVER

Service: ALL

Firewall/Network Options

Inspection mode: Flow-based Proxy-based

NAT: ☐

Protocol options: default

Security Profiles

AntiVirus: ☐

Web filter: ☐

DNS filter: ☐

Application control: ☐

IPS: ☒ WEBSERVER

File filter: ☐

SSL inspection: SSL certificate-inspection



Using full SSL inspection would significantly increase the time required to complete this lab. Therefore, for the purposes of this exercise, you will not configure full SSL inspection.

5. Click **OK**.

Generate Attacks From the Linux Server

You will run a Perl script to generate attacks from the Linux server located in front of HQ-NGFW-1.

To generate attacks from the Linux server

1. On the HQ-PC-1 VM, open PuTTY, and then connect over SSH to the LINUX-ISP saved session.
2. Log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
3. Enter the following command to start the attacks:
`nikto.pl -host 100.65.0.200`
4. Leave the PuTTY session open (you can minimize it) so that traffic continues to generate.



Do not close the LINUX PuTTY session or traffic will stop generating.

To increase the verbosity of the command, you may enter `nikto.pl -host 100.65.0.200 -Display V`.

Monitor the IPS

You will check the IPS logs to monitor for known attacks that HQ-NGFW-1 is detecting and dropping.

To monitor the IPS

1. Return to the HQ-NGFW-1 GUI, and then click **Log & Report > Security Events > Intrusion Prevention**.
2. Locate and review the relevant log entries for the detected and dropped attacks.



FortiGate creates an intrusion prevention log entry for the following:

- Detected attack without blocking it
- Dropped attack with blocking it

3. Click a log entry, and then click **Details**.
4. In the **Attack Name** field, click the link.

Summary									
Date/Time	Severity	Source	Protocol	User	Action	Count	Attack Name		
2024/11/15 12:44:26	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		
2024/11/15 12:44:25	Medium	100.65.0.254	6		detected		Apache.Expect.Header.XSS		
2024/11/15 12:41:23	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		
2024/11/15 12:41:13	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		
2024/11/15 12:41:03	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		
2024/11/15 12:40:53	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		
2024/11/15 12:40:43	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		
2024/11/15 12:40:33	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		
2024/11/15 12:40:32	Medium	100.65.0.254	6		detected		Apache.Expect.Header.XSS		
2024/11/15 12:39:32	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		
2024/11/15 12:39:22	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		
2024/11/15 12:39:21	Medium	100.65.0.254	6		detected		Apache.Expect.Header.XSS		
2024/11/15 12:25:33	High	100.65.0.254	6		dropped		HTTPURLSQLInjection		

Log Details

Security

Level: High

Threat Level: High

Threat Score: 30

Intrusion Prevention

Profile: WUISS-RVIR

Attack Name: HTTPURLSQLInjection

Attack ID: 15621

Reference: <https://fortiguant.fortinet.com/encyclopedia/15621>

Incident Serial: 20,566,168

Direction: outgoing

Severity: High

Message: web_misc: HTTPURLSQLInjection

5. Review the FortiGuard Labs *Threat Encyclopedia* for the signatures.

The FortiGuard Labs *Threat Encyclopedia* provides information about signatures, such as severity, coverage, affected products, impact, and recommended actions that you can take.



Are the signatures matching the application that is currently installed on the HQ-PC-1 VM? This information is important to make a note of before you tune the **WEBSERVER** IPS sensor. If the signatures do not apply to your application, is it really necessary to inspect those packets?

Troubleshoot IPS Activity

You will troubleshoot and monitor IPS activity.

To troubleshoot IPS activity

1. Connect to the HQ-NGFW-1 CLI, and then log in with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Enter the following command:

```
diagnose test application ipsmonitor 1
```

The output should be similar to the following example:

```
HQ-NGFW-1 # diag test app ipsmonitor 1
pid = 2008, engine count = 1 (+1)
0 - pid:2048:2048 cfg:1 master:0 run:1
1 - pid:2579:2579 cfg:0 master:1 run:1

pid: 2579 index:1 master
version: 07006000FLEN07600-00007.00006.01014-2407022157
up time: 0 days 0 hours 33 minutes
init time: 1 seconds
socket size: 256(MB)
database: ipsetdb isdb
bypass: disable
```

3. Enter the following command to enable IPS bypass mode:

`diagnose test application ipsmonitor 5`



If you then enter the `diagnose test application ipsmonitor 1` command, the last line shows the new bypass status of `enable`.

On the HQ-NGFW-1 GUI, you can also verify in **Log & Report > Security Events > Intrusion Prevention** that no new log entry is generated.

4. Enter the following command to restart the IPS-related engines:

```
diagnose test application ipsmonitor 99
```

5. Enter the following command to verify the status:

```
diagnose test application ipsmonitor 1
```

The output should be similar to the following example:

```
HQ-NGFW-1 # diag test app ipsmonitor 1
pid = 2678, engine count = 1 (+1)
0 - pid:2680:2680 cfg:1 master:0 run:1
1 - pid:2689:2689 cfg:0 master:1 run:1

pid:      2689 index:1 master
version:   07006000FLEN07600-00007.00006.01014-2407022157
up time:   0 days 0 hours 0 minutes
init time: 0 seconds
socket size: 256(MB)
database:  ipsetdb isdb
bypass:    disable
```



Because you have restarted the IPS engine, the corresponding process ID has changed and the bypass mode resets to the `disable` status.

6. Close the HQ-NGFW-1 CLI and GUI sessions.
7. Close the LINUX PuTTY session.

Exercise 2: Controlling Application Traffic

In this exercise, you will create a profile-based application control profile in flow-based inspection mode. Flow-based and proxy-based inspection modes share identical configuration steps for application control.

Next, you will view the application control logs to confirm that FortiGate identifies applications. Finally, you will monitor the traffic that matches the application control profile.

Configure Filter Overrides

The configuration file for this exercise has the application control categories set to **Monitor** (except for **Unknown Applications**). This allows the applications to pass, but also records a log message.

You will configure filter overrides.

To configure filter overrides

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Security Profiles > Application Control**.
3. Double-click the **default** application control profile to edit it.

93 Cloud Applications require deep inspection.
0 policies are using this profile.

Name: default

Comments: Monitor all applications. 25/255

Categories

Mixed All Categories

Business (179, 6)	Cloud/IT (31)
Collaboration (293, 6)	Email (87, 12)
Game (124)	General Interest (241, 9)
Mobile (3)	Network Service (332)
P2P (85)	Proxy (106)
Remote Access (91)	Social Media (150, 31)
Storage/Backup (296, 16)	Update (48)
Video/Audio (206, 13)	VoIP (31)
Web Client (18)	Unknown Applications



There are 93 cloud-based application signatures available in the application control signatures database that require deep inspection. The number of cloud-based application signatures can vary.

The number beside the cloud icon in each category represents the number of cloud application signatures in a specific category. The number of cloud applications increases as new applications are added to this list.

4. In the **Application and Filter Overrides** section, click **Create New** to add a filter override.
5. On the **Add New Override** page, in the **Type** field, select **Filter**.
6. In the **Filter** field, click **+** to add a filter.
7. In the **Select Entries** section, expand **Behavior**, and then click **Excessive-Bandwidth**.

Name	Category	Technology	Popularity
Application signature 1,262			
1kxun	Video/Audio	Client-Server	★★★★☆
1und1.Mail	Email	Browser-Based	★★★★☆



The **Excessive-Bandwidth** setting blocks many applications that are known to be bandwidth intensive. Applications can belong to different categories, but they may be part of this behavior filter if they are bandwidth intensive.

8. Click **OK**.
- The configuration should look similar to the following image with the **Action** set to **Block**.

Priority	Details	Type	Action
1	BHVR Excessive-Bandwidth	Filter	Block

9. Click **OK**.

Apply the Application Control Profile to the Firewall Policy

Now that you have configured the application control profile, you will apply it to the firewall policy.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI, edit the existing **Application_Control** firewall policy and do the following:

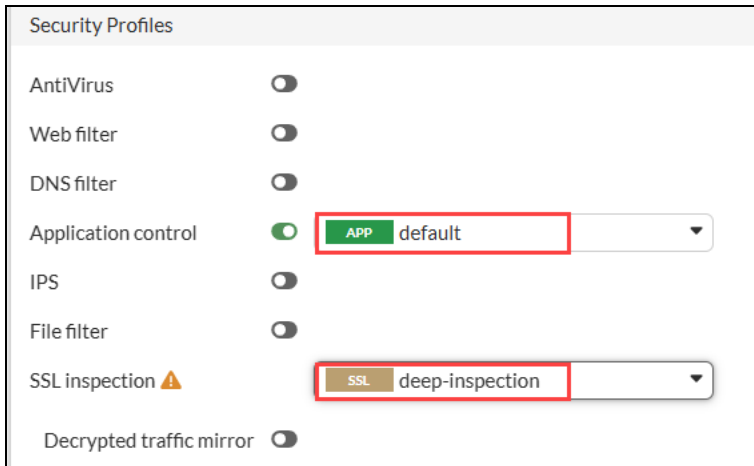
- Enable the **default** application control profile.
- Enable **deep-inspection** in the SSL/SSH inspection profile.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Test the Application Control Profile on page 153](#).

To apply the application control profile to the firewall policy

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Double-click the **Application_Control** firewall policy to edit it.
3. In the **Security Profiles** section, enable **Application control**, and then select **default**.
4. In the **SSL inspection** field, select **deep-inspection**.



Security Profiles	
AntiVirus	☐
Web filter	☐
DNS filter	☐
Application control	☒ APP default
IPS	☐
File filter	☐
SSL inspection ⚠	☒ SSL deep-inspection
Decrypted traffic mirror	☐



If the FortiGate self-signed, full-inspection certificate has not been imported into the browser, end users see a certificate warning message. In this lab environment, the FortiGate self-signed SSL inspection certificate has been imported into the browser.

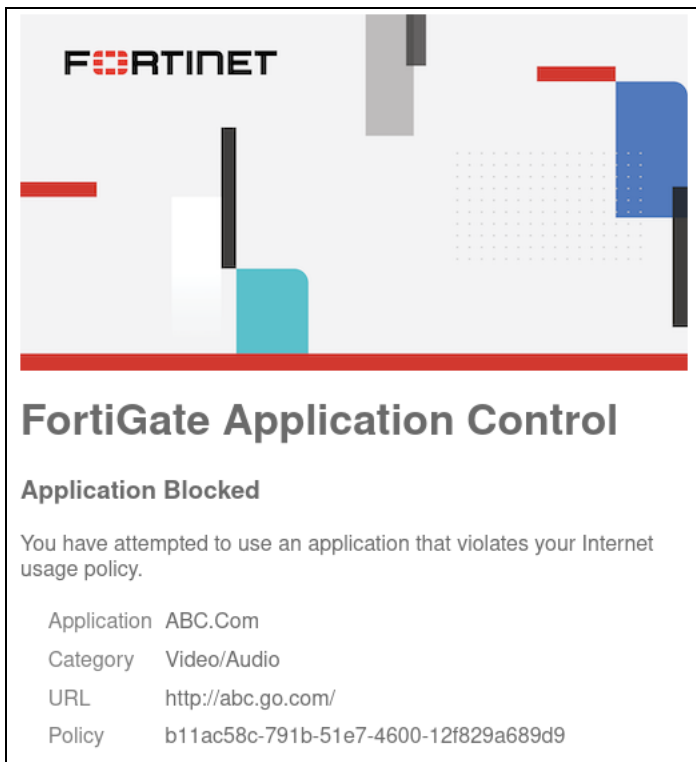
5. Click **OK** to save the changes.
6. Click **OK** to confirm.

Test the Application Control Profile

You will test the application control profile by going to the application that you blocked in the application override configuration.

To test the application control profile

1. On the HQ-PC-1 VM, open a new browser tab, and then go to the following URL: `http://abc.go.com`.
You should see that you cannot connect to this site—it times out.
2. Return to the HQ-NGFW-1 GUI, and then click **Security Profiles > Application Control**.
3. Double-click the **default** application sensor.
4. At the bottom of the page, in the **Options** section, enable **Replacement Messages for HTTP-based Applications**.
5. Click **OK**.
6. On the HQ-PC-1 VM, open a new browser tab, and then go to the following URL: `http://abc.go.com`.
FortiGate should display a block message—it can take up to 2 minutes for the **Application Blocked** page to appear because of the change in configuration.



If the **Application Blocked** page does not appear after 2 minutes, close all browser tabs, and then restart the browser.

Configure Application Overrides

You will configure application overrides. The application overrides take precedence over filter overrides and application categories.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI, complete the following:

- Modify the **default** application control profile.
- Add **Application Overrides** for the **ABC.Com** application signature, and set the action to **Allow**.

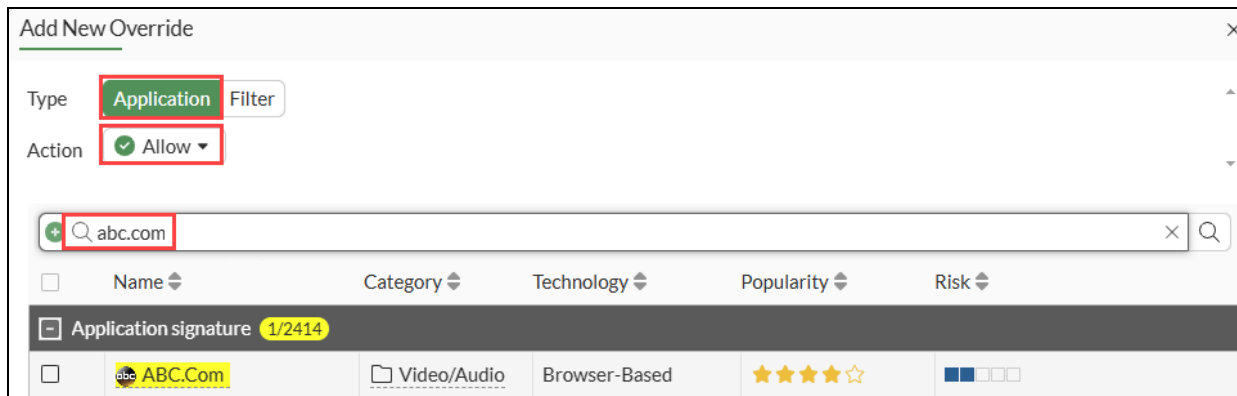
If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Test Application Overrides on page 155](#).

To configure application overrides

1. Return to the browser tab where you are logged in to the HQ-NGFW-1 GUI, and then click **Security Profiles > Application Control**.
2. Double-click the **default** application sensor.
3. In the **Application and Filter Overrides** section, click **Create New**.
4. On the **Add New Override** page, in the **Type** field, select **Application**.
5. In the **Action** field, select **Allow**.
6. In the search field, type `abc.com`, and then press **Enter**.

FortiGate returns a signature.



7. Click **ABC.Com** to select it.
8. Click **OK**.
9. Drag the **ABC.Com** application filter and place it above the **Excessive-Bandwidth** filter.

The configuration should look like the following image:

Application and Filter Overrides

[+ Create New](#) [Edit](#) [Delete](#)

Priority	Details	Type	Action
1	 ABC.Com	Application	 Allow
2	 Excessive-Bandwidth	Filter	 Block
2			

10. Click **OK**.



This application control profile is already applied to a firewall policy that is scanning all outbound traffic. You do not need to reapply the application control profile for the changes to take effect.

Test Application Overrides

You will test the application control profile by going to the application that you allowed.

To test the application control profile

1. On the HQ-PC-1 VM, open a new browser tab, and then go to the following URL: `http://abc.go.com`. FortiGate allows the website to load correctly.

View Logs and Traffic Matching With the Application Control Profile

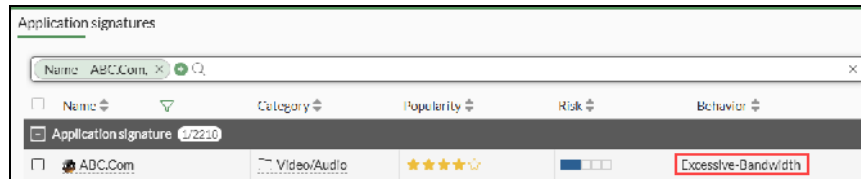
You will view the logs and traffic that matched the test you just performed.

To view logs

1. Return to the HQ-NGFW-1 GUI, and then click **Log & Report > Security Events**.
2. On the **Summary** tab, click **Application Control**.
3. Using the **Application Name** log filter, search for **ABC.Com**.
You will see log messages with the action set to **block**. This is the log for **ABC.Com** before applying the application override.

Why does the log show that **ABC.Com** is blocked?

The log entry is from blocking the excessive bandwidth behavior.



Why is there no log entry for when **ABC.Com** was added as an application override?

The action for **ABC.Com** in **Application Overrides** is set to **Allow**, which does not generate a security log entry. To allow and generate a security log entry for **ABC.Com**, you must set the action to **Monitor**.

4. Double-click a log to view more details.

The details include the application sensor name, application name, category, policy ID, and action that FortiGate took.

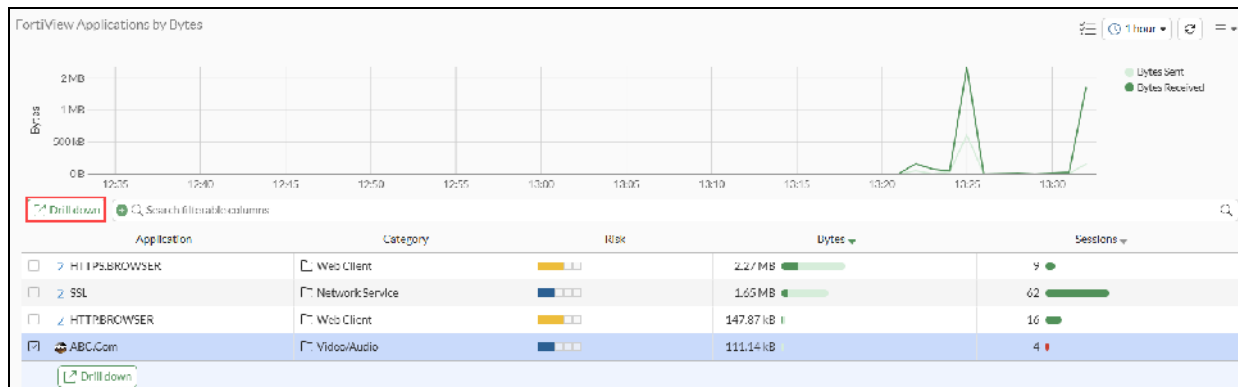
5. Click **Log & Report > Forward Traffic**, and then search and view the log information for **ABC.Com**.

You can see more details about the log, including translated IP address, bytes sent, bytes received, action, and application.

To view the traffic that matched the application control profile

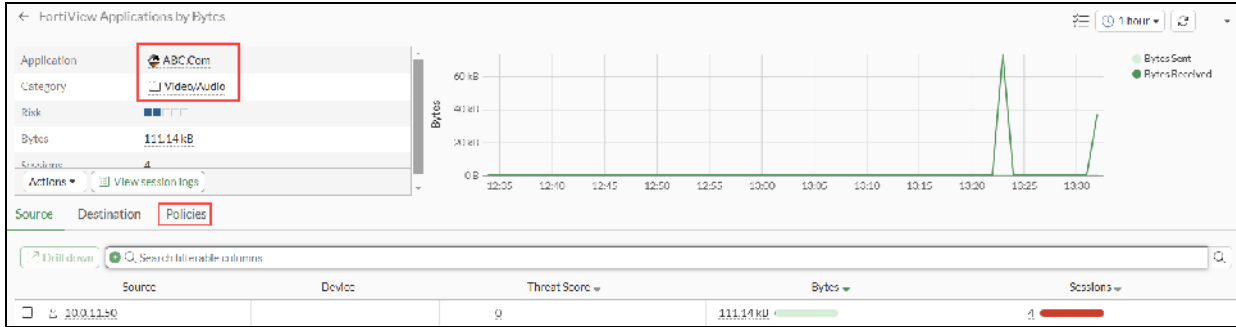
1. On the HQ-NGFW-1 GUI, click **Dashboard > FortiView Applications**.

The page should look similar to the following example:



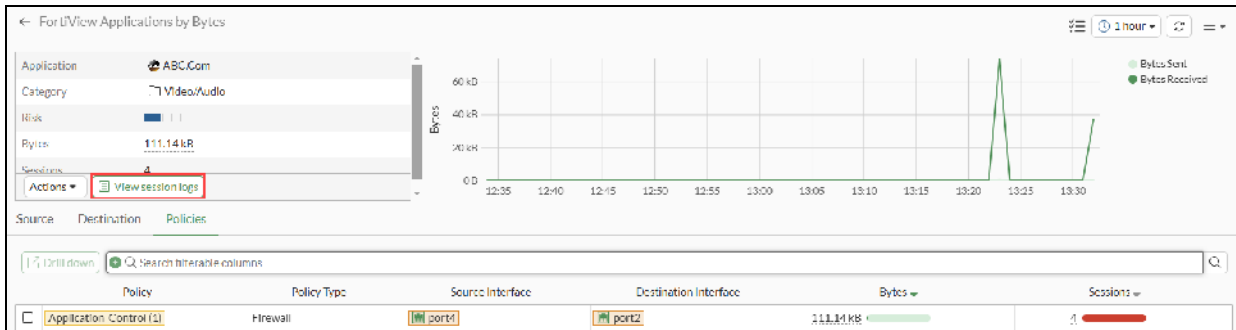
2. Click the line that includes **ABC.Com** to select it, and then click **Drill down**.

The page should look similar to the following example:



3. Click **Policies**.

The page should look similar to the following example:



4. Click **View session logs**.

The page should look similar to the following example:

Sessions: HQ-NGFW-1						
Date/Time	Source	Destination	Application Name	Security Action	Sent / Received	
2024/11/15 13:32:30	10.0.11.50	52.71.104.197 (abcgo.com)	ABC.Com	block	216 B / 36.77 kB	
2024/11/15 13:23:50	10.0.11.50	18.233.14.105 (abcgo.com)	ABC.Com	block	112 B / 60 B	
2024/11/15 13:23:18	10.0.11.50	52.71.104.197 (abcgo.com)	ABC.Com	block	216 B / 36.78 kB	
2024/11/15 13:23:18	10.0.11.50	52.71.104.197 (abcgo.com)	ABC.Com	block	216 B / 36.77 kB	



You now have the information (bytes, sessions, and policy ID) regarding the traffic that matched the **ABC.Com** application.

5. Close the HQ-NGFW-1 GUI session.

Lab 10: SSL VPN

In this lab, you will examine how to configure an SSL VPN connection in tunnel mode. You will also manage user groups and portals for an SSL VPN.

Objectives

- Configure and connect to an SSL VPN
- Enable authentication security
- Configure a firewall policy for SSL VPN users to access private network resources
- Configure FortiClient for the SSL VPN connection in tunnel mode

Time to Complete

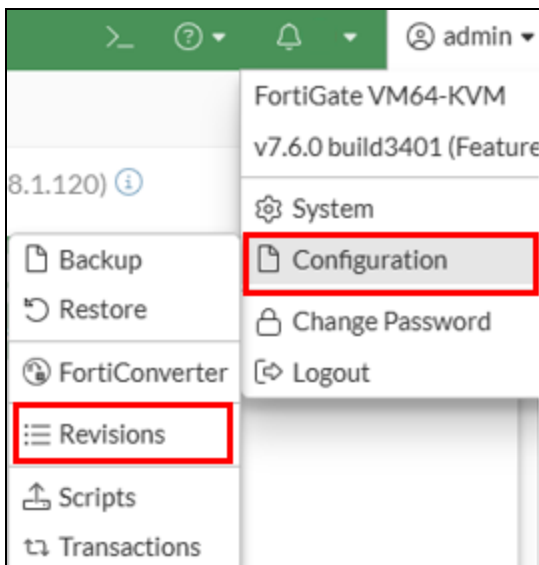
Estimated: 30 minutes

Prerequisites

Before you begin this lab, you must restore a configuration file to HQ-NGFW-1.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.
3. Click **+** to expand the list.



4. Select the configuration with the comment **HQ-SSL-VPN**, and then click **Revert**.

Delete

Details

Diff

Revert

Save Changes

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

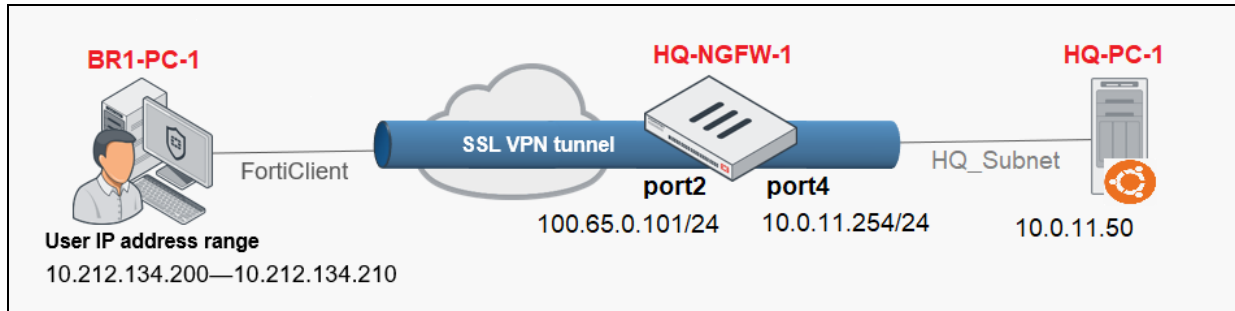
5. Click **OK** to reboot.

Exercise 1: Configuring SSL VPN Tunnel Mode

In this exercise, you will examine how to change the SSL VPN settings to allow remote access to the resources in the local subnet (10.0.11.0/24). You will also create a connection in tunnel mode from the BR1-PC-1 VM.

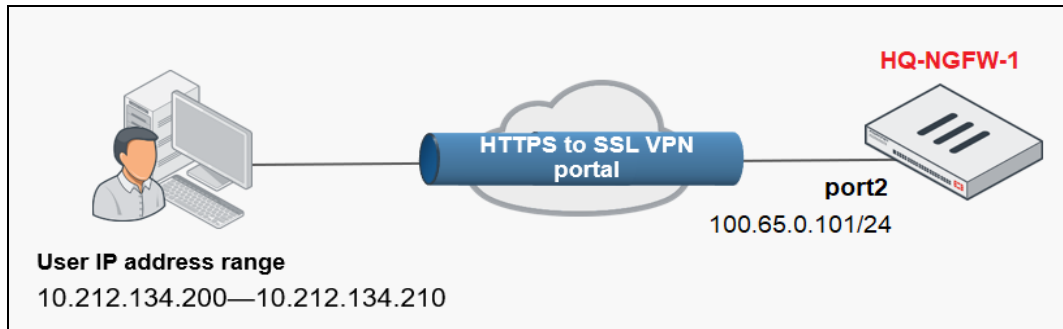
You will use the remote access module of FortiClient, which supports the Fortinet SSL VPN client.

FortiClient is already installed on the BR1-PC-1 VM.



Configure the SSL VPN Settings

You will configure the SSL VPN settings to allow the remote connection that is shown in the following image:



By default, SSL VPN tunnel mode settings and the **VPN > SSL-VPN** menus are hidden on the GUI in FortiOS. To enable the GUI menu, enter the following CLI commands:

```
config system settings
set gui-sslvpn enable
end
```

The configuration file is preconfigured for you to show the SSL VPN menus.

To create a user for SSL VPN connections

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:

- Username: admin
 - Password: Fortinet1!
2. Click **User & Authentication > User Definition**.
 3. Click **Create New**.
 4. Click **Local User**, and then click **Next**.
 5. Type the following credentials for the remote user, and then click **Next**:

Username	user1
Password	Fortinet1!

6. Leave the contact information field empty, and then click **Next**.
7. In the **Account Status** field, click **Next**.
8. Enable **User Group**, click **+**, and then in the section on the right, select **SSL_VPN_USERS**.
9. Click **Submit**.



The **SSL_VPN_USERS** group was preconfigured for this lab.

To review the settings of this group, click **User & Authentication > User Groups**.

To configure the SSL VPN settings for access

1. Continuing on the HQ-NGFW-1 GUI, click **VPN > SSL-VPN Settings**.
2. In the **Connection Settings** section, configure the following settings:

Field	Value
Listen on Interface(s)	port2
Listen on Port	10443
Server Certificate	Fortinet_Factory
Restrict Access	Allow access from any host
Inactive For	3000 seconds

3. In the **Tunnel Mode Client Settings** section, verify the following setting:

Field	Value
Address Range	Automatically assign addresses

4. In the **Authentication/Portal Mapping** section, select **All Other Users/Groups**, and then click **Edit**.

Users/Groups	Portal
All Other Users/Groups	Not Set

5. In the **Portal** field, select **tunnel-access**, and then click **OK**.
6. Click **Apply** to save the changes.

Configure the Routing for Tunnel Mode

You will establish the routing address to use in tunnel mode.

In tunnel mode, FortiClient establishes one or more routes on the SSL VPN user's host after the tunnel is connected. Traffic destined to the internal subnets is correctly routed through the tunnel.

To configure the routing for tunnel mode

1. Continuing on the HQ-NGFW-1 GUI, click **VPN > SSL-VPN Portals**.
2. Select the **tunnel-access** portal, and then click **Edit**.
3. In the **Tunnel Mode** section, in the **Routing Address Override** field, select **HQ_SUBNET**.

☒ Tunnel Mode

Split tunneling

- ☐ Disabled
All client traffic will be directed over the SSL-VPN tunnel.
- ☒ Enabled Based on Policy Destination
Only client traffic in which the destination matches the destination of the configured firewall policies will be directed over the SSL-VPN tunnel.
- ☐ Enabled for Trusted Destinations
Only client traffic which does not match explicitly trusted destinations will be directed over the SSL-VPN tunnel.

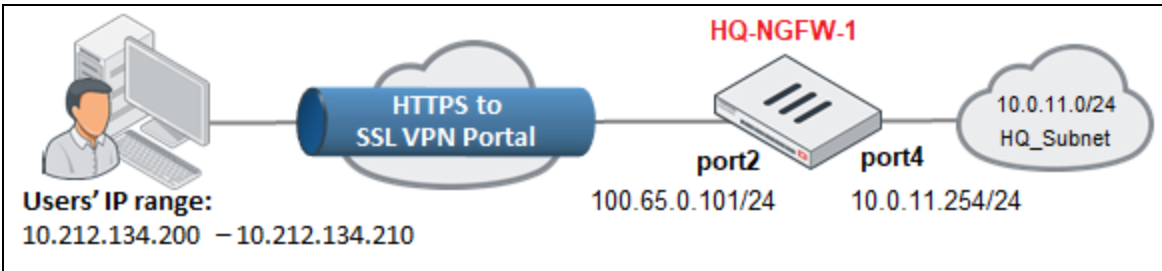
Routing Address Override: HQ_SUBNET

Source IP Pools: SSLVPN_TUNNEL_ADDR1

4. Click **OK**.

Create a Firewall Policy for SSL VPN

You will create a firewall policy that allows traffic to the local subnet (10.0.11.0/24) from remote users connected to the SSL VPN.



To create a firewall policy for SSL VPN

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Click **Create New**, and then configure the following firewall policy settings:

Field	Value
Name	SSL-VPN-Access
Schedule	always
Action	ACCEPT
Incoming Interface	SSL-VPN tunnel interface (ssl.root)
Outgoing Interface	port4
Source	SSLVPN_TUNNEL_ADDR1
User/group	SSL_VPN_USERS
Destination	HQ_SUBNET
Service	ALL
Inspection mode	Flow-based
NAT	Disabled

3. Click **OK** to save the configuration.

Configure FortiClient for SSL VPN Connections

SSL VPN connections in tunnel mode require FortiClient. You will use FortiClient, which is installed on the BR1-PC-1 VM, to test your configuration.

To configure FortiClient for SSL VPN in tunnel mode

1. Connect to the BR1-PC-1 VM with the following credentials:
 - Username: Administrator
 - Password: Fortinet1!
2. Click **Desktop** > **forticlientsslvpn** > **64bit**, and then double-click **forticlientsslvpn** to configure SSL VPN client settings.
3. Configure the following settings for the FortiClient SSL VPN application:

Field	Value
Server	100.65.0.101
Customize port	10443
User	user1
Password	Fortinet1!



4. Click **Connect**.
5. Click **Continue** to accept the certificate.
The tunnel is connected.



To test the tunnel

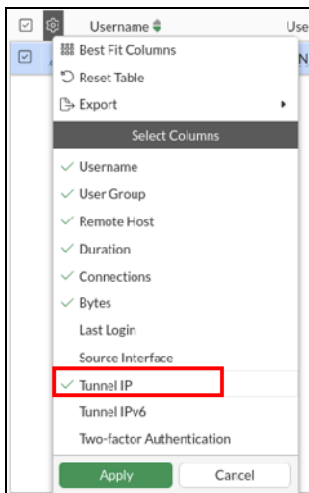
1. Continuing on the BR1-PC-1 VM, open Firefox, and then access the following URL:
`http://10.0.11.50`
2. Look at the URL.
You are connected to the web server URL as if you were based in the local subnet (10.0.11.0/24).

Monitor an SSL VPN User

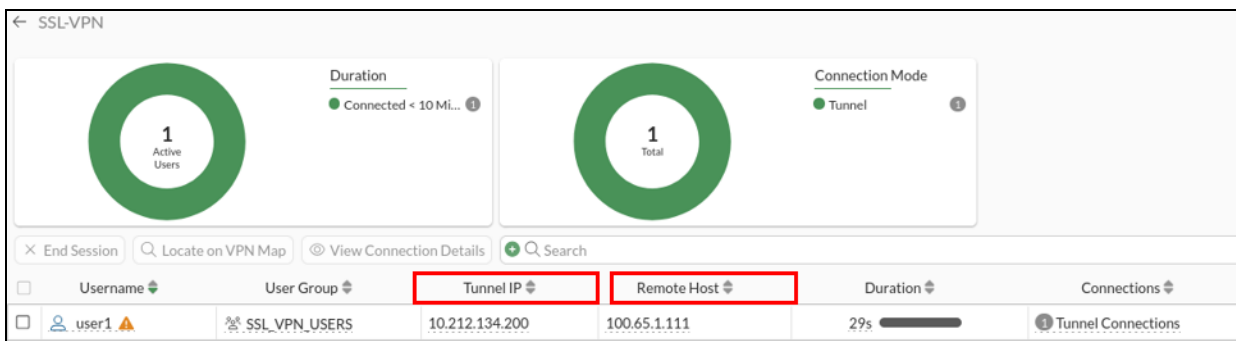
You will monitor and disconnect an SSL VPN user from the FortiGate GUI.

To monitor and disconnect an SSL VPN user

1. Return to the HQ-NGFW-1 GUI.
2. Click **Dashboard > Network**, and then view the **SSL-VPN** widget.
3. To the left of the username, click the gear icon, and then select **Tunnel IP**.

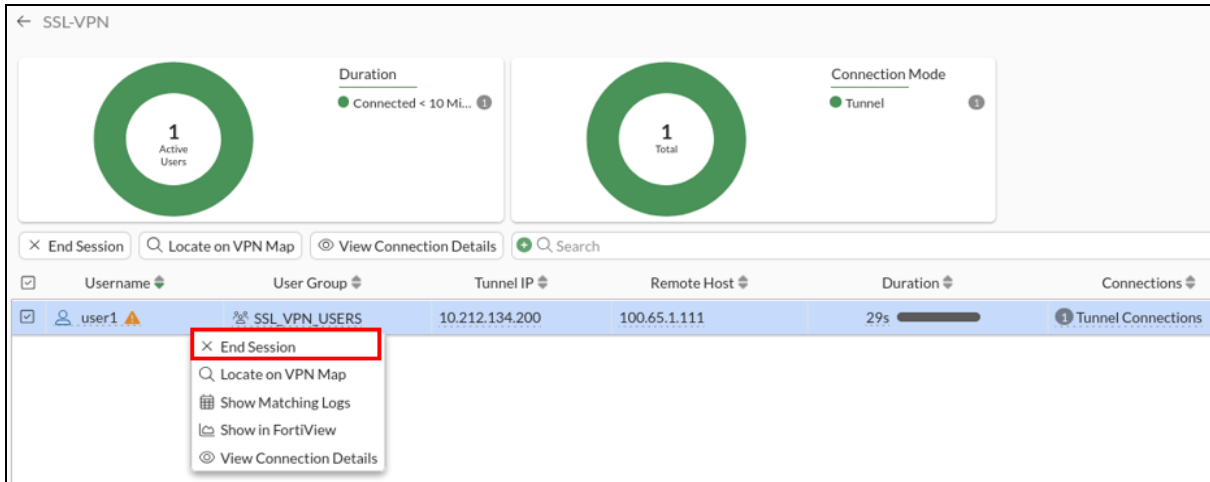


4. Click **Apply**.
The user1 user is connecting from the remote host 100.65.1.111.



This log displays two IP addresses:

- **Remote Host:** IP address of the remote user's gateway (egress interface)
 - **Tunnel IP:** IP address FortiGate assigned to the virtual network adapter `fortissl`
5. Right-click **user1**, and then select **End Session**.
 6. Click **OK**.



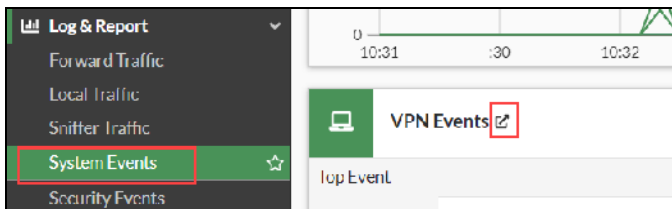
The user1 user no longer appears in the SSL VPN monitor.

Review VPN Events

You will review the VPN events for the SSL VPN connection you performed in this lab.

To review VPN events for the SSL VPN connection

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **Log & Report > System Events**, and then expand the **VPN Events** widget to view the logs.



3. View the log details of the **tunnel-up** log you see.
Hint: Use your log filters to filter on **Action = tunnel-up**.

The **tunnel-up** log, in the VPN event list, shows the SSL VPN connection in tunnel mode through FortiClient.

Date/Time	Level	Action	Status	Message	VPN Tunnel
2024/11/28 14:03:19	Error	ssl-alert		SSL alerts	
2024/11/28 14:03:18	Information	ssl-new-con		SSL new connection	
2024/11/28 14:03:18	Information	tunnel-up		SSL tunnel established	
2024/11/28 14:03:18	Information	ssl-new-con		SSL new connection	
2024/11/28 14:02:46	Information	ssl-new-con		SSL new connection	
2024/11/28 14:02:46	Information	tunnel-down		SSL tunnel shutdown	
2024/11/28 14:02:46	Information	tunnel down		SSL tunnel shutdown	
2024/11/28 14:02:46	Information	tunnel down		SSL tunnel shutdown	
2024/11/28 14:01:03	Error	ssl-alert		SSL alerts	
2024/11/28 14:00:42	Information	ssl-new-con		SSL new connection	
2024/11/28 14:00:42	Information	tunnel-up		SSL tunnel established	
2024/11/28 14:00:42	Information	ssl-new-con		SSL new connection	
2024/11/28 13:55:44	Information	info		User changed SSL setting	

LogDetails

Source

Source Country/RegionReserved
Useruser1
GroupSSL_VPN_USERS

Destination

Destination HostN/A

Action

Actiontunnel-up
Reasonlogin successfully

Security

LevelInformation

Event

Remote IP100.65.1.111
Tunnel ID406.B31.B29

Lab 11: IPsec VPN Configuration

In this lab, you will configure site-to-site IPsec VPN tunnels between two FortiGate devices. First, you will configure a dial-up tunnel, and then a static tunnel.

Objectives

- Deploy a site-to-site VPN between two FortiGate devices
- Configure dial-up and static remote gateways

Time to Complete

Estimated: 40 minutes

Prerequisites

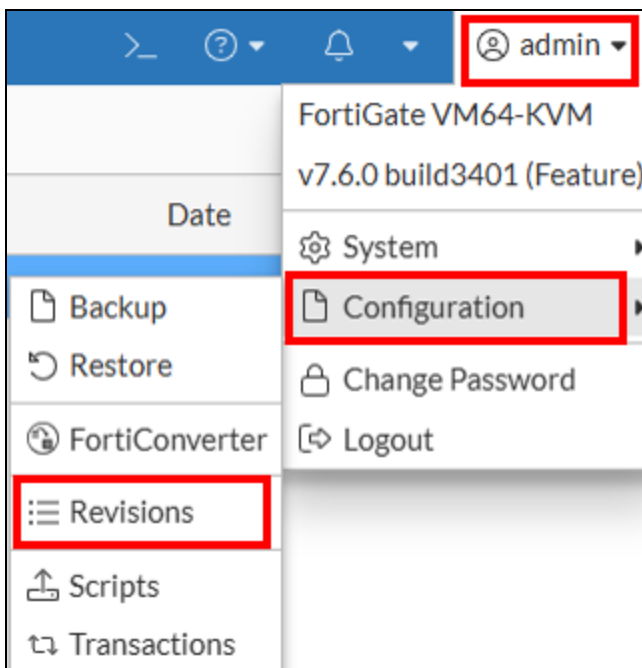
Before you begin this lab, you must restore a configuration file on BR1-FGT and HQ-NGFW-1.



Make sure that you restore the correct configuration file on each FortiGate, using the following steps. Failure to restore the correct configuration on each FortiGate will prevent you from doing the lab exercises.

To restore the BR1-FGT configuration file

1. Connect to the BR1-FGT GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



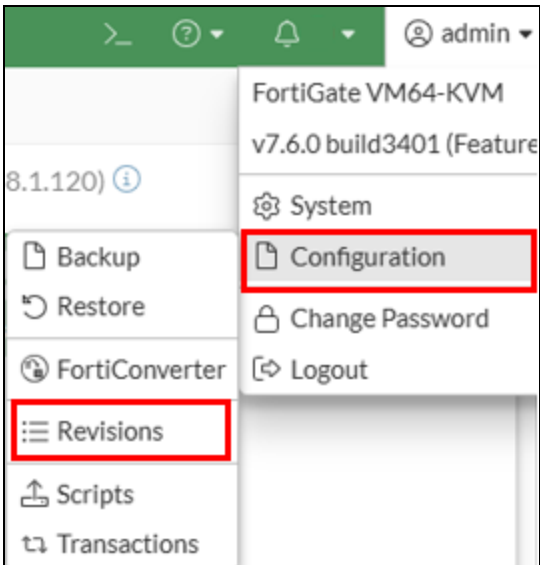
3. Click + to expand the list.
4. Select the configuration with the comment **BR-dialup-ipsec**, and then click **Revert**.

 Delete	 Details	 Diff	 Revert	 Save Changes 
Config ID	Username	Date	Comments	
7.6.0 build 3401 3				
3	admin	2024/12/13 17:56:40	BR1-SF	
2	admin	2024/12/13 17:54:11	BR-dialup-ipsec	
1	admin	2024/12/13 17:51:50	BR1-FGT-initial	

5. Click **OK** to reboot.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click + to expand the list.
4. Select the configuration with the comment **HQ-dialup-ipsec** , and then click **Revert**.

Delete Details Diff Revert Save Changes ⚠			
Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Configuring a Dial-Up IPsec VPN Between Two FortiGate Devices

In this exercise, you will configure a dial-up VPN between HQ-NGFW-1 and BR1-FGT. HQ-NGFW-1 will act as the dial-up server and BR1-FGT will act as the dial-up client.

Create Phase 1 and Phase 2 Negotiations on HQ-NGFW-1 (Dial-Up Server)

You will configure the IPsec VPN by creating phase 1 and phase 2 negotiations.

To create phase 1 and phase 2 negotiations

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **VPN > VPN Tunnels**, and then click **Create New > Custom IPsec tunnel**.
3. Configure the following setting:

Field	Value
Name	ToBranch

4. In the **Network** section, configure the following settings:

Field	Value
Remote Gateway	Dialup User
Interface	port2
Dead Peer Detection	On Idle

5. In the **Authentication** section, configure the following settings:

Field	Value
Method	Pre-shared Key
Pre-shared Key	fortinet
Version	1
Mode	Aggressive

Field	Value
Accepted peer ID	Specific peer ID
Peer ID	BR1-FGT



Setting a peer ID is useful when the FortiGate, acting as the dial-up server, has multiple dial-up tunnels, and you want dial-up clients to connect to a specific tunnel.

- In the **Phase 2 Selectors** section, click **Create new**.
- In the **Name** field, type `ToBranch`.
- In the **Local address** field, select **Subnet Address**, and then type `10.0.11.0/24`.

New Phase 2 Selector

Name:

Comments: 0/255

Encapsulation: ☒ Tunnel Mode ☐ Transport Mode

IP version: ☒ IPv4 ☐ IPv6

Named address: ☐

Local address:

Remote address:

- Keep the default values for the remaining settings.
- Click **OK**, and then click **OK** again to save the IPsec VPN configuration.



You do not need to add a static route because it is a dial-up VPN. FortiGate dynamically adds or removes appropriate routes to each dial-up peer, each time the peer VPN is trying to connect.

Even though you could have configured `172.20.1.0/24` as the **Remote Address** instead of `0.0.0.0/0`, it is more convenient to use the latter for scalability purposes. That is, when you have multiple remote peers, each with different remote subnets, using `0.0.0.0/0` as the remote subnet results in the dial-up server accepting any subnet during the tunnel negotiation. This allows multiple remote peers to use the same phase 2 selector configuration.

This exercise has only one remote peer (BR1-FGT). HQ-NGFW-1 then learns about the remote subnet `172.20.1.0/24` when BR1-FGT connects to the tunnel. However, if there are more remote peers with different remote subnets, you do not need to change the existing dial-up server configuration for the additional remote peers to be able to connect.

Create Firewall Policies for VPN Traffic on HQ-NGFW-1 (Dial-Up Server)

You will create two firewall policies between **port4** and **To Remote**—one for each traffic direction.

To create firewall policies for VPN traffic

1. On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Click **Create New**.
3. Configure the following settings:

Field	Value
Name	Branch_out
Schedule	always
Action	ACCEPT
Incoming Interface	port4
Outgoing Interface	ToBranch
Source	HQ_SUBNET
Destination	BR1_SUBNET
Service	ALL

4. In the **Firewall/Network Options** section, disable **NAT**.
5. Click **OK**.
6. Click **Create New** again.
7. Configure the following settings:

Field	Value
Name	Branch_in
Schedule	always
Action	ACCEPT
Incoming Interface	ToBranch
Outgoing Interface	port4
Source	BR1_SUBNET
Destination	HQ_SUBNET
Service	ALL

8. In the **Firewall/Network Options** section, disable **NAT**.
9. Click **OK**.

Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT
port4 → 1							
port4 → 1							
☐ Branch_out (2)	HQ_SUBNET	BR1_SUBNET	always	ALL	✓ ACCEPT		✗ Disabled
ToBranch → 1							
☐ Branch_in (3)	BR1_SUBNET	HQ_SUBNET	always	ALL	✓ ACCEPT		✗ Disabled

Create Phase 1 and Phase 2 on BR1-FGT (Dial-Up Client)

You will create phase 1 and phase 2 on BR1-FGT.

To create phase 1 and phase 2

1. Connect to the BR1-FGT GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **VPN > VPN Tunnels**, and then click **Create New > Custom IPsec Tunnel**.
3. Configure the following setting:

Field	Value
Name	ToHQ

4. In the **Network** section, configure the following settings:

Field	Value
Remote Gateway	Static IP Address
IP Address	100.65.0.101
Interface	port2
Dead Peer Detection	On Idle

5. In the **Authentication** section, configure the following settings:

Field	Value
Method	Pre-shared Key
Pre-shared Key	fortinet
Version	1
Mode	Aggressive
Accepted peer ID	Any peer ID

6. In the **Phase 1 proposal** section, configure the following setting:

Field	Value
Local ID	BR1-FGT

Phase 1 proposal

Encryption - authentication

AES128

SHA256

×

AES256

SHA256

×

AES128

SHA1

×

AES256

SHA1

×

+

Diffie-Hellman groups

☐ 1

☒ 14

☐ 17

☐ 20

☐ 28

☐ 31

☐ 2

☐ 15

☐ 18

☐ 21

☐ 29

☐ 32

☒ 5

☐ 16

☐ 19

☐ 27

☐ 30

Key lifetime

86400

second(s)

Local ID

BR1-FGT



The local ID should be the same as the peer ID that you configured on HQ-NGFW-1, which is acting as the dial-up server.

Note that the **Peer ID** and **Local ID** fields are case sensitive.

- 7. In the **Phase 2 Selectors** section, click **Create new**.
- 8. Configure the following settings:

Field	Value
Name	ToHQ
Local address	Select Subnet Address , and then type 172.20.1.0/24.
Remote address	Select Subnet Address , and then type 10.0.11.0/24.

New Phase 2 Selector

Name

ToHQ

Comments

Comments

0/255

Encapsulation

Tunnel Mode

Transport Mode

IP version

IPv4

IPv6

Named address

☐

Local address

IP Address

Subnet Address

IP Range

172.20.1.0/24

Remote address

IP Address

Subnet Address

IP Range

10.0.11.0/24

- 9. Keep the default values for the remaining settings.
- 10. Click **OK**, and then click **OK** again to save the IPsec VPN configuration.



Except for the **Local Address** and **Remote Address** settings, all phase 1 and phase 2 settings on both VPN peers mirror each other. For dial-up IPsec VPN, the local and remote addresses do not have to mirror each other for the tunnel to come up.

Create a Static Route for VPN Traffic on BR1-FGT (Dial-Up Client)

You will create one static route on BR1-FGT. This step was not necessary on HQ-NGFW-1 because, as the dial-up server, it automatically adds the route for the remote network after the tunnel comes up.

To create a static route for VPN traffic on BR1-FGT

1. Continuing on the BR1-FGT GUI, click **Network > Static Routes**.
2. Click **Create New**.
3. Configure the following settings:

Field	Value
Destination	Select Subnet, and then type 10.0.11.0/24.
Interface	ToHQ

New Static Route

Destination ⓘ

SubnetInternet Service

10.0.11.0/24

Interface

ToHQ

×

Administrative Distance ⓘ

10

Comments

Write a comment...0/255

Status

Enabled

Disabled

⊕ Advanced Options

4. Click **OK**.

Create the Firewall Policies for VPN Traffic on BR1-FGT (Dial-Up Client)

You will create two firewall policies between **port4** and **ToHQ**—one for each traffic direction.

To create firewall policies for VPN traffic

1. On the BR1-FGT GUI, click **Policy & Objects > Firewall Policy**.
2. Click **Create New**.
3. Configure the following settings:

Field	Value
Name	HQ_out
Schedule	always
Action	ACCEPT
Incoming Interface	port4
Outgoing Interface	ToHQ
Source	BR1_SUBNET
Destination	HQ_SUBNET
Service	ALL

- 4. In the **Firewall/Network Options** section, disable **NAT**.
- 5. Click **OK**.
- 6. Click **Create New** again.
- 7. Configure the following settings:

Field	Value
Name	HQ_in
Schedule	always
Action	ACCEPT
Incoming Interface	ToHQ
Outgoing Interface	port4
Source	HQ_SUBNET
Destination	BR1_SUBNET
Service	ALL

- 8. In the **Firewall/Network Options** section, disable **NAT**.
- 9. Click **OK**.

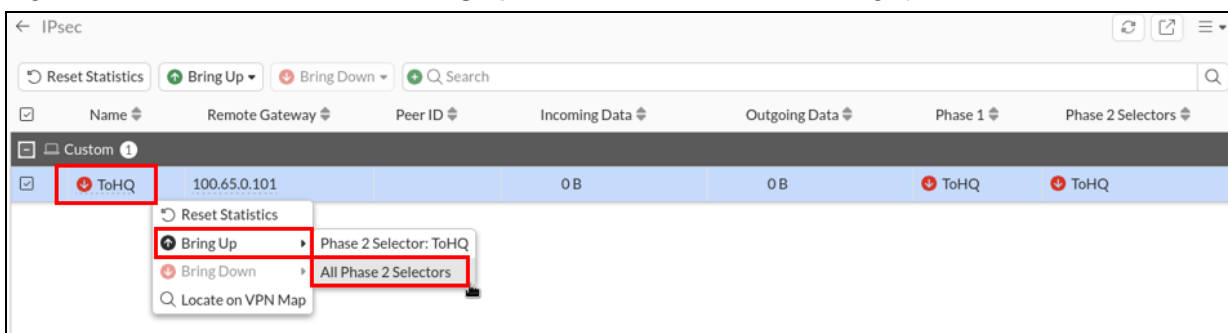
Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT
port4 → port2 1							
port4 → ToHQ 1							
HQ_out (2)	BR1_SUBNET	HQ_SUBNET	always	ALL	✓ ACCEPT		✗ Disabled
ToHQ → port4 1							
HQ_in (3)	HQ_SUBNET	BR1_SUBNET	always	ALL	✓ ACCEPT		✗ Disabled

Test and Monitor the VPN

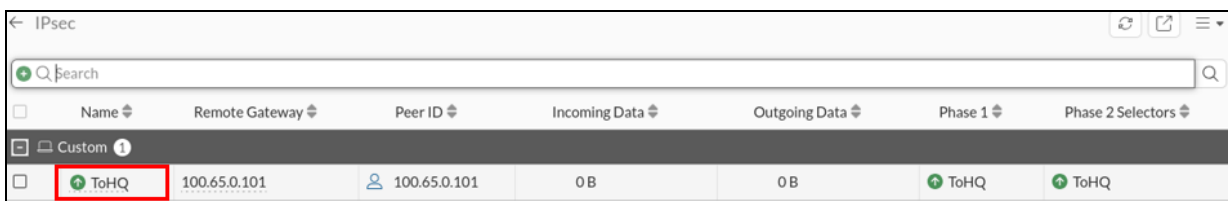
Now that you configured the VPN on both FortiGate devices, you will test the VPN.

To test the VPN

1. On the BR1-FGT GUI, click **Dashboard > Network > IPsec**.
2. Expand **Custom** to see the custom VPN tunnel entry.
The **ToHQ** VPN is currently down.
3. Right-click the VPN, and then select **Bring Up > All Phase 2 Selectors** to bring up the tunnel.



The **Name** column of the VPN now contains a green up arrow, which indicates that the tunnel is up. If required, click the refresh icon in the upper-right corner to refresh the widget information.



Stop and think!

Do you always have to manually bring up the tunnel after you create it?

No. With the current configuration, the tunnel will stay down until you manually bring it up, or there is traffic that should be routed through the tunnel. Because you are not generating traffic between the 172.20.1.0/24 and 10.0.11.0/24 subnets yet, the tunnel is still down. If you had generated the required traffic while the tunnel was down, it would have come up automatically.

You can initiate a tunnel only from BR1-FGT because it is the dial-up client.

4. On the BR1-PC-1 VM, open a terminal window, and then enter the following command to ping the HQ-PC-1 VM:

```
ping 10.0.11.50
```


The ping should work.
5. On the BR1-FGT GUI, click **Dashboard > Network > IPsec**.
6. In the upper-right corner, click the refresh icon multiple times to refresh the widget information.

You will notice that the counters in the **Incoming Data** and **Outgoing Data** columns increase over time. This indicates that the traffic between 10.0.11.50 and 172.20.1.60 is being encrypted successfully and routed through the tunnel.

IPsec						
Search						
	Name	Remote Gateway	Peer ID	Incoming Data	Outgoing Data	Phase 1
Custom						
	ToHQ	100.65.0.101	100.65.0.101	11.77 kB	27.73 kB	ToHQ

- On the HQ-NGFW-1 GUI, click **Dashboard > Network > Static & Dynamic Routing**.
- Find the static route that was dynamically added to the FortiGate.
- View the route details.
- Notice the address listed in the **Gateway IP** column for that route.

	Network	Gateway IP	Interfaces	Distance	Type
	0.0.0.0/0	100.65.0.254	port2	10	Static
	10.0.11.0/24	0.0.0.0	port1	0	Connected
	10.0.12.0/24	0.0.0.0	port5	0	Connected
	10.0.13.0/24	0.0.0.0	port6	0	Connected
	100.65.0.0/24	0.0.0.0	port2	0	Connected
	100.66.0.0/24	0.0.0.0	port3	0	Connected
	172.20.1.0/24	ToBranch		15	Static
	192.168.0.0/16	0.0.0.0	port1	0	Connected

- On the BR1-PC-1 VM, press **Ctrl+C** to stop the ping.

Exercise 2: Configuring a Static IPsec VPN Between Two FortiGate Devices

In this exercise, you will configure a static VPN between HQ-NGFW-1 and BR1-FGT. You will also configure a static route on HQ-NGFW-1 for VPN traffic.

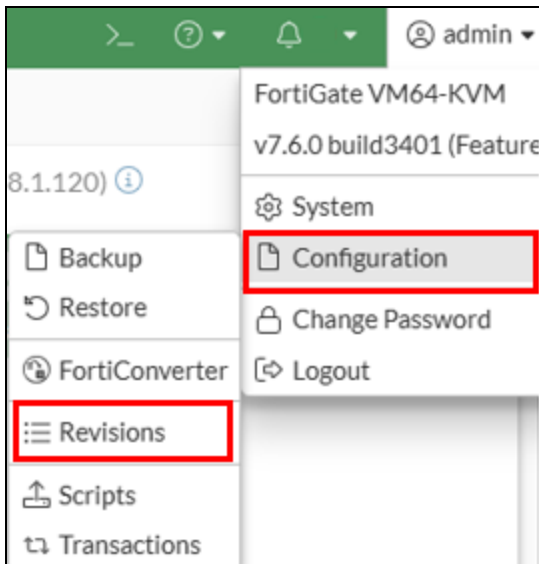
Before you begin this lab, you must restore a configuration file on HQ-NGFW-1.



Make sure that you restore the correct configuration on HQ-NGFW-1, using the following steps. Failure to restore the correct configuration on HQ-NGFW-1 will prevent you from doing the lab exercise.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **HQ-static-ipsec**, and then click **Revert**.

✕ Delete

i Details

Diff

↶ Revert

💾 Save Changes ⚠

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Create Phase 1 and Phase 2 Negotiations on HQ-NGFW-1

You will configure the IPsec VPN by creating phase 1 and phase 2 negotiations.

To create phase 1 and phase 2 negotiations

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
- Username: admin

• Password: Fortinet1!
2. Click **VPN > VPN Tunnels**, and then click **Create New > Custom IPsec tunnel**.
3. Configure the following setting:

Field	Value
Name	ToBranch

4. In the **Network** section, configure the following settings:

Field	Value
Remote Gateway	Static IP Address
IP Address	100.65.1.111
Interface	port2
Dead Peer Detection	On Idle

5. In the **Authentication** section, configure the following settings:

Field	Value
Method	Pre-shared Key
Pre-shared Key	fortinet
Version	1
Mode	Aggressive
Accepted peer ID	Any peer ID

6. In the **Phase 2 Selectors** section, click **Create new**.

7. Configure the following settings:

Field	Value
Name	ToBranch
Local address	Select Subnet Address , and then type 10.0.11.0/24.
Remote address	Select Subnet Address , and then type 172.20.1.0/24.

New Phase 2 Selector

Name: ToBranch

Comments: 0/255

Encapsulation: Tunnel Mode | Transport Mode

IP version: IPv4 | IPv6

Named address: ☐

Local address: IP Address | **Subnet Address** | IP Range
10.0.11.0/24

Remote address: IP Address | **Subnet Address** | IP Range
172.20.1.0/24

8. Keep the default values for the remaining settings.
9. Click **OK**, and then click **OK** to save the IPsec VPN configuration.

Create a Static Route for VPN Traffic on HQ-NGFW-1

You will create one static route on HQ-NGFW-1.

To create a static route for VPN traffic on HQ-NGFW-1

1. Continuing on the HQ-NGFW-1 GUI, click **Network > Static Routes**.
2. Click **Create New**.
3. Configure the following settings:

Field	Value
Destination	Select Subnet , and then type 172.20.1.0/24.
Interface	ToBranch

New Static Route

Destination ⓘ

Subnet

Internet Service

172.20.1.0/24

Interface

ToBranch

+

Administrative Distance ⓘ

10

Comments

Write a comment... 0/255

Status

Enabled

Disabled

Advanced Options

4. Click **OK**.

Create Firewall Policies for VPN Traffic on HQ-NGFW-1

You will create two firewall policies between **port4** and **ToBranch**—one for each traffic direction.

To create firewall policies for VPN traffic

- 1. On the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
- 2. Click **Create New**.
- 3. Configure the following settings:

Field	Value
Name	Branch_out
Schedule	always
Action	ACCEPT
Incoming Interface	port4
Outgoing Interface	ToBranch
Source	HQ_SUBNET
Destination	BR1_SUBNET
Service	ALL

4. In the **Firewall/Network Options** section, disable **NAT**.
5. Click **OK**.
6. Click **Create New** again.
7. Configure the following settings:

Field	Value
Name	Branch_in
Schedule	always
Action	ACCEPT
Incoming Interface	ToBranch
Outgoing Interface	port4
Source	BR1_SUBNET
Destination	HQ_SUBNET
Service	ALL

8. In the **Firewall/Network Options** section, disable **NAT**.
9. Click **OK**.

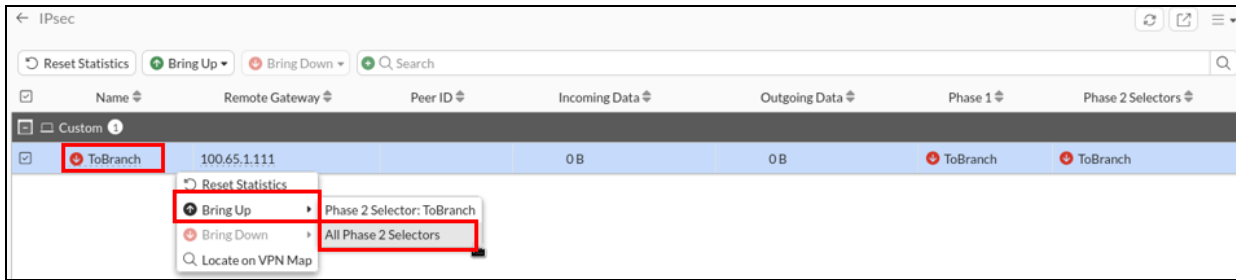
Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT
port4 → port2 1							
port4 → ToBranch 1							
☐ Branch_out (2)	HQ_SUBNET	BR1_SUBNET	always	ALL	✓ ACCEPT		✗ Disabled
ToBranch → port4 1							
☐ Branch_in (3)	BR1_SUBNET	HQ_SUBNET	always	ALL	✓ ACCEPT		✗ Disabled

Test and Monitor the VPN

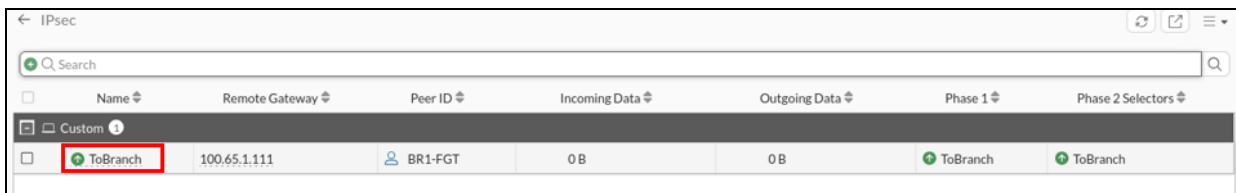
You will test the VPN and monitor its status.

To test the VPN

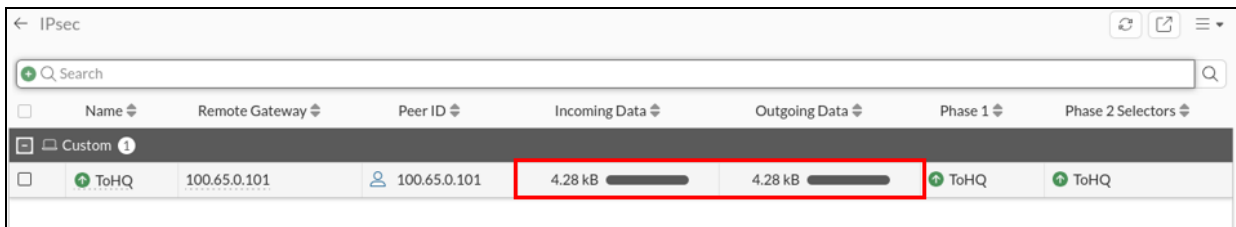
1. On the HQ-NGFW-1 GUI, click **Dashboard > Network > IPsec**.
2. Expand **Custom** to show the custom VPN tunnel entry.
The **ToBranch** VPN is currently down.
3. Right-click the VPN, and then select **Bring Up > All Phase 2 Selectors**.



4. In the upper-right corner, click the refresh icon to refresh the widget information.
The **Name** column of the VPN now contains a green up arrow, which indicates that the tunnel is up.



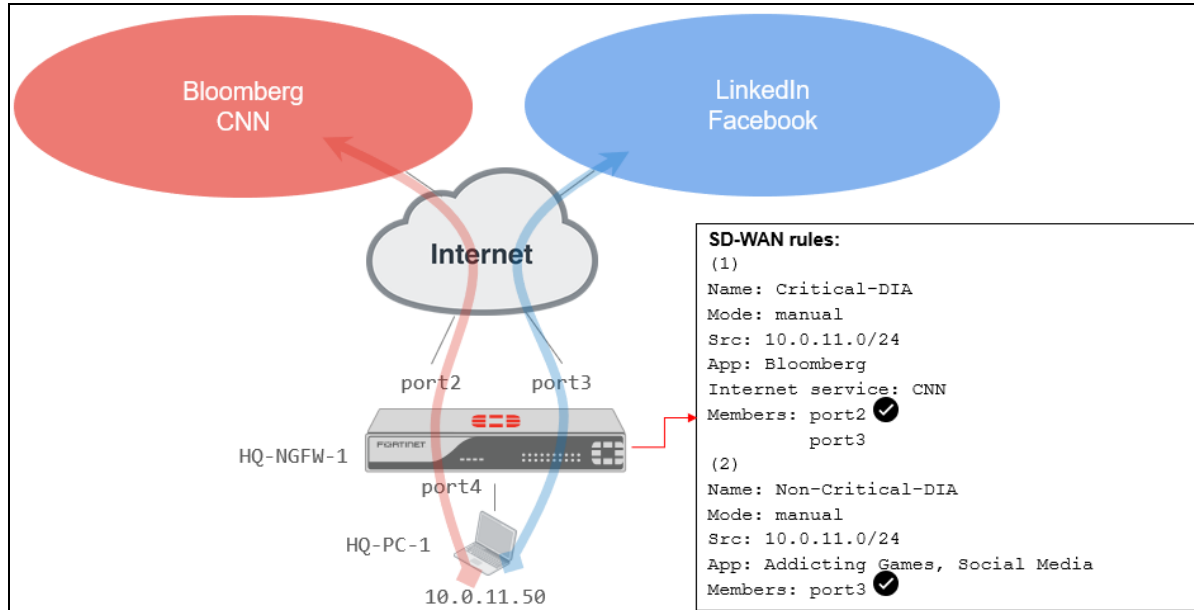
5. On the BR1-PC-1 VM, open a terminal window, and then enter the following command to ping the HQ-PC-1 VM:
ping 10.0.11.50
The ping should work.
6. On the HQ-NGFW-1 GUI, click **Dashboard > Network > IPsec**.
7. In the upper-right corner, click the refresh icon multiple times to refresh the widget information.
The counters in the **Incoming Data** and **Outgoing Data** columns increase over time. This indicates that the traffic between 10.0.11.50 and 172.20.1.60 is being encrypted successfully and routed through the tunnel.



8. On the BR1-PC-1 VM, press **Ctrl+C** to stop the ping.

Lab 12: SD-WAN Configuration

In this lab, you will configure the following basic SD-WAN direct internet access (DIA) setup on HQ-NGFW-1.



Next, you will generate internet traffic on HQ-PC-1 and monitor the traffic distribution and events on the FortiGate GUI.

Objectives

- Configure SD-WAN members and an SD-WAN zone
- Configure routes
- Configure SD-WAN rules for an internet service
- Configure firewall policies
- Verify SD-WAN traffic distribution and events

Time to Complete

Estimated: 30 minutes

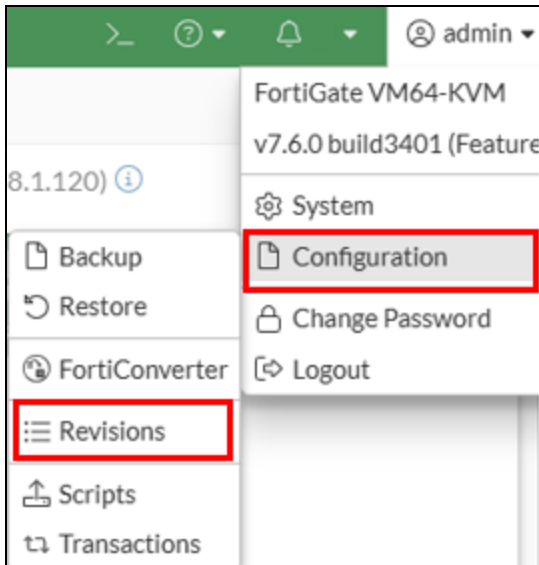
Prerequisites

Before you begin this lab, you must restore a configuration file on HQ-NGFW-1.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!

1. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **HQ-SD-WAN**, and then click **Revert**.

☐ Delete	☐ Details	☐ Diff	☐ Revert	☐ Save Changes ⚠
Config ID	Username	Date	Comments	
7.6.0 build 3401 16				
16	admin	2024/12/13 16:17:45	HQ-diagnostics	
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA	
14	admin	2024/12/13 16:12:36	HQ-SF	
13	admin	2024/12/13 16:08:57	HQ-SD-WAN	
12	admin	2024/12/13 16:06:23	HQ-static-ipsec	
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec	
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN	
9	admin	2024/12/13 15:57:06	HQ-app-control	
8	admin	2024/12/13 15:53:55	HQ-web-filtering	
7	admin	2024/12/13 15:45:29	HQ-AV	
6	admin	2024/12/13 15:41:55	HQ-Certificate	
5	admin	2024/12/13 15:39:27	HQ-FSSO	
4	admin	2024/12/13 15:36:07	HQ-FW-Auth	
3	admin	2024/12/13 15:33:19	HQ-routing	
2	admin	2024/12/13 15:30:41	HQ-firewall-policy	
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial	

5. Click **OK** to reboot.

Exercise 1: Configuring SD-WAN

In this exercise, you will configure a basic DIA setup using the FortiGate GUI. You will create a zone for port2 and port3 on HQ-NGFW-1, and then configure SD-WAN rules to steer traffic for critical and non-critical internet applications.

Remove Interface References

Before you can add port2 and port3 as SD-WAN member interfaces, you must remove all firewall policies that reference the two interfaces.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI (admin/Fortinet1!), remove all firewall policies that reference **port2** and **port3**.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Configure a Zone and Members for DIA](#) on page 190.

To remove interface references

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Policy & Objects > Firewall Policy**.
3. Select the **Internet** policy, and then click **Delete**.



4. Click **OK** to confirm the deletion.

Configure a Zone and Members for DIA

You will configure the underlay zone, and then add port2 and port3 as members.

Take the Expert Challenge!

On the HQ-NGFW-1 GUI (admin/Fortinet1!), complete the following:

- Configure an **SD-WAN Zone** named Underlay.
- Configure SD-WAN members with the following configuration, and add them to the **Underlay** zone:
 - **port2** with **Gateway** 100.65.0.254
 - **port3** with **Gateway** 100.66.0.254

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Configure a Performance SLA on page 192](#)

To create an SD-WAN zone

1. Continuing on the HQ-NGFW-1 GUI, click **Network > SD-WAN > SD-WAN Zones**.
2. Click **Create new > SD-WAN Zone**.
3. In the **Name** field, type Underlay.



New SD-WAN Zone

Name Underlay

Interface members +

4. Click **OK**.
5. Click **Create new > SD-WAN Member**.
6. Configure the following settings:

Field	Value
Interface	port2
SD-WAN Zone	Underlay
Gateway	100.65.0.254
Status	Enabled

7. Click **OK**.
8. Click **Create new > SD-WAN Member**.
9. Configure the following settings:

Field	Value
Interface	port3

Field	Value
SD-WAN Zone	Underlay
Gateway	100.66.0.254
Status	Enabled

- Click **OK**.
- On the **SD-WAN Zones** tab, expand the **Underlay** zone.
The page should look similar to the following example:

+ Create new	+ Search				
☐	Interface	Gateway	Cost	Download	Upload
☐	virtual-wan-link				
☐	Underlay				
☐	port2	100.65.0.254	0	391 bps	118 bps
☐	port3	100.66.0.254	0	0 bps	0 bps



port2 and **port3** are members of the **Underlay** zone.

The default SD-WAN zone **virtual-wan-link** has no members, and therefore is represented with a red icon.

Configure a Performance SLA

You will configure a performance SLA for monitoring the health of port2 and port3.

To configure a performance SLA

- Continuing on the HQ-NGFW-1 GUI, click **Network > SD-WAN**, and then click the **Performance SLAs** tab.
- Click **Create new** to add a performance SLA.
- Configure the following settings:

Field	Value
Name	Level3_DNS
Server	4.2.2.1 Click + , and then type 4 . 2 . 2 . 2 .
Participants	Click Specify , and then select port2 and port3 .

New Performance SLA

Name:

Probe mode: ☒ Active ☐ Passive ☐ Prefer Passive

Protocol: ☒ Ping ☐ HTTP ☐ DNS

Servers:

Participants: ☐ All SD-WAN Members ☒ Specify

☒ port2 ☒ port3

SLA Target: ☐

Link Status

Check interval: ms

Failures before inactive:

Restore link after: check(s)

Actions when Inactive

Update static route: ☒

- Click **OK** to save the settings.
- Click **Network > SD-WAN**, and then click the **Performance SLAs** tab to refresh the page.
The page should show that port2 and port3 are up (green up arrow).

Name	Detect Server	Packet Loss	Latency	Jitter
IPv4 1				
Level3_DNS	4.2.2.1	port2: 17.86%	port2: 17.91ms	port2: 0.18ms
	4.2.2.2	port3: 0%	port3: 18.34ms	port3: 0.76ms

FortiGate can reach the detect server through port2 and port3, and displays a green arrow for **Packet Loss**, **Latency**, and **Jitter**.

- To check the routing table, click **Dashboard > Network**, and then click the **Static & Dynamic Routing** widget.
The page should look similar to the following example:

Network	Gateway IP	Interfaces	Distance	Type
10.0.11.0/24	0.0.0.0	port4	0	Connected
10.0.12.0/24	0.0.0.0	port5	0	Connected
10.0.13.0/24	0.0.0.0	port6	0	Connected
100.65.0.0/24	0.0.0.0	port2	0	Connected
100.66.0.0/24	0.0.0.0	port3	0	Connected
192.168.0.0/16	0.0.0.0	port1	0	Connected
172.20.1.0/24	100.65.0.254	port2	10	Static

Stop and think!

There are no routes to 4.2.2.1 and 4.2.2.2, yet the performance SLA shows that port2 and port3 are up. Why?

To route the health check probes, FortiOS installs special routes in the forwarding information base (FIB) using the gateway information of members. These routes are not displayed in the routing table, but you can enter the `get router info kernel` CLI command to see them.

Configure Rules

You will configure two SD-WAN rules. One rule will be used to steer the traffic of critical applications. The other rule will be used to steer the traffic of non-critical applications. Both rules will use manual mode.



By default, application detection for SD-WAN rules is not visible on the GUI. This feature has been enabled for you on the HQ-NGFW-1 GUI.

The commands to enable the visibility of application detection for SD-WAN rules are:

```
config system global
    set gui-app-detection-sdwan enable
end
```

To configure rules

1. Continuing on the HQ-NGFW-1 GUI, click **Network > SD-WAN**, and then click the **SD-WAN Rules** tab.
2. Click **Create new** to add a rule.
3. Configure the following settings:

Field	Value
Name	Critical-DIA

Field	Value
Source address	HQ_SUBNET
Internet service	Select CNN-Web .
Application	Select Bloomberg .
Outgoing Interfaces	Select Manual .
Interface preference	Select port2 , and then select port3 .



You can combine the traffic detection for **Internet service** and **Application** in the same rule.

When you select **Internet service**, FortiGate uses the Internet Service Database (ISDB) list of IP addresses and protocols for each application. The FortiGuard server regularly updates and loads this list on FortiGate.

When you select **Application**, FortiGate detects the application according to the first packets exchanged with the initial session.

The page should look similar to the following example. Note that you might not be able to view the application icons at this stage.

Priority Rule

Name

Critical-DIA

Status

Enabled

Disabled

Comment

Source

Address

HQ_SUBNET

User group

Destination

Address

Internet service

CNN-Web

Application

Bloomberg

Outgoing Interfaces

Interface selection strategy

Manual

Best quality

Lowest cost (SLA)

Interface preference

port2

port3

4. Click **OK** to save the settings.
5. Repeat the previous steps to configure a rule for non-critical traffic using the following settings:

Field	Value
Name	Non-Critical-DIA
Source address	HQ_SUBNET
Application	Select Apps , and then select Addicting.Games . Select Category , and then select Social.Media .
Outgoing Interfaces	Select Manual .
Interface preference	Select port3 .

Priority Rule

Name:

Status: ☒ Enabled ☐ Disabled

Comment:

Source:

Address:

User group:

Destination:

Address:

Internet service:

Application: ☒ Addicting.Games ☒ Social.Media

Outgoing interfaces:

Interface selection strategy: ☒ Manual ☐ Best quality ☐ Lowest cost (SLA)

Interface preference:

- Click **OK** to save the settings.
 - Continuing on the **SD-WAN Rules** tab, double-click the implicit **sd-wan** rule.
 - Select the **Source-Destination IP** load balancing algorithm.
 - Click **OK** to save the settings.
 - Click **Network > SD-WAN**, and then click the **SD-WAN Rules** tab to refresh the page.
- The page should look similar to the following example:

ID	Name	Comments	Source	Destination	Criteria	Members
IPv4 2						
1	Critical-DIA		HQ_SUBNET	CNN-Web Bloomberg		port2 ☒ port3
2	Non-Critical-DIA		HQ_SUBNET	Addicting.Games Social.Media		port3 ☒
Implicit 1						
	sd-wan		all	all	Source-Destination IP	☐ any

Stop and think!

port2 is the preferred member for rule 1, and **port3** is the preferred member for rule 2. Why?

port2 is the preferred member for rule 1 because it is configured first in the list and it is alive. **port3** is the preferred member for rule 2 because it is the only member for this rule and it is alive.

Configure a Static Route and Firewall Policy

You will configure a static route and firewall policy for routing and allowing SD-WAN traffic. Both objects will reference the underlay zone.

To create a static route for SD-WAN

1. Continuing on the HQ-NGFW-1 GUI, click **Network > Static Routes**.
2. Click **Create new** to add a static route.
3. Configure the following settings:

Field	Value
Destination	Subnet 0.0.0.0/0.0.0.0
Interface	Underlay

4. Click **OK**.

The page should look similar to the following example:

Destination	Gateway IP	Interface	Status
172.20.1.0/24	100.65.0.254	port2	Enabled
0.0.0.0/0		Underlay	Enabled



You don't need to configure a gateway when you use an SD-WAN zone as the outgoing interface for a static route. FortiGate automatically uses the gateway configured for each interface of the SD-WAN zone.

Stop and think!

The static route table shows two routes. The default route through the SD-WAN zone underlay that you have configured, and a static route through port2 that was already configured. Is this a valid configuration?

Yes, this configuration is valid. You can configure static routes for each SD-WAN zone and routes for each SD-WAN member for additional granularity.

To create a firewall policy to allow DIA traffic

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Click **Create new**.
3. Configure the following settings:

Field	Value
Name	DIA
Schedule	always
Action	Accept
Incoming Interface	port4
Outgoing Interface	Underlay
Source	HQ_SUBNET
Destination	all
Service	ALL
NAT	Enable
Security Profiles	Enable Application Control , and then select default . For SSL Inspection , select certificate-inspection .
Logging Options	Enable Log Allowed Traffic , and then select All Sessions .

Name  DIA

Schedule  always

Action ☒ ACCEPT ☐ DENY

Incoming interface  port4

Outgoing interface  Underlay

Source & Destination  Show logic

Source  HQ_SIBNFT

User/group

Destination  all

Service  ALL

Firewall/Network Options

Inspection mode ☒ Flow based ☐ Proxy based

NAT ☒

IP pool configuration ☒ Use Outgoing Interface Address ☐ Use Dynamic IP Pool

Preserve source port ☐

Passive health check ☐

Protocol options ☒ default

Security Profiles

AntiVirus ☐

Web filter ☐

DNS filter ☐

Application control ☒ ☒ default

IPS ☐

File filter ☐

SSL inspection ☒ ☒ certificate-inspection

Logging Options

Log allowed traffic: ☒ Security events ☒ All sessions

Generate logs when session starts ☐

Comments

0/1023

Enable this policy ☒

4. Click **OK** to save the settings.

To review the routing table

1. Open an SSH session to HQ-NGFW-1.
2. Log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
3. Enter the following command to verify the list of active routes in the routing table:
`get router info routing-table all`
4. Verify that both default routes, through **port2** and **port3**, have the same distance value and are active in the routing table.

```
HQ-NGFW-1 # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4
       * - candidate default

Routing table for VRF=0
S*  0.0.0.0/0 [1/0] via 100.65.0.254, port2, [1/0]
    [1/0] via 100.66.0.254, port3, [1/0]
C    10.0.11.0/24 is directly connected, port4
C    10.0.12.0/24 is directly connected, port5
C    10.0.13.0/24 is directly connected, port6
C    100.65.0.0/24 is directly connected, port2
C    100.66.0.0/24 is directly connected, port3
S    172.20.1.0/24 [1/0] via 100.65.0.254, port2, [1/0]
C    192.168.0.0/16 is directly connected, port1
```



After you create a static route for an SD-WAN zone, FortiGate automatically adds individual routes, with the same distance value, for all member interfaces. This ensures that all routes are active in the routing table, which makes them eligible for traffic steering.

Exercise 2: Monitoring the SD-WAN Setup

In this exercise, you will generate internet traffic from the HQ-PC-1 VM. Next, you will monitor DIA traffic distribution and logs using the SD-WAN tools available on the FortiGate GUI.

Generate Internet Traffic From the HQ-PC-1 VM

You will generate internet traffic from the HQ-PC-1 VM using a script.

To generate internet traffic from the HQ-PC-1 VM

1. On the HQ-PC-1 VM, log in with the following credentials:
 - Username: Administrator
 - Password: Fortinet1!
2. Open a terminal window, and then enter the following commands:

```
cd Desktop
sh traffic-generation.sh
```

The output should look similar to the following example:

```
Administrator@ubuntu-2204-desktop:~$ cd Desktop/
Administrator@ubuntu-2204-desktop:~/Desktop$ sh traffic-generation.sh
```



The HQ-PC-1 VM is now generating various types of internet traffic.

3. Open a browser, and then visit a few internet websites including:

```
www.linkedin.com
www.facebook.com
www.cnn.com
```

Monitor DIA Traffic Distribution

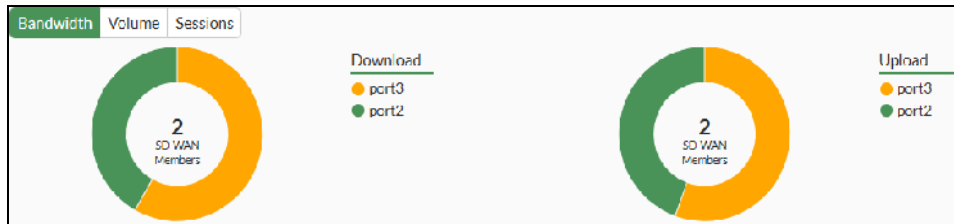
You will use the SD-WAN page on the FortiGate GUI to monitor the DIA traffic distribution. Next, you will view the traffic logs to obtain additional details.

To monitor DIA traffic distribution

1. On the HQ-NGFW-1 GUI, log in with the following credentials:

- Username: admin
 - Password: Fortinet1!
2. Click **Network > SD-WAN**, and then click the **SD-WAN Zones** tab.
 3. Click **Bandwidth** to display SD-WAN distribution graphs based on bandwidth.

The page should look similar to the following example:



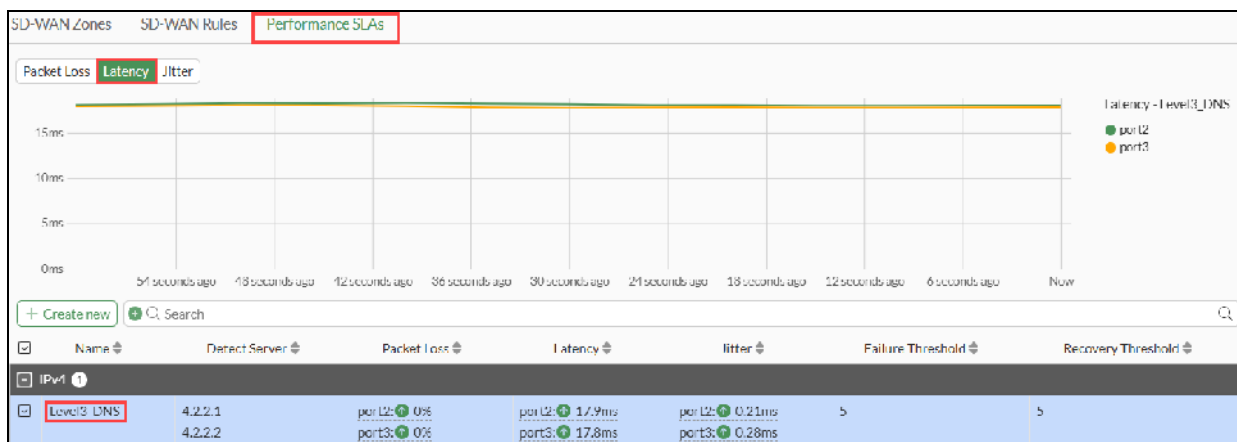
The traffic distribution in this example may be different from yours.

4. Hover over each graph to display the bandwidth that each member (port2 and port3) uses.
5. Click the **Volume** and **Sessions** graphs to explore them.

To monitor the performance SLAs

1. Click the **Performance SLAs** tab.
2. Click **Latency** to display a graph that shows the latency for the performance SLA **Level3_DNS** monitored over the last minute.

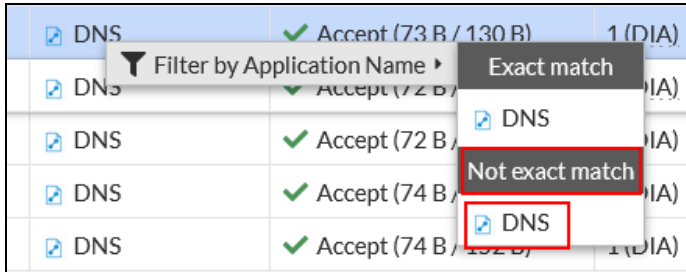
The page should look similar to the following example:



3. Notice the green arrows beside port2 and port3 in the **Packet Loss**, **Latency**, and **Jitter** columns. They indicate that the port is up and that the measured performances are within acceptable values.
4. Click **Packet Loss** and **Jitter** to explore them.

To view SD-WAN traffic logs

1. On the HQ-NGFW-1 GUI, click **Log & Report > Forward Traffic**.
2. Hover over the upper-left corner of the log table to display the table column settings icon.
A gear icon is displayed.
3. Click the gear icon, select **Destination Interface**, select **SD-WAN Quality**, and then select **SD-WAN Rule Name**.
4. Click **Apply**.
5. To simplify the view, beside a DNS log message, right-click **DNS**, and then select the **Not exact match** filter.



The page should look similar to the following example:

Destination	Application Name	Result	Policy ID	Destination...	SD-WAN Quality	SD-WAN Rule Name
31.13.00.36 (facebook.com)	Facebook	✓ Accept (UTM Allowed)	DIA (1)	port3	Seq_num(2 port3 Underlay), alive, selected	Non-Critical-DIA
142.251.32.66 (pagespeed2.googleapis.com)	HTTPBROWSER	✓ Accept (UTM Allowed)	DIA (1)	port3	Seq_num(1 port2 Underlay), alive, selected	Critical-DIA
151.101.67.5 (www.cnn.com)	HTTPBROWSER	✓ Accept (UTM Allowed)	DIA (1)	port2	Seq_num(1 port2 Underlay), alive, selected	Critical-DIA
34.49.241.109 (logsoptimizely.com)	HTTPBROWSER	✓ Accept (UTM Allowed)	DIA (1)	port2		

6. Browse the log table, and then confirm the following:
 - **Bloomberg** and **CNN** traffic matches the **Critical-DIA** rule, and uses **port2**.
 - **Social-Media—Facebook** and **LinkedIn**— traffic matches the **Non-Critical-DIA** rule, and uses **port3**.

Stop and think!

Logs for all other traffic (**DNS**, **HTTP_BROWSER**, and so on) show no information in the **SD-WAN Quality** and **SD-WAN Rule Name** columns. Why?

The traffic doesn't match any of the configured SD-WAN rules. As a result, it matches the implicit SD-WAN rule. The SD-WAN implicit rule load balances sessions based on the source and destination IP addresses and the FIB contents.

Check SD-WAN System Events

You will use the **System Events** log page on the FortiGate GUI to review some messages related to SD-WAN activity.

To monitor SD-WAN system events

1. Continuing on the HQ-NGFW-1 GUI, click **Network > Interfaces**.
2. Right-click **port2**, and then select **Set Status > Down**.
3. Wait about 20 seconds, and then enable **port2**.
4. Click **Log & Reports > System Events**.

- In the upper-right corner, click the time selection list, and then select **1 hour** to display a summary of events in the past hour.



- On the **Summary** page, scroll down, and then click the **SD-WAN Events** summary widget to expand it.

Date/Time	Level	Message	Log Description
2024/11/06 03:34:57	Notice	Service will be redirected in sequence order.	SDWAN status
2024/11/06 03:34:57	Notice	Member link is available. Start forwarding traffic.	SDWAN status
2024/11/06 03:34:56	Notice	SD-WAN health-check member changed state.	SDWAN SLA notification
2024/11/06 03:34:37	Notice	Service will be redirected in sequence order.	SDWAN status
2024/11/06 03:34:37	Notice	Member link is unreachable or miss threshold. Stop forwarding traffic.	SDWAN status
2024/11/06 03:34:36	Warning	SD-WAN health-check member changed state.	SDWAN SLA information warning

- Double-click a few messages to view the details.
- Double-click the first **Service will be redirected in sequence order** message received, and then review the details.

Message	Log Description	Log Details
Service will be redirected in sequence order.	SDWAN status	Other
Member link is available. Start forwarding traffic.	SDWAN status	Log event original timestamp: 17308922176531303200
SD-WAN health-check member changed state.	SDWAN SLA notification	Timezone: -0800
Service will be redirected in sequence order.	SDWAN status	Log ID: 0113022923
Member link is unreachable or miss threshold. Stop forwarding traffic.	SDWAN status	Type: event
SD-WAN health-check member changed state.	SDWAN SLA information warning	Sub Type: sdwan
		Event type: Service
		Service ID: 1
		Sequence: 2

You can see that FortiGate directs the traffic only to member 2 after you switched off port2.

- Double-click the second **Service will be redirected in sequence order** message received, and then review the details.

Message	Log Description	Log Details
Service will be redirected in sequence order.	SDWAN status	Other
Member link is available. Start forwarding traffic.	SDWAN status	Log event original timestamp: 1730892896725911000
SD-WAN health-check member changed state.	SDWAN SLA notification	Timezone: -0800
Service will be redirected in sequence order.	SDWAN status	Log ID: 0113022923
Member link is unreachable or miss threshold. Stop forwarding traffic.	SDWAN status	Type: event
SD-WAN health-check member changed state.	SDWAN SLA information warning	Sub Type: sdwan
		Event type: Service
		Service ID: 1
		Sequence: 1,2

When both interfaces are back up, the member sequence order corresponds to the configured order—member 1, and then member 2.

- In the HQ-NGFW-1 terminal window, press **Ctrl+C** to stop `traffic-generation.sh`.
- Close the HQ-NGFW-1 GUI session and the terminal window on the HQ-PC-1 VM.

Lab 13: Security Fabric

In this lab, you will learn how to configure the Fortinet Security Fabric. After you configure the Security Fabric, you will access the physical and logical topology views.

Objectives

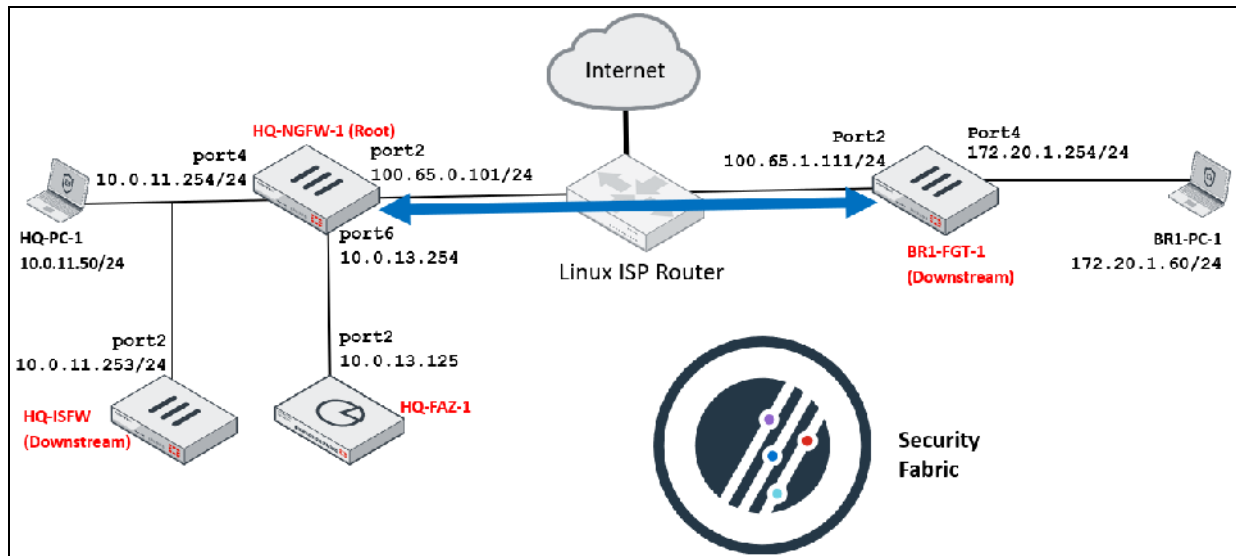
- Configure the Security Fabric on HQ-NGFW-1 (root) and HQ-ISFW (downstream)
- Configure the Security Fabric on HQ-NGFW-1 (root) and BR1-FGT-1 (downstream)

Time to Complete

Estimated: 30 minutes

Topology

In this lab, you will learn how to configure the Security Fabric on all FortiGate devices in the topology. HQ-NGFW-1 and BR1-FGT-1 are connected through an IPsec tunnel. HQ-NGFW-1 is the root FortiGate in the Security Fabric, and BR1-FGT-1 and HQ-ISFW are downstream FortiGate devices. FortiAnalyzer is behind HQ-NGFW-1 and will be used in the Security Fabric.



Prerequisites

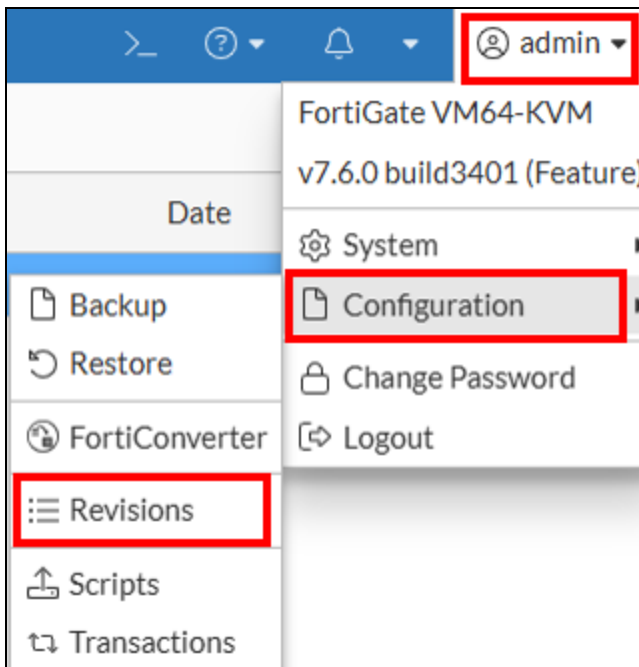
Before you begin this lab, you must restore configuration files on BR1-FGT-1, HQ-NGFW-1, and HQ-ISFW.



Make sure you restore the correct configuration file on each FortiGate, using the following steps. Failure to restore the correct configuration file on each FortiGate will prevent you from doing the lab exercises.

To restore the BR1-FGT-1 configuration file

1. Connect to the BR1-FGT-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



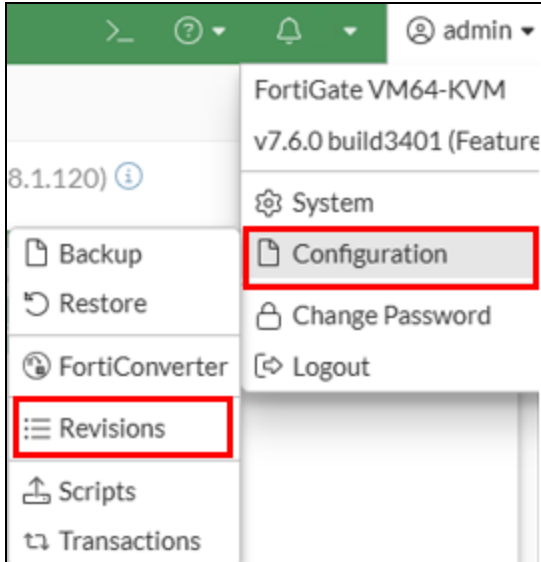
3. Click **+** to expand the list.
4. Select the configuration with the comment **BR1-SF**, and then click **Revert**.

Delete	Details	Diff	Revert	Save Changes
Config ID	Username	Date	Comments	
7.6.0 build 3401 3				
3	admin	2024/12/13 17:56:40	BR1-SF	
2	admin	2024/12/13 17:54:11	BR-dialup-ipsec	
1	admin	2024/12/13 17:51:50	BR1-FGT-initial	

5. Click **OK** to reboot.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



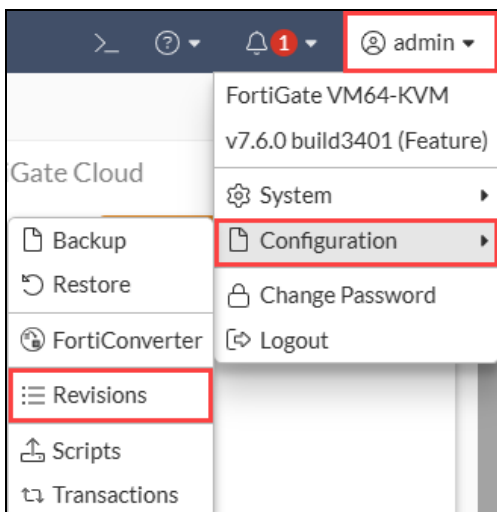
3. Click + to expand the list.
4. Select the configuration with the comment **HQ-SF**, and then click **Revert**.

Delete Details Diff Revert Save Changes ⚠			
Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

To restore the HQ-ISFW configuration file

1. Connect to the HQ-ISFW GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **ISFW-SF**, and then click **Revert**.

Delete Details Diff Revert Save Changes			
Config ID	Username	Date	Comments
7.6.0 build 3401 2			
2	admin	2024/12/16 10:35:45	ISFW-SF
1	admin	2024/12/16 10:33:24	HQ-ISFW-initial

5. Click **OK** to reboot.

Exercise 1: Configuring the Security Fabric on HQ-NGFW-1 and HQ-ISFW

In this exercise, you will configure the Security Fabric between HQ-NGFW-1 (root) and HQ-ISFW (downstream).

Configure FortiAnalyzer Logging on HQ-NGFW-1 (Root)

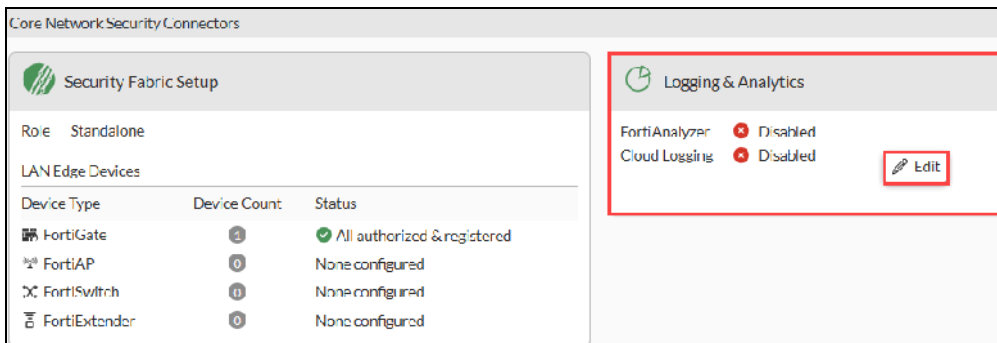
You will configure the root of the Security Fabric to send all logs to FortiAnalyzer. These settings will be automatically replicated to all downstream devices when they become members of the Security Fabric.



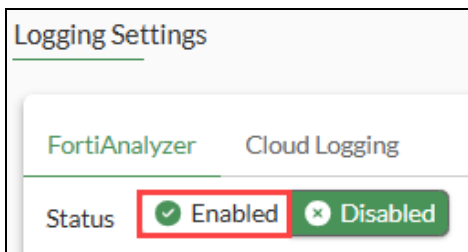
For this lab, FortiAnalyzer is already preconfigured to accept the registration requests that originate from all FortiGate devices in the topology.

To configure HQ-NGFW-1 to send logs to FortiAnalyzer

1. Log in to the HQ-NGFW-1 GUI with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **Security Fabric > Fabric Connectors**.
3. Select **Logging & Analytics**, and then click **Edit**.



4. In the **Logging Settings** window, on the **FortiAnalyzer** tab, in the **Status** field, select **Enabled**.



5. Edit the settings so they match the following image:

Logging Settings

FortiAnalyzer Cloud Logging

Status ☒ Enabled ☐ Disabled

Server 10.0.13.125

Connection status  Refresh

Upload option ☒ Real Time ☐ Every Minute ☐ Every 5 Minutes

Allow access to FortiGate REST API ☒

Verify FortiAnalyzer certificate ☒

6. Click **OK**.
7. In the verification window that appears, click **Accept**.

Verify FortiAnalyzer Serial Number

The FortiAnalyzer's access to the FortiGate's REST API will be authenticated by the FortiAnalyzer certificate. The serial number from the certificate must match the serial number observed on the FortiAnalyzer.

⚠ The obtained serial number from the FortiAnalyzer certificate is: **FAZ-VMTM24012176**

Do you wish to accept the serial number and certificate—verifying that they match the correct FortiAnalyzer?

Accept Deny

8. In **Security Fabric > Fabric Connectors > Logging & Analytics**, ensure that the value in the **FortiAnalyzer** field is **Connected**.

Logging & Analytics

FortiAnalyzer ☒ Connected

Cloud Logging ☐ Disabled

Configure the Security Fabric on HQ-NGFW-1 (Root)

You will configure HQ-NGFW-1 as the root FortiGate.

To enable the Security Fabric on HQ-NGFW-1

1. On the HQ-NGFW-1 GUI, log in with the following credentials:

- Username: admin
 - Password: Fortinet1!
2. Click **Security Fabric > Fabric Connectors**.
 3. Click **Security Fabric Setup**, and then click **Edit**.
 4. In the **Security Fabric Settings** section, in the **Security Fabric role** field, select **Serve as Fabric Root**.
 5. Configure the following settings:

Field	Value
Allow other Security Fabric devices to join (ensure both interfaces are selected)	Enabled port4, To-BR1-FGT-1
Fabric name	fortinet

The configuration should look like the following example:

Security Fabric Settings

Security Fabric role: Standalone **Serve as Fabric Root** Join Existing Fabric

Allow other Security Fabric devices to join: ☒ port4 To-BR1-FGT-1

Fabric name: fortinet

Allow downstream device REST API access: ☐

Fabric global object: ☒

Management IP/FQDN: Use WAN IP Specify

Management port: Use Admin Port Specify 443

SAML SSO Settings

SAML Single Sign-On: ☐ Advanced Options

6. Click **OK**.

To verify the Security Fabric connection on HQ-NGFW-1 interfaces

1. On the HQ-NGFW-1 GUI, click **Network > Interfaces**.
2. Click **port4**, and then click **Edit**.
3. In the **Administrative Access** section, confirm that the **Security Fabric Connection** checkbox is selected.
4. In the **Network** section, enable **Device detection**.

The configuration should look like the following example:

Administrative Access

IPv4

- ☐ SCIM
- ☒ PING
- ☐ SNMP
- ☒ Security Fabric Connection ⓘ
- ☒ HTTPS
- ☐ FMG-Access
- ☐ FTM
- ☐ Speed Test
- ☒ HTTP ⓘ
- ☒ SSH
- ☐ RADIUS Accounting

Receive LLDP ⓘ Use VDOM Setting Enable Disable

Transmit LLDP ⓘ Use VDOM Setting Enable Disable

☐ DHCP Server

Network

Device detection ⓘ ☒

5. Click **OK**.
6. Click **Network > Interfaces**, and then expand **port2**.
7. Click the **To-BR1-FGT-1** interface, and then click **Edit**.
8. In the **Administrative Access** section, confirm that the **Security Fabric Connection** checkbox is selected.
9. Click **OK**.

Stop and think!

Why is the **Security Fabric Connection** checkbox already selected on the **port4** and **To-BR1-FGT-1** interfaces on HQ-NGFW-1?

When you add the interfaces in the **Allow other Security Fabric devices to join** field, the **Security Fabric Connection** checkbox is automatically selected on those interfaces.

Configure the Security Fabric on HQ-ISFW

You will configure HQ-ISFW to join the Security Fabric as a downstream FortiGate.

Take the Expert Challenge!

On the HQ-ISFW GUI, enable the **Security Fabric** settings, and connect ISFW to the Security Fabric that has HQ-NGFW-1 as its root device. After, you enable the **Security Fabric** settings, enable **Network Device detection** on port4, and verify that **Security Fabric Connection** is selected on the port.

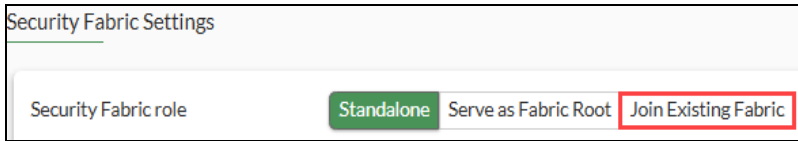
If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Authorize HQ-ISFW \(Downstream\) on HQ-NGFW-1 \(Root\) on page 215](#).

To enable the Security Fabric on HQ-ISFW (downstream)

1. On the HQ-ISFW GUI, log in with the following credentials:

- Username: admin
 - Password: Fortinet1!
2. Click **Security Fabric > Fabric Connectors**.
 3. Click **Security Fabric Setup**, and then click **Edit**.
 4. In the **Security Fabric Settings** section, in the **Security Fabric role** field, select **Join Existing Fabric**.



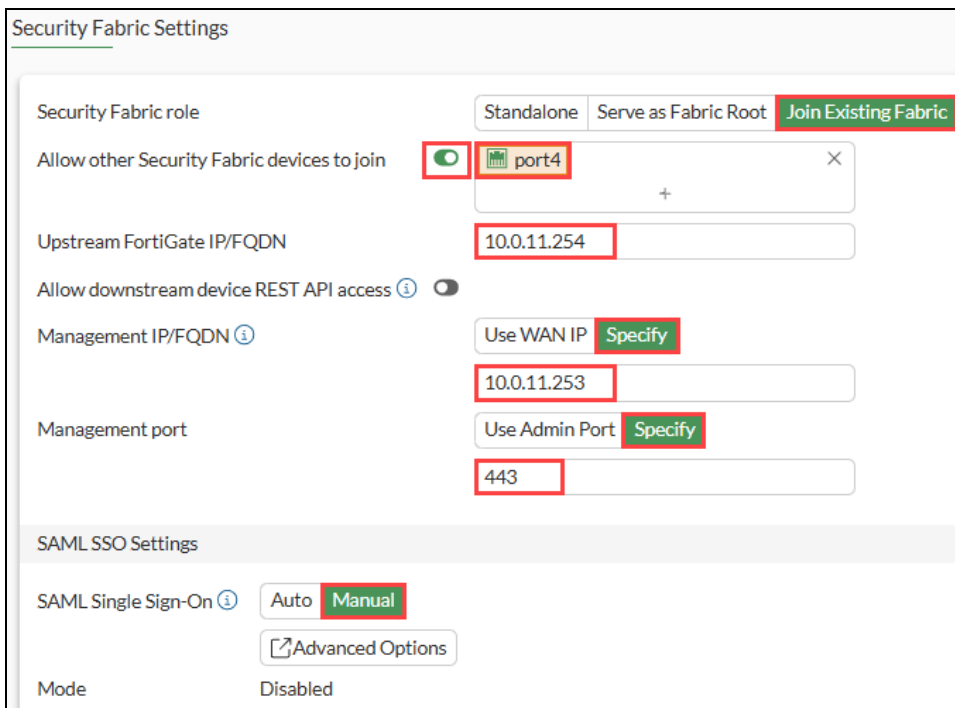
Security Fabric Settings

Security Fabric role

Standalone Serve as Fabric Root **Join Existing Fabric**

5. Enable **Allow other Security Fabric devices to join**.
6. In the **Allow other Security Fabric devices to join** field, select **port4**.
7. In the **Upstream FortiGate IP/FQDN** field, type 10.0.11.254.
8. In the **Management IP/FQDN** field, select **Specify**, and then type 10.0.11.253.
9. In the **Management port** field, select **Specify**, and then type 443.
10. In the **SAML Single Sign-On** field, select **Manual**.

The configuration should look like the following example:



Security Fabric Settings

Security Fabric role

Standalone Serve as Fabric Root **Join Existing Fabric**

Allow other Security Fabric devices to join ☒ **port4** X

Upstream FortiGate IP/FQDN **10.0.11.254**

Allow downstream device REST API access ☐

Management IP/FQDN **Specify**

10.0.11.253

Management port **Specify**

443

SAML SSO Settings

SAML Single Sign-On **Manual**

Mode Disabled

11. Click **OK**.
12. Click **OK** to confirm the settings.



FortiAnalyzer logging is enabled on HQ-ISFW after the Security Fabric is enabled. Downstream FortiGate devices retrieve FortiAnalyzer settings from the root FortiGate when they join the Security Fabric.

To verify the Security Fabric connection on the HQ-ISFW interface

1. On the HQ-ISFW GUI, click **Network > Interfaces**.
2. Click **port4**, and then click **Edit**.
3. In the **Administrative Access** section, confirm that the **Security Fabric Connection** checkbox is selected.
4. In the **Network** section, enable **Device detection**.

Administrative Access

IPv4

☐ SCIM

☒ PING

☐ SNMP

☒ Security Fabric Connection ⓘ

☒ HTTPS

☐ FMG-Access

☐ FTM

☐ Speed Test

☒ HTTP ⓘ

☒ SSH

☐ RADIUS Accounting

Receive LLDP ⓘ

Use VDOM Setting

Enable

Disable

Transmit LLDP ⓘ

Use VDOM Setting

Enable

Disable

☐ DHCP Server

Network

Device detection ⓘ ☒

5. Click **OK**.

Authorize HQ-ISFW (Downstream) on HQ-NGFW-1 (Root)

You will authorize HQ-ISFW on HQ-NGFW-1 to join the Security Fabric.

To authorize HQ-ISFW on HQ-NGFW-1

1. On the HQ-NGFW-1 GUI, click **System > Firmware & Registration**.
It may take a few minutes to appear. You may need to refresh the page, or log out of the HQ-NGFW-1 GUI and log in again.
2. Click the HQ-ISFW FortiGate serial number, and then click **Authorize**.

	Device	Status
☐	HQ-NGFW-1	Online
☒	FGVM02TM24013502	Unauthorized

Upgrade

Register

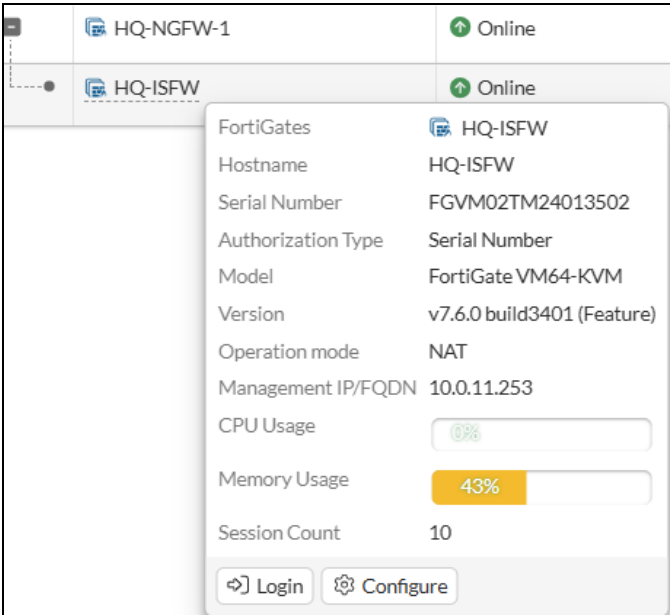
Authorize

It may take a few minutes to authorize. You may need to refresh the page, or log out of the HQ-NGFW-1 GUI and log in again.



After authorization, HQ-ISFW appears in the Security Fabric topology section, which means HQ-ISFW joined the Security Fabric successfully.

3. Hover over the **HQ-ISFW** icon to display a summary of the firewall settings, and then verify that it is correctly registered in the Security Fabric.

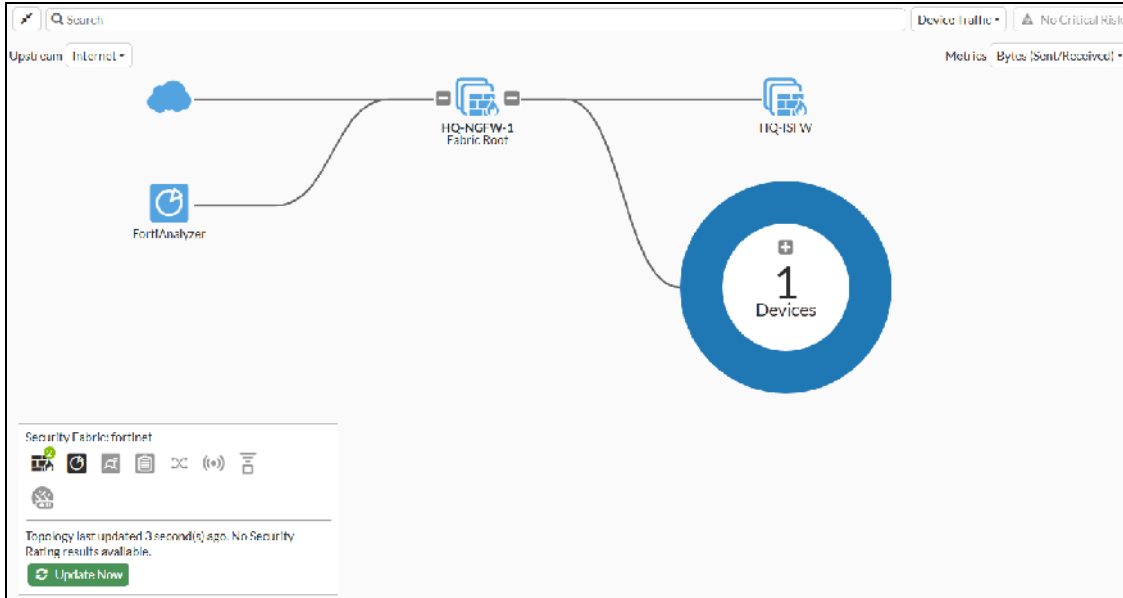


Check the Security Fabric Deployment Result

You will check the Security Fabric deployment result on HQ-NGFW-1 (root).

To check the Security Fabric on HQ-NGFW-1

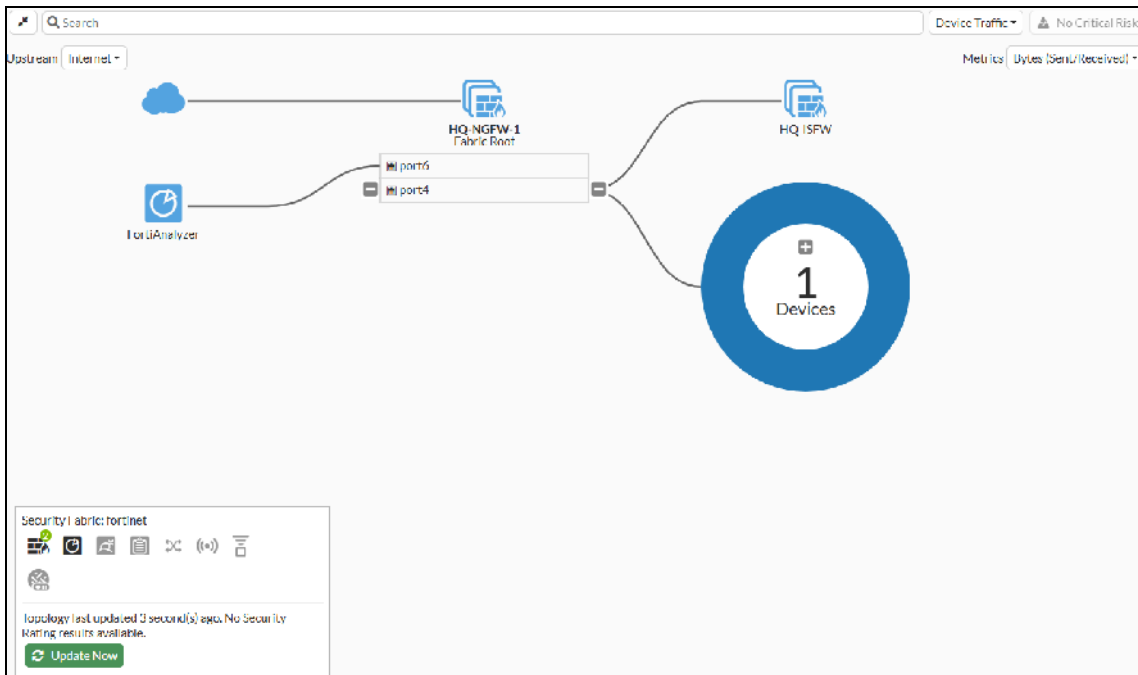
1. On the HQ-PC-1 VM, open a new browser, and then go to <https://www.fortinet.com>.
This is to generate some traffic from the HQ-PC-1 VM so it is included in the topology views.
2. On the HQ-NGFW-1 GUI, click **Security Fabric > Physical Topology**.
3. Click **Update Now** to update the topology view.
This page shows a visualization of access layer devices in the Security Fabric.



Your topology view might not match exactly what is shown in this example.

4. On the HQ-NGFW-1 GUI, click **Security Fabric > Logical Topology**.

This dashboard displays information about the interfaces that connect each device in the Security Fabric.



Exercise 2: Configuring the Security Fabric on HQ-NGFW-1 and BR1-FGT-1

In this exercise, you will add another FortiGate to the Security Fabric. In this topology, the downstream BR1-FGT-1 connects to the root HQ-NGFW-1 over IPsec VPN to join the Security Fabric.

Take the Expert Challenge!

On the BR1-FGT-1 GUI, enable the **Security Fabric** settings, and connect BR1-FGT-1 to the Security Fabric using the tunnel IP address 10.10.10.1. After you enable the **Security Fabric** settings, verify that **Security Fabric Connection** is selected on port4.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Authorize BR1-FGT-1 \(Downstream\) on HQ-NGFW-1 \(Root\)](#) on page 219.

Configure the Security Fabric on BR1-FGT-1 (Downstream)

You will configure BR1-FGT-1 to join the Security Fabric as a downstream FortiGate over the IPsec VPN.

To enable the Security Fabric on BR1-FGT-1

1. On the BR1-FGT-1 GUI, log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **Security Fabric > Fabric Connectors**.
3. Click **Security Fabric Setup**, and then click **Edit**.
4. In the **Security Fabric role** field, select **Join Existing Fabric**.
5. Enable **Allow other Security Fabric devices to join**.
6. In the **Allow other Security Fabric devices to join** field, select **port4**.
7. In the **Upstream FortiGate IP/FQDN** field, type 10.10.10.1.
8. In the **Management IP/FQDN** field, click **Specify**, and then type 10.10.10.3.
9. In the **Management port** field, select **Specify**, and then type 443.
10. In the **SAML Single Sign-On** field, select **Manual**.

The configuration should look like the following example:

Security Fabric Settings

Security Fabric role: Standalone | Serve as Fabric Root | **Join Existing Fabric**

Allow other Security Fabric devices to join: ☒ **port4**

Upstream FortiGate IP/FQDN: **10.10.10.1**

Allow downstream device REST API access: ☐

Management IP/FQDN: Use WAN IP | **Specify** | **10.10.10.3**

Management port: Use Admin Port | **Specify** | **443**

SAML SSO Settings

SAML Single Sign-On: Auto | **Manual**

Advanced Options

Mode: Disabled

11. Click **OK**.
12. Click **OK** to confirm.

To verify the Security Fabric connection on the BR1-FGT-1 interface

1. Click **Network > Interfaces**.
2. Click **port4**, and then click **Edit**.
3. In the **Administrative Access** section, confirm that the **Security Fabric Connection** checkbox is selected.
4. In the **Network** section, enable **Device detection**.
5. Click **OK**.

Authorize BR1-FGT-1 (Downstream) on HQ-NGFW-1 (Root)

You will authorize BR1-FGT-1 on HQ-NGFW-1 to join the Security Fabric.

To authorize BR1-FGT-1 on HQ-NGFW-1

1. On the HQ-NGFW-1 GUI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Click **System > Firmware & Registration**.
3. Click the BR1-FGT-1 serial number, and then click **Authorize**.

	HQ-NGFW-1	Online
	HQ-ISFW	Online
	FGVM02TM24013504	Unauthorized

It may take a few minutes to authorize. You may need to refresh the page, or log out of the HQ-NGFW-1 GUI and log in again.

After authorization, BR1-FGT-1 appears in the topology. Now, both HQ-ISFW and BR1-FGT-1 are shown as downstream devices of the root, HQ-NGFW-1. The configuration should look like the following example:



Device	Status	Registration Status
HQ-NGFW-1	Online	Registered
HQ-ISFW	Online	Registered
BR1-FGT-1	Online	Registered

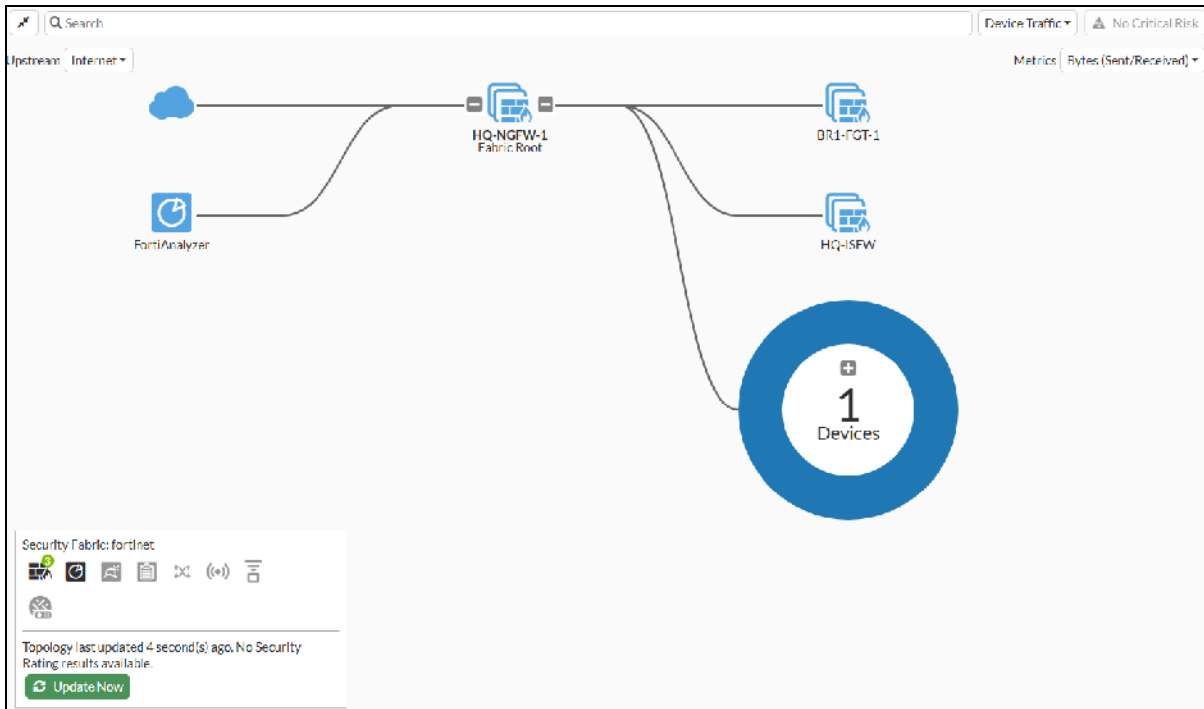
You may need to refresh the page to match the image above.

Check the Security Fabric Deployment Result

You will check the Security Fabric deployment result on the root, HQ-NGFW-1.

To check the Security Fabric on HQ-NGFW-1

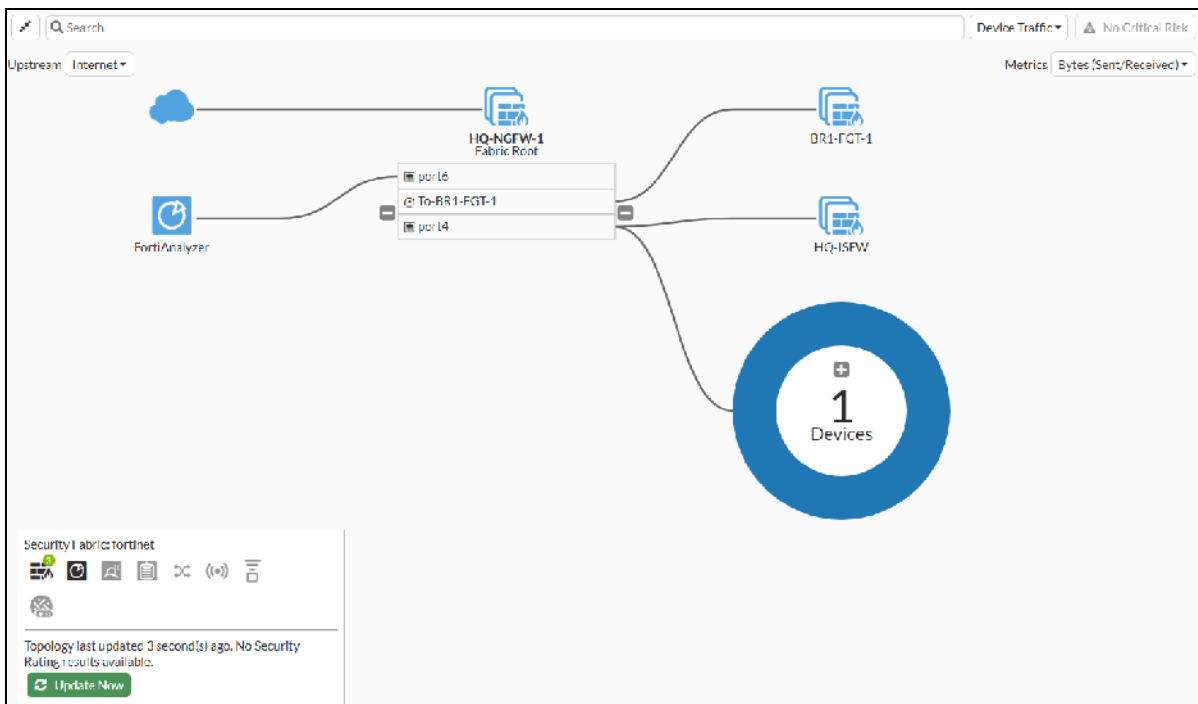
- On the HQ-NGFW-1 GUI, click **Security Fabric > Physical Topology**.
This page shows a visualization of access layer devices in the Security Fabric.



You may need to click the **Update Now** button to refresh the topology. Your topology view might not match what is shown in this example.

2. Click **Security Fabric > Logical Topology**.

This dashboard displays information about the interfaces that each device in the Security Fabric connects to.



You may need to click the **Update Now** button to refresh the topology.



Your topology view might not match what is shown in this example. At a minimum, you should see HQ-NGFW-1, BR1-FGT-1, and HQ-ISFW in the topology view.

You can generate some traffic from the BR1-PC-1 VM to have it shown in the topology.

Lab 14: High Availability

In this lab, you will examine how to set up a FortiGate Clustering Protocol (FGCP) high availability (HA) cluster of FortiGate devices. You will explore active-passive HA mode and observe FortiGate HA behavior. You will also perform an HA failover and use diagnostic commands to observe the election of a new primary device in the cluster. Finally, you will configure management ports on FortiGate devices to reach each FortiGate individually for management purposes.

Objectives

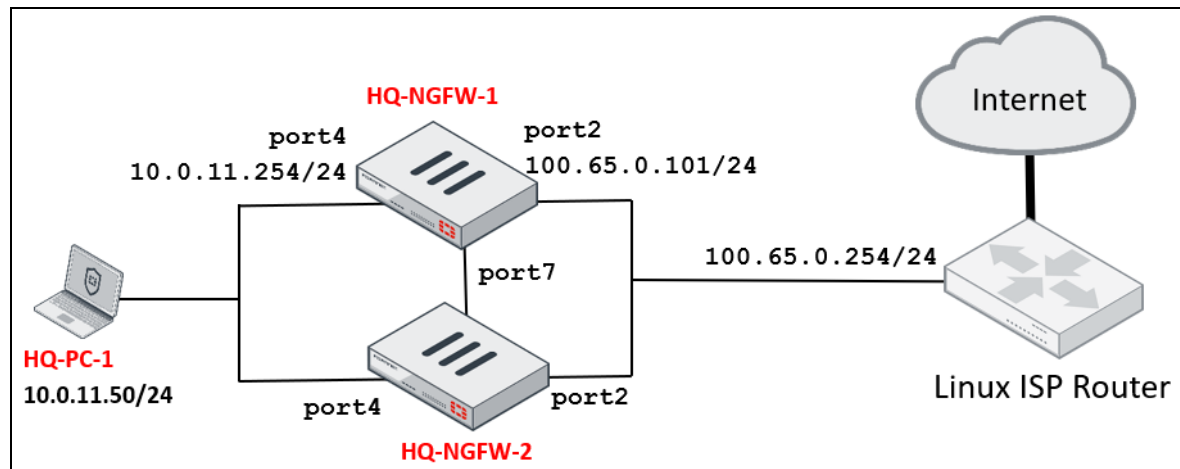
- Set up an HA cluster using FortiGate devices
- Observe HA synchronization and interpret diagnostic output
- Perform an HA failover
- Manage individual cluster members by configuring a reserved management interface

Time to Complete

Estimated: 40 minutes

Lab HA Topology

After you upload the required configurations to each FortiGate, the logical topology will change to the following:



Prerequisites

Before you begin this lab, you must restore a configuration file to HQ-NGFW-1 and HQ-NGFW-2.

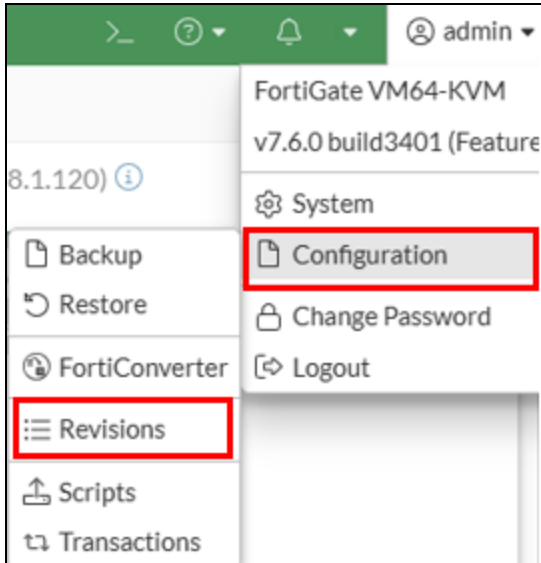


You *must* restore the configuration file on HQ-NGFW-1 *first*, and then on HQ-NGFW-2.

If you restore the configuration file on HQ-NGFW-2 first, it will produce unintended results and may prevent you from doing the lab exercises.

To restore the HQ-NGFW-1 configuration file

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click + to expand the list.
4. Select the configuration with the comment **HQ-NGFW-1-HA**, and then click **Revert**.

Delete

Details

Diff

Revert

Save Changes

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

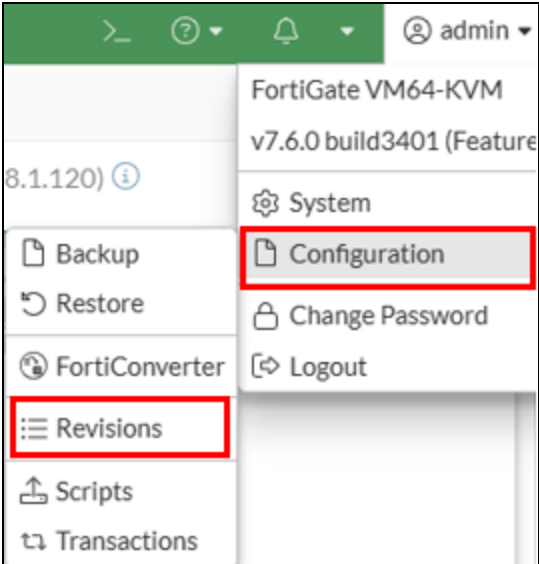
5. Click **OK** to reboot.



You must wait until HQ-NGFW-1 finishes rebooting before you can restore the configuration on HQ-NGFW-2 in the next step.

To restore the HQ-NGFW-2 configuration file

1. Connect to the HQ-NGFW-2 GUI, and then log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click + to expand the list.
4. Select the configuration with the comment **HQ-NGFW-2-HA**, and then click **Revert**.

✕ Delete i Details Diff ↺ Revert Save Changes ⚠			
Config ID	Username	Date	Comments
7.6.0 build 3401 2			
2	admin	2024/12/16 11:52:52	HQ-NGFW-2-HA
1	admin	2024/12/16 11:49:59	HQ-NGFW-2-initial

5. Click **OK** to reboot.

Exercise 1: Configuring HA

FortiGate HA uses FGCP, which uses a heartbeat link for HA-related communications to discover other FortiGate devices in the same HA group, elect a primary device, synchronize configuration, and detect failed devices in an HA cluster.

In this exercise, you will examine how to configure HA settings on both FortiGate devices. You will observe the HA synchronization status, and use `diagnose` commands to verify that the configuration is in sync on both FortiGate devices.



Unless instructed otherwise, always use the console connection of HQ-NGFW-1 and HQ-NGFW-2 to access the CLI. This ensures that you can access the CLI of the device regardless of the HA role.

Configure HA Settings on HQ-NGFW-1

You will configure HA-related settings using the HQ-NGFW-1 GUI.

To configure HA settings on HQ-NGFW-1

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **System** > **HA**, and then configure the following HA settings:

Field	Value
Mode	Active-Passive
Device priority	200
Group ID	5
Group name	Training
Password	Fortinet Tip: Click Change , and then type the password.
Session pickup	<enable>
Monitor interfaces	Click X to remove any ports that are selected.
Heartbeat interfaces	Click X to remove port4 , and then select port7 .

The configuration should look like the following example:

High Availability

Mode: Active-Passive

Device priority: 200

Increase priority effect: ☐

Cluster Settings

Group ID: 5

Group name: Training

Password: Fortinet

Session pickup: ☒

Monitor interfaces: +

Heartbeat interfaces: port7

☐ Management Interface Reservation

☐ Unicast Heartbeat

3. Click **OK**.

Configure HA Settings on HQ-NGFW-2

You will configure HA-related settings on HQ-NGFW-2, using the console.

To configure HA settings on HQ-NGFW-2

1. Connect to the HQ-NGFW-2 CLI, and then log in with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Enter the following commands:

```
config system ha
  set mode a-p
  set group-name Training
  set group-id 5
  set password Fortinet
  set hbdev port7 0
  set session-pickup enable
  set override disable
  set priority 100
end
```

Observe and Verify the HA Synchronization Status

Now that you have configured HA on both FortiGate devices, you will verify that HA is established and that the configurations are fully synchronized.

The checksums for all cluster members must match for the FortiGate devices to be synchronized.

To observe and verify the HA synchronization status

1. On the HQ-NGFW-2 CLI, notice the debug messages about the HA synchronization process.

These messages sometimes display useful status change information.

2. Wait 4–5 minutes for the FortiGate devices to synchronize.

After the FortiGate devices are synchronized, the HQ-NGFW-2 device logs out all admin users.

```
secondary starts to sync with primary
logout all admin users
```

3. When prompted, log back in to the NGFW-2 CLI with the following credentials:

- Username: admin
- Password: Fortinet1!

4. Enter the following command to check the HA synchronization status:

```
diagnose sys ha checksum show
```

5. On the HQ-NGFW-1 CLI, enter the following command to check the HA synchronization status:

```
diagnose sys ha checksum show
```

6. Compare the output from both FortiGate devices.

If both FortiGate devices are synchronized, the checksums match.

7. Alternatively, you can run the following CLI command on any member to view the checksums of all members:

```
diagnose sys ha checksum cluster
```

Verify FortiGate Roles in an HA Cluster

After the checksums of both FortiGate devices match, you will verify the cluster member roles to confirm the primary and secondary devices.

To verify FortiGate roles in an HA cluster

1. On both the HQ-NGFW-1 CLI and HQ-NGFW-2 CLI, enter the following command to verify that the HA cluster is established:

```
get system status
```

2. On both FortiGate devices, view the `Current HA mode` line, and then write down the device serial number (Serial-Number).

Notice that HQ-NGFW-1 is a-p, primary and HQ-NGFW-2 is a-p, secondary.

Stop and think!

Why was HQ-NGFW-1 elected as the primary?

In the primary election process, FGCP first checks the number of connected monitored ports. Because you didn't configure monitored ports, FGCP then checks the next criterion.

As the override setting is disabled, FGCP checks the HA uptime next. Because you enabled HA on both devices about the same time, the HA uptime difference is less than 5 minutes.

Therefore, FGCP checks the next criterion, which is priority.

HQ-NGFW-1 has a priority of 200, which is greater than HQ-NGFW-2, which has a priority of 100. The result is that FGCP elects HQ-NGFW-1 as the primary.

3. On the HQ-NGFW-1 CLI, enter the following command to confirm the reason for the primary election:

```
get system ha status
```

4. In the output, look for the `Primary selected using` section to identify the reason for the latest primary election event.

The output should look similar to the following example:

Primary selected using:

```
<2024/11/08 07:30:37> vcluster-1: FGVM02TM24013423 is selected as the primary because its override priority is larger than peer member FGVM02TM24013501.
```



The output confirms that FGCP elected HQ-NGFW-1 as the primary because of its higher priority.

If you see that the election reason is a higher uptime, then that is probably because you rebooted one of the members, and as a result, the HA uptime of that device was reset. The reboot then caused the HA uptime difference to be more than 5 minutes.

Exercise 2: Triggering an HA Failover

You set up an HA cluster. In this exercise, you will examine how to trigger an HA failover, and observe the renegotiation among devices to elect a new primary device.



Unless instructed otherwise, always use the console connection of HQ-NGFW-1 and HQ-NGFW-2 to access the CLI. This ensures that you can access the device CLI regardless of the HA role.

Trigger a Failover by Rebooting the Primary FortiGate

You will reboot the primary FortiGate in the cluster to trigger a failover.

Take the Expert Challenge!

1. On the HQ-PC-1 VM, complete the following:
 - Play a long video (more than 5 minutes long) on <https://www.youtube.com>.
 - Run a continuous ping to IP address 4.2.2.2.
2. On the HQ-NGFW-1 CLI, reboot HQ-NGFW-1.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you have performed these steps, see [Verify the HA Failover and FortiGate Roles on page 232](#).

To trigger a failover by rebooting the primary FortiGate

1. On the HQ-PC-1 VM, open a browser, and then visit the following URL:
<https://www.youtube.com>
2. Play a long video (more than 5 minutes long).
3. While the video is playing, open a terminal, and then enter the following command to run a continuous ping to a public IP address:

```
ping 4.2.2.2
```
4. On the HQ-NGFW-1 CLI, enter the following command to reboot HQ-NGFW-1 and trigger a failover:

```
execute reboot
```
5. Press **Y** to confirm that you want to reboot HQ-NGFW-1.

Verify the HA Failover and FortiGate Roles

You will verify the HA failover, and check the roles of FortiGate in an HA cluster.

To verify the HA failover and FortiGate roles

1. On the HQ-PC-1 VM, check the video and terminal that you started earlier.
Because of the failover, HQ-NGFW-2 is now the primary processor of traffic. The ping and video should still be running.
2. Close the browser tab with the video.
3. Return to the terminal, and then press `Ctrl+C` to stop the ping.
4. On the HQ-NGFW-2 CLI, enter the following command to verify that HQ-NGFW-2 is acting as the primary device in the HA cluster:

```
get system status
```

Stop and think!

When HQ-NGFW-1 finishes rebooting and rejoins the cluster, does it rejoin as the *secondary* device, or resume its initial role of the *primary* device?

5. On any FortiGate in the cluster, enter the following command to see the status of all cluster members:

```
get system ha status
```

You should see that HQ-NGFW-1 rejoins the cluster as a secondary device. It lost its role as the primary device.

```
HQ-NGFW-2      , FGVM02TM24013501, HA cluster index = 0
HQ-NGFW-1      , FGVM02TM24013423, HA cluster index = 1
number of vcluster: 1
vcluster 1: work 169.254.0.1
Primary: FGVM02TM24013501, HA operating index = 0
Secondary: FGVM02TM24013423, HA operating index = 1
```



HQ-NGFW-1 becomes the secondary device in the cluster because it has a lower HA uptime than HQ-NGFW-2. In addition, the HA uptime difference between the members is more than 5 minutes.

Also, the override setting is disabled, so priority does not take precedence over uptime.

Trigger an HA Failover by Resetting the HA Uptime

You will trigger a failover by resetting the HA uptime on the current primary FortiGate—which should be HQ-NGFW-2—and then you will verify the role of HQ-NGFW-2 in the HA cluster.

To trigger an HA failover by resetting the HA uptime on FortiGate

1. On the HQ-NGFW-2 CLI console, enter the following command:

```
diagnose sys ha reset-uptime
```



After you reset the HA uptime on HQ-NGFW-2, HQ-NGFW-1 becomes the member with the highest HA uptime. Because the HA uptime difference between the members is more than 5 minutes, HQ-NGFW-1 is elected as the new primary.

HQ-NGFW-2 now has the *secondary* role in the cluster.

2. On the HQ-NGFW-2 CLI, enter the following command to verify this:

```
get system status
```

Observe HA Leave and Join Messages Using Diagnostic Commands

The HA synchronization process is responsible for FGCP packets that communicate cluster status and build the cluster. You will use real-time diagnostic commands to observe this process.

To observe HA failover using diagnostic commands

1. On the HQ-NGFW-1 CLI, enter the following commands:

```
diagnose debug enable
diagnose debug application hataalk 0
diagnose debug application hataalk 255
```



The `diagnose debug application hataalk 0` command stops the debug. You will use this command later.

2. On the HQ-NGFW-2 CLI, enter the following command to reboot HQ-NGFW-2:

```
execute reboot
```

3. Press `Y` to confirm that you want to reboot HQ-NGFW-2.

4. On the HQ-NGFW-1 CLI, view the output while the secondary device reboots and starts communicating with the cluster.

```
HQ-NGFW-1 # <hataalk> vcluster 1: ha prio=0(primary), state/chg time/now=2(work)/1731224894/1731225124
<hataalk> member 'FGVM02TM24013501' lost heartbeat on hbdev 'port7': now=65057, last hb jiffies+timeout=64656+400=65
<hataalk> lost member 'FGVM02TM24013501' heartbeat, delete it
<hataalk> deleting gmember 'FGVM02TM24013501'
<hataalk> vcluster_1: deleting vmember 'FGVM02TM24013501'
<hataalk> vcluster_1: reelect=1, delete-vmember
<hataalk> cfg_changed is set to 1: hataalk_del_member
<hataalk> vcluster_1: reelect=0, hataalk_vcluster_timer_func
<hataalk> vcluster 1: 'FGVM02TM24013423' is elected as the cluster primary of 1 members
```

```
<hatalk> parse options for 'FGVM02TM24013501', packet version=1
<hatalk> new member 'FGVM02TM24013501' is added into group
<hatalk> [hatalk_gmember_update_last_hb_jiffies:200] recv hb packet from 'FGVM02TM24013501' on hbdev='port7' since
<hatalk> vcluster_1: uptime is updated, vmember-uptime-reset-count changed
<hatalk> cfg_changed is set to 1: vmember-uptime-reset-count changed
<hatalk> vcluster_1: vmember 'FGVM02TM24013501' updated, override=0, usr_priority=100, mondev/pingsvr=0/0, uptime/r
<hatalk> cfg_changed is set to 1: hatalk_vcluster_add_vmmember
<hatalk> cfg_changed is set to 0: hatalk_packet_setup_heartbeat
<hatalk> setup new heartbeat packet: nbddev= port7 , packet_version=9
<hatalk> options buf is small: opt_type=41(DEVINFO), opt_sz=13806, buf_sz=1178
<hatalk> pack compressed dev info: dev nr=8, orig sz=13800, z_len=100
<hatalk> heartbeat packet is set on hbdev 'port7'
<hatalk> vcluster_1: reelect=1, vmember updated
<hatalk> reap child 2611 (0)
<hatalk> reap child 2612 (0)
<hatalk> [hatalk_timer_func:1242] need_set_haip=1
<hatalk> 'port7' is selected as 'port_ha'
<hatalk> change port_ha ip to 169.254.0.2
<hatalk> parse options for 'FGVM02TM24013501', packet version=2
<hatalk> vcluster_1: vmember 'FGVM02TM24013501' updated, override=0, usr_priority=100, mondev/pingsvr=0/0, uptime/r
<hatalk> vcluster_1: reelect=1, vmember updated
<hatalk> vcluster_1: 'FGVM02TM24013423' is elected as the cluster primary of 2 members
<hatalk> vcluster_1: state changed, 2(work)->2(work)
<hatalk> vcluster_1: work_as_primary immediately
<hatalk> root: ha_mode=2200, mode=2, state=2(0x2000), ha_prio=0(0x0).
<hatalk:ERROR> 'FGVM02TM24013501' is not found in vcluster_bk
<hatalk> 'port7' is selected as 'port_ha'
<hatalk> vcluster_1: start sending garps
```

The sanitized output shows that the current primary FortiGate is sending heartbeat packets and trying to synchronize its configuration with the configuration of the secondary FortiGate.

5. Press the up arrow key twice, select the second-last command (in this case, `diagnose debug application hatalk 0`), and then press `Enter` to stop the debug output on HQ-NGFW-1.

Exercise 3: Configuring the HA Management Interface

In this exercise, you will examine how to configure a spare interface in the cluster as a reserved HA management interface. This allows both FortiGate devices to be reachable for management purposes regardless of the member role.

If you don't configure a reserved HA management interface, the primary FortiGate handles your cluster management connections. However, you can access the CLI of the secondary FortiGate from the primary FortiGate CLI, or by using the console connection of the secondary FortiGate.

You can also configure an in-band HA management interface, which is an alternative to the reserved HA management interface, and does *not* require reserving an interface that is only for management access.



Unless instructed otherwise, always use the console connection of HQ-NGFW-1 and HQ-NGFW-2 to access the CLI. This ensures that you can access the device CLI regardless of the HA role.

Access the Secondary FortiGate CLI Through the Primary FortiGate CLI

You will connect to the secondary FortiGate CLI through the primary FortiGate CLI.

To access the secondary FortiGate CLI through the primary FortiGate CLI

1. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Enter the following command to access the secondary FortiGate CLI through the primary FortiGate heartbeat interface:
`execute ha manage <id> admin`



Use ? to list the values for <id>.

```
HQ-NGFW-1 # execute ha manage
<id>      please input peer box index.
<0>      Subsidiary unit FGVM02TM24013501

HQ-NGFW-1 # execute ha manage 0 admin
```

3. When prompted, enter the password Fortinet1! to log in to HQ-NGFW-2.

```
HQ-NGFW-1 # execute ha manage 0 admin
admin@169.254.0.1's password:
HQ-NGFW-2 #
```

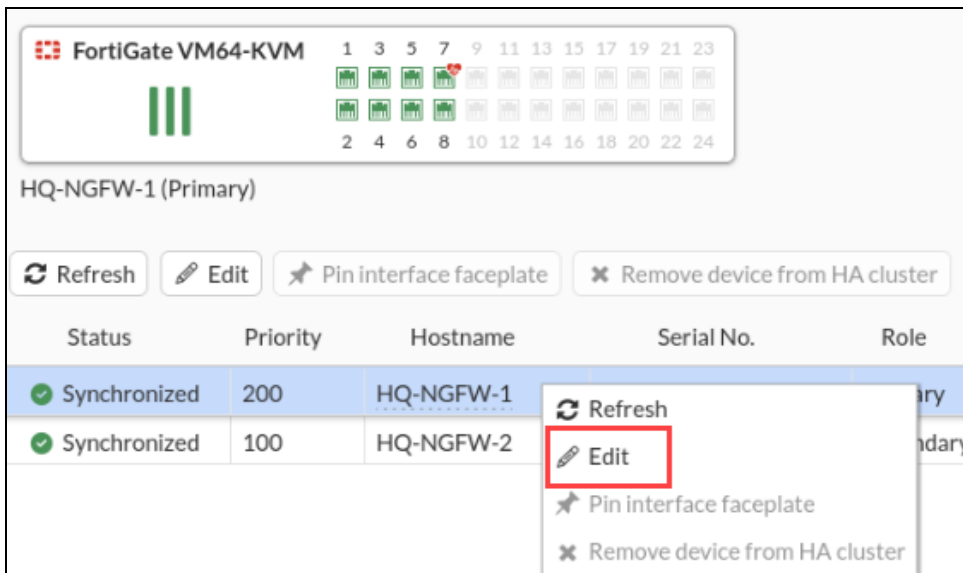
4. Enter the following command to get the status of the secondary FortiGate:
`get system status`
5. View the `Current HA mode` line.
You will notice that HQ-NGFW-2 is `a-p, secondary`.
6. Enter the following command to return to the HQ-NGFW-1 CLI:
`exit`

Set Up a Reserved HA Management Interface

You will use an unused interface on the FortiGate devices in an HA cluster to configure a reserved HA management interface and a unique IP address for each member. This way, you can access each member directly, regardless of its role.

To set up a reserved HA management interface

1. On the HQ-PC-1 VM, open a browser, and then log in to the HQ-NGFW-1 GUI at `10.0.11.254` with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **System > HA**.
3. Right-click **HQ-NGFW-1**, and then click **Edit**.



4. Enable **Management Interface Reservation**, and then in the **Interface** field, select **port8**.
5. Click **OK**.



port8 connects to the same LAN segment as port4.

Configure and Access the Primary FortiGate Using the Reserved HA Management Interface

You will configure and verify access to the primary FortiGate using the reserved HA management interface.

To configure and verify access to the primary FortiGate using the reserved HA management interface

1. On the HQ-NGFW-1 CLI, log in with the following credentials:

- Username: admin
- Password: Fortinet1!

2. Enter the following commands to configure port8:

```
config system interface
  edit port8
  set ip 10.0.11.251/24
  set allowaccess ping ssh snmp https
end
```



Even though this address overlaps with port4, which is not allowed by default (FortiGate does not allow overlapped subnets by default), it is allowed here because the routing entries for the reserved HA management interface are excluded from the routing table.

3. On the HQ-PC-1 VM, open a browser, and then log in to the HQ-NGFW-1 GUI at 10.0.11.251 (note the IP address) with the following credentials:

- Username: admin
- Password: Fortinet1!

This verifies connectivity to port8.

Configure and Access the Secondary FortiGate Using the Reserved HA Management Interface

You will configure and verify access to the secondary FortiGate using the reserved HA management interface.

Take the Expert Challenge!

1. On the HQ-NGFW-2 CLI, complete the following:
 - Verify that the reserved HA management interface is synchronized with the secondary device.

```
show system ha
```
 - Verify that **port8** has no configuration, and then configure **port8 IP/Netmask** as `10.0.11.252/24` with the same `allowaccess` configured for HQ-NGFW-1 **port8**.
2. On the HQ-PC-1 VM, log in to the HQ-NGFW-2 GUI (`admin/Fortinet1!`) using the port8 IP address to verify connectivity.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After the configuration is ready, see [Disconnect HQ-NGFW-2 From the Cluster on page 238](#).

To configure and verify access to the secondary FortiGate using the management interface

1. On the HQ-NGFW-2 CLI, enter the following command to verify that the reserved HA management interface is synchronized with the secondary device:

```
show system ha
```

Look for `ha-mgmt-status` and `config ha-mgmt-interfaces`. These should already be configured.

2. Enter the following command to verify that port8 has no configuration:

```
show system interface port8
```

3. Configure port8, using the following commands:

```
config system interface
  edit port8
    set ip 10.0.11.252/24
    set allowaccess ping ssh snmp https
  next
end
```

4. On the HQ-PC1-1 VM, open a browser, and then log in to the HQ-NGFW-2 GUI at `10.0.11.252` (note the IP address) with the following credentials:

- Username: `admin`
- Password: `Fortinet1!`

This will verify connectivity to port8.

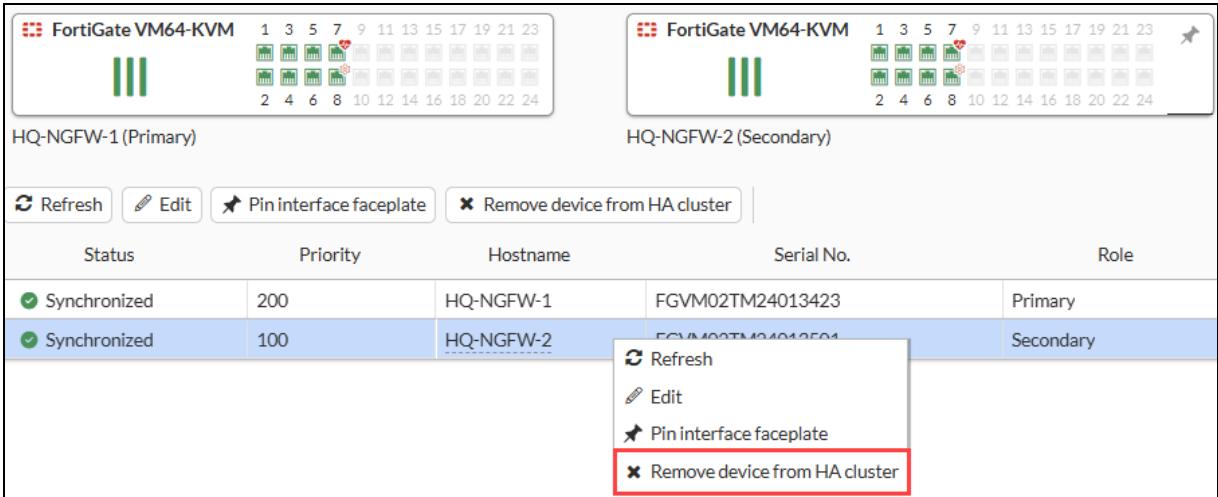
Each device in the cluster now has its own management IP address for monitoring purposes.

Disconnect HQ-NGFW-2 From the Cluster

You will disconnect HQ-NGFW-2 from the cluster. HQ-NGFW-2 will prompt you to configure an IP address on any port on HQ-NGFW-2 so that you can access it after the disconnection.

To disconnect HQ-NGFW-2 from the cluster

- On the HQ-PC-1 VM, open a browser, and then log in to the HQ-NGFW-1 GUI at 10.0.11.254 with the following credentials:
 - Username: admin
 - Password: Fortinet1!
- Click **System > HA**.
- Right-click **HQ-NGFW-2**, and then click **Remove device from HA cluster**.



- When prompted, configure the following settings:

Field	Value
Interface	port4
IP/Netmask	10.0.11.250/24

- Click **OK**.
This removes the FortiGate from the HA cluster.

Restore the HQ-NGFW-2 Configuration

You will restore the HQ-NGFW-2 configuration, so that you can use HQ-NGFW-2 in the next labs.



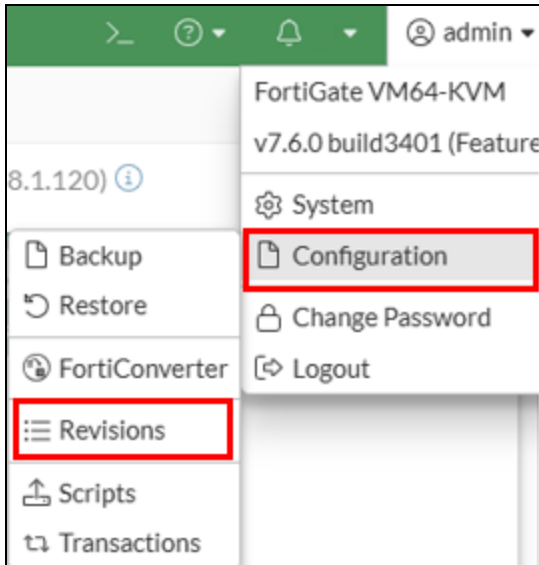
Failure to perform these steps will prevent you from doing the next exercises.

To restore the HQ-NGFW-2 configuration file

- On the HQ-PC-1 VM, open a browser, and then log in to the HQ-NGFW-2 GUI at 10.0.11.250 with the following credentials:

- Username: admin
- Password: Fortinet1!

2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **initial**, and then click **Revert**.

✕ Delete i Details Diff ↺ Revert Save Changes ⚠			
Config ID	Username	Date	Comments
7.6.0 build 3401 2			
2	admin	2024/12/16 11:52:52	HQ-NGFW-2-HA
1	admin	2024/12/16 11:49:59	HQ-NGFW-2-initial

5. Click **OK** to reboot.



Failure to perform these steps will prevent you from doing the next exercises.

Lab 15: Diagnostics Performance

In this lab, you will run diagnostic commands to learn about the current status of FortiGate. You will also use the sniffer and debug flow tools to troubleshoot and fix a connectivity problem.

Objectives

- Monitor for abnormal behavior, such as traffic spikes
- Diagnose problems at the physical and network layers
- Diagnose connectivity problems using the GUI debug flow tool
- Diagnose resource problems, such as high CPU or memory usage

Time to Complete

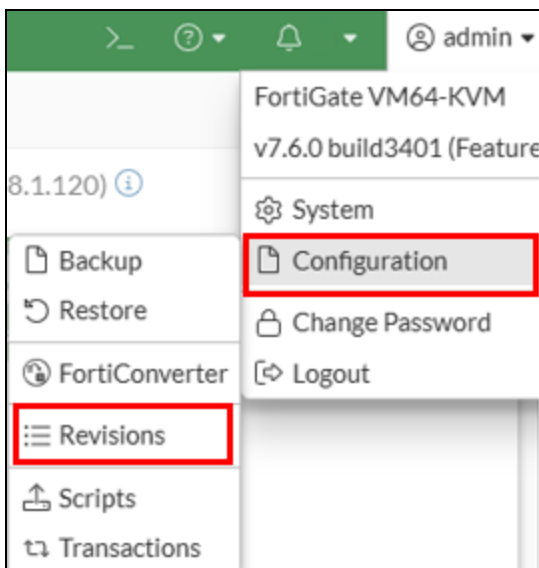
Estimated: 30 minutes

Prerequisites

Before you begin this lab, you must restore a configuration file on HQ-NGFW-1.

To restore the HQ-NGFW-1 configuration file

1. On the HQ-NGFW-1 GUI, log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. In the upper-right corner, click **admin**, and then click **Configuration > Revisions**.



3. Click **+** to expand the list.
4. Select the configuration with the comment **HQ-diagnostics**, and then click **Revert**.

Delete

Details

Diff

Revert

Save Changes

Config ID	Username	Date	Comments
7.6.0 build 3401 16			
16	admin	2024/12/13 16:17:45	HQ-diagnostics
15	admin	2024/12/13 16:15:04	HQ-NFGW-1-HA
14	admin	2024/12/13 16:12:36	HQ-SF
13	admin	2024/12/13 16:08:57	HQ-SD-WAN
12	admin	2024/12/13 16:06:23	HQ-static-ipsec
11	admin	2024/12/13 16:02:48	HQ-dialup-ipsec
10	admin	2024/12/13 15:59:48	HQ-SSL-VPN
9	admin	2024/12/13 15:57:06	HQ-app-control
8	admin	2024/12/13 15:53:55	HQ-web-filtering
7	admin	2024/12/13 15:45:29	HQ-AV
6	admin	2024/12/13 15:41:55	HQ-Certificate
5	admin	2024/12/13 15:39:27	HQ-FSSO
4	admin	2024/12/13 15:36:07	HQ-FW-Auth
3	admin	2024/12/13 15:33:19	HQ-routing
2	admin	2024/12/13 15:30:41	HQ-firewall-policy
1	admin	2024/12/13 15:28:26	HQ-NGFW-1-initial

5. Click **OK** to reboot.

Exercise 1: Determining What Is Happening Now

In this exercise, you will use CLI commands to get information about FortiGate, such as traffic volume, CPU usage, memory usage, and the ARP table.

Run Diagnostic Commands

You will run some diagnostic commands and make a note of some of the information that is displayed.

To run diagnostic commands

1. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Find the following information and write down your answers in the space provided—refer to the list of commands that follows to get the answers:

Field	Value
Firmware branch point	
Current HA mode	
Host name	
CPU utilization	
Memory utilization	
Average network usage	
Average session setup rate	
Negotiated speed and duplex mode for interface port2	
MTU for port2	
MAC address for the IP address 100.65.0.254	
Name of the process consuming the most CPU (if any)	
Name of the process consuming the most memory	

Enter the following CLI commands to find the information requested above:



```
get system status
get system performance status
get hardware nic port1
get system arp
diagnose sys top 1
```

(Press **Shift+P** to order the processes by CPU usage, **Shift+M** to order them by memory usage, or **Q** to stop.)

3. Close the HQ-NGFW-1 CLI session.

Exercise 2: Troubleshooting a Connectivity Problem

In this exercise, you will use the sniffer and debug flow tool to troubleshoot a network connectivity problem.

Identify the Problem

There is a network connectivity problem between the HQ-PC-1 VM and the Linux ISP server.

To identify the problem

1. On the HQ-PC-1 VM, open a terminal window.
2. Enter the following command to start a continuous ping to the Linux ISP server (IP address 100.65.0.254):

```
ping 100.65.0.254
```

The ping is failing. You will use the sniffer and debug flow tool on HQ-NGFW-1 to find out why.

3. Do not close the terminal window—keep the ping running.

Use the Sniffer

Take the Expert Challenge!

Now that you understand what the problem is, try to fix it without looking at the FortiGate configuration. Use the built-in sniffer and debug flow tool to troubleshoot the problem.

If you require assistance, or to verify your work, use the step-by-step instructions that follow.

After you complete the challenge, see [Test the Fix on page 248](#).

You will start troubleshooting by sniffing the ICMP traffic going to the Linux ISP server.

To use the sniffer

1. On the HQ-NGFW-1 CLI, log in with the following credentials:
 - Username: admin
 - Password: Fortinet1!
2. Enter the following command to sniff the ICMP traffic to 100.65.0.254:

```
diagnose sniffer packet any "icmp and host 100.65.0.254" 4
```
3. Observe the output.

```
interfaces=[any]  
filters=[icmp and host 100.65.0.254]
```

```
5.439019 port4 in 10.0.11.50 -> 100.65.0.254: icmp: echo request
10.442347 port4 in 10.0.11.50 -> 100.65.0.254: icmp: echo request
15.444343 port4 in 10.0.11.50 -> 100.65.0.254: icmp: echo request
20.545397 port4 in 10.0.11.50 -> 100.65.0.254: icmp: echo request
```

The packets are arriving on FortiGate, but FortiGate is not routing them.

4. Press **Ctrl+C** to stop the sniffer.

Use the GUI Debug Flow Tool

You will run the GUI debug flow tool to get information about why FortiGate is dropping the packets.

To use the GUI debug flow tool

1. Connect to the HQ-NGFW-1 GUI, and then log in with the following credentials:
 - Username: `admin`
 - Password: `Fortinet1!`
2. Click **Network > Diagnostics**, and then click the **Debug Flow** tab.
3. In the **Number of packets** field, change the value to 3 packets.
4. Enable **Filters**, and then configure the following settings:

Field	Value
IP address	100.65.0.254
Protocol	ICMP

5. Click **Start debug flow**.
The output should be similar to the following example:

Time	Message
Packet Trace #1 14	
06:39:29	vd-root:0 received a packet(proto=1, 10.0.11.50:3->100.65.0.254:2048) tun_id=0.0.0.0 from port4. type=8, code=0, id=3, seq=168.
06:39:29	allocate a new session-00000ec6
06:39:29	in-[port4], out-[]
06:39:29	len=0
06:39:29	result: skb_flags-02000000, vid-0, ret-no-match, act-accept, flag-00000000
06:39:29	find a route: flag=00000000 gw-0.0.0.0 via port2
06:39:29	in-[port4], out-[port2], skb_flags-02000000, vid-0, app_id: 0, url_cat_id: 0
06:39:29	gnum-100004, use addr/intf hash, len=1
06:39:29	checked gnum-100004 policy-0, ret-matched, act-accept
06:39:29	ret-matched
06:39:29	policy-0 is matched, act-drop
06:39:29	after iprobe_captive_check(): is_captive-0, ret-matched, act-drop, idx-0
06:39:29	after iprobe_captive_check(): is_captive-0, ret-matched, act-drop, idx-0
06:39:29	Denied by forward policy check (policy 0)

FortiGate receives the ICMP packet from 10.0.11.50 to 100.65.0.254 from port4.

```
vd-root:0 received a packet(proto=1, 10.0.11.50:3->100.65.0.254:2048) tun_id=0.0.0.0
  from port4. type=8, code=0, id=3, seq=168.
```

It creates a new session.

```
allocate a new session-00000ec6
```

It finds a route for the destination 100.65.0.254 through port2.

```
find a route: flag=00000000 gw-0.0.0.0 via port2
```

It drops the packet. The debug flow shows the error message.

```
Denied by forward policy check (policy 0)
```

The Denied by forward policy check message indicates that a firewall policy denied the traffic. It could be either a denied policy that the administrator explicitly configured, or the implicit denied policy for traffic that does not match a configured policy.

policy 0 indicates that the default implicit policy denied the traffic. If an explicitly configured policy blocked the traffic, its policy ID number would be indicated in this output, instead of 0.

Fix the Problem

Now that you have found the cause of the problem, you will fix it.

To fix the problem

1. Continuing on the HQ-NGFW-1 GUI, click **Policy & Objects > Firewall Policy**.
2. Look at the firewall policies.
The **Internet** firewall policy does not allow ICMP traffic (only HTTP)—this is why FortiGate is dropping the ping packets.
3. Edit the **Internet** firewall policy.
4. Change the service from **HTTP** to **ALL**.
5. Click **OK**.

Test the Fix

You will test to confirm that the configuration change fixed the problem.

To test the fix

1. On the HQ-PC-1 VM, check the terminal window to see if the continuous ping is working now.
2. Press **Ctrl+C** to stop the ping, but leave the terminal open.
3. On the HQ-NGFW-1 CLI session where you are running debug commands, clear all the ICMP sessions from the session table, using the following commands:

```
diagnose sys session filter clear
diagnose sys session filter proto 1
diagnose sys session clear
```
4. Continuing on the HQ-NGFW-1 GUI, click **Network > Diagnostics**, and then click the **Debug Flow** tab.
5. In the **Number of packets** field, change the value to 3 packets.
6. Enable **Filters**, and then configure the following settings:

Field	Value
IP address	100.65.0.254
Protocol	ICMP

7. Click **Start debug flow**.
There should not be any output yet, because the ping is not running.
8. Return to the terminal window, and then start the ping again.

```
ping 100.65.0.254
```
9. Check the debug flow output.

Packet Trace #4 27	
06:48:58	vd-root:0 received a packet(proto=1, 10.0.11.50:4->100.65.0.254:2048) tun_id=0.0.0.0 from port4. type=8, code=0, id=4, seq=1.
06:48:58	allocate a new session-0000170b
06:48:58	in-[port4], out-[]
06:48:58	len=0
06:48:58	result: skb_flags-02000000, vid-0, ret-no-match, act-accept, flag-00000000
06:48:58	find a route: flag=00000000 gw-0.0.0.0 via port2
06:48:58	in-[port4], out-[port2], skb_flags-02000000, vid-0, app_id: 0, url_cat_id: 0
06:48:58	gnum-100004, use addr/intf hash, len=2
06:48:58	checked gnum-100004 policy-1, ret-matched, act-accept
06:48:58	ret-matched
06:48:58	gnum-4e20, check-fffffffa002c9c7
06:48:58	checked gnum-4e20 policy-6, ret-no-match, act-accept
06:48:58	checked gnum-4e20 policy-6, ret-no-match, act-accept
06:48:58	checked gnum-4e20 policy-6, ret-no-match, act-accept
06:48:58	gnum-4e20 check result: ret-no-match, act-accept, flag-00000000, flag2-00000000
06:48:58	gnum-4e20 check result: ret-no-match, act-accept, flag-00000000, flag2-00000000
06:48:58	gnum-4e20 check result: ret-no-match, act-accept, flag-00000000, flag2-00000000
06:48:58	find SNAT: IP-100.65.0.101(from IPPOOL), port-5121
06:48:58	policy-1 is matched, act-accept
06:48:58	after iprope_captive_check(): is_captive-0, ret-matched, act-accept, idx-1
06:48:58	after iprope_captive_check(): is_captive-0, ret-matched, act-accept, idx-1
06:48:58	after iprope_captive_check(): is_captive-0, ret-matched, act-accept, idx-1
06:48:58	in-[port4], out-[port2], skb_flags-02000000, vid-0
06:48:58	len=0
06:48:58	Allowed by Policy-1: SNAT
06:48:58	npu_state=0x100, hook=4
06:48:58	SNAT 10.0.11.50->100.65.0.101:5121

The output is a bit different now. The error message is not displayed and you can see a few new logs.

The firewall policy with the ID 1 is allowing traffic.

Allowed by Policy-1: SNAT

FortiGate applies source NAT (SNAT).

SNAT 10.0.10.50->100.65.0.101:5121

Additionally, you can see the debug flow logs from the return (ping reply) packets.

```
vd-root:0 received a packet(proto=1, 100.65.0.254:5121->100.65.0.101:0) tun_
id=0.0.0.0 from port2. type=0, code=0, id=5121, seq=1.
Find an existing session, id-0000170b, reply direction
DNAT 100.65.0.101:0->10.0.10.50:4
find a route: flag=00000000 gw-0.0.0.0 via port4
```



The procedure in this exercise describes what you should usually do when troubleshooting connectivity problems on FortiGate. Sniff the traffic first to check that the packets are arriving on FortiGate and that FortiGate is routing them correctly. If the sniffer shows that FortiGate is dropping the traffic, use the debug flow tool to find out why.

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